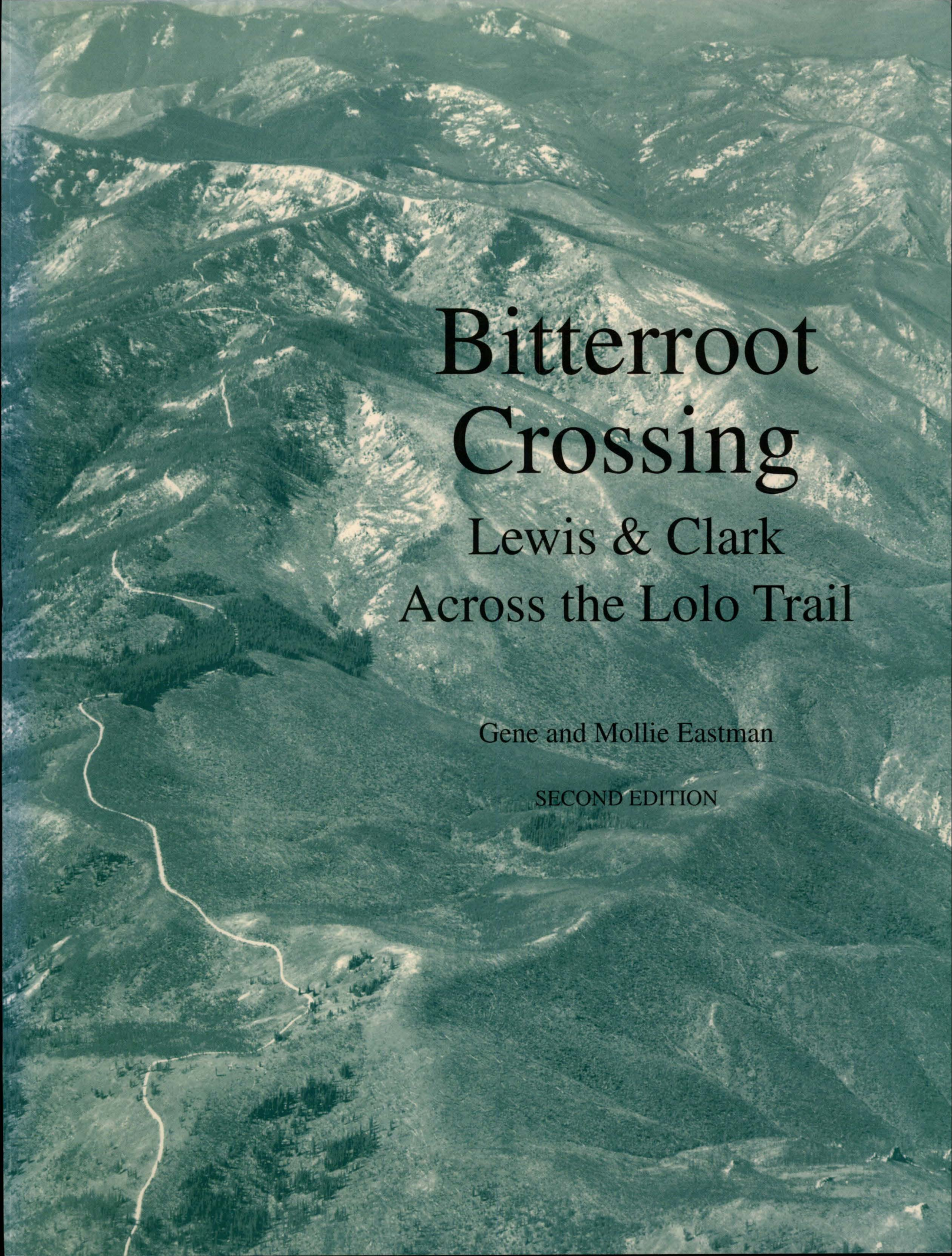

Bitterroot Crossing

Lewis & Clark Across the Lolo Trail

Gene and Mollie Eastman

SECOND EDITION

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Lewis & Clark Across the Lolo Trail

Gene and Mollie Eastman

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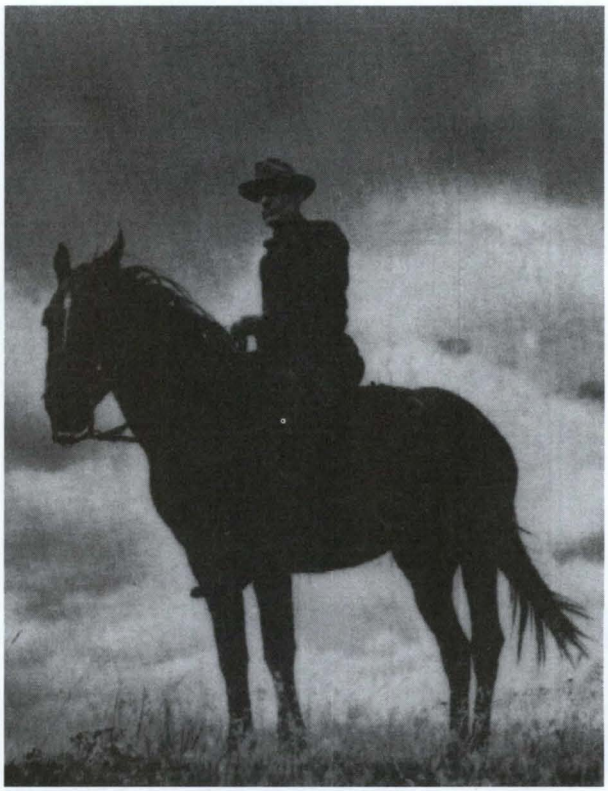
Dedication

Dedicated to the Nez Perce, who befriended the white man beginning with Lewis and Clark;
And to the old Spanish mustang horse—without him, we'd have never tamed the West;
And to Teddy Roosevelt and Elers Koch—who wanted to leave the West wild.

"An Indian never gets lost. It could be so dark a night the bats stay home; or smoke from a wild fire that clouds the sky; but the Indian can still travel in a straight path."

Charles M. Russell, *Trails Plowed Under* (1927)

Indian Youth at the Pendleton Roundup in 1920 (Photo courtesy of the National Park Service, Nez Perce National Historical Park).



Elers Koch: 1880-1954 (Photo courtesy of Mrs. Peter Koch).

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This report would not have been possible without the help and dedication of the following past Forest Service individuals who left a written record of their stewardship and concerns for the Lewis and Clark Trail: Elers Koch (1940s), who located the Lewis and Clark Trail and campsites, Ralph Space (1960s), who also mapped and located the trail and campsites and protected them from logging, and Andy Arvish (1970s), who mapped the trail and campsites along with making recommendations that the trail tread be protected and a corridor be protected from logging and road building.

The following dedicated Forest Service employees are presently managing the historic trails on the Clearwater National Forest and are doing an outstanding job. Our sincere gratitude goes to James Caswell, former Clearwater National Forest Supervisor; Cindy Lane, Ranger for the Pierce and Lochsa Districts; Doug Gober, Ranger for the North Fork District; Pete Minard, Technical Service Manager; Mark Hill, Zone Archaeologist; Robbin Johnston, Archaeologist; Jack Stratton, Archaeologist; Duane Annis, Recreation; Roland Kehlet, Trails Coordinator; and Steve Staub, Engineering.

Cindy Lane and Jim Caswell are to be commended for their efforts in keeping the Hungery Creek and Fish Creek portions of the Lewis and Clark Trail protected from off road vehicles. This area is also proposed for the Lewis and Clark Wilderness area, which would leave the trails in their historic condition.

The Clearwater National Forest had done a commendable job of managing the Lewis and Clark Trail with the knowledge that they have. It is the hope of this researcher that the research presented in this book will give the managers a new view of the Lewis and Clark Trail location, its past history, and a basis to make more precise decisions that will both protect and make available to the public this national heritage.

This report was a team effort

Special thanks to Larry Jones (Idaho State Historical Society) and Dennis Baird (Head of Reference, University of Idaho Library) who provided historical research information and historical maps for this report, and to Donna K. Smith (Reference Assistant, University of Idaho Library) for her tireless editing of this manuscript. Special thanks also go to Robbin Johnston, Ph.D., for his guidance and suggestions on how to locate and define old historic trails; to Mark Hill (zone archaeologist, Clearwater and Nez Perce National Forests) who provided the old maps from the map drawer; to Andy Arvish, for furnishing his recommendations and reports on the Lolo Trail; and to Ralph Space, who cared for the Lewis and Clark Trail and published his findings.

And finally, in memory of Elers Koch, the foundation researcher who located and wrote about the Lolo Trail from 1915 to 1940.

> "*A nation which does not remember what it was yesterday, does not know what it is today, nor what it is trying to do. We are trying to do a futile thing if we do not know where we came from or what we have been about.*"
>
> President Woodrow Wilson, 1911.

Preface

The geographic theme of this book is the beautiful mountainous region of Idaho stretching between Lolo Pass and the Clearwater River valley near modern day Orofino. This landscape was used by the Nez Perce people for thousands of years before Lewis and Clark reached the area in 1805. While modified by humans in the succeeding two hundred years, the Clearwater Mountains retain many scenic and historical treasures little changed by modern development.

Within the context of the Lewis and Clark Bicentennial observations, the authors of this book seek to examine the historical resources of this region, the trail remnants in particular. In doing so, they have applied a fine mix of archival research, cartographic and air photo sleuthing, all combined with a thorough and remarkable knowledge of the ground itself. The Eastmans have also been successful in collecting much of the oral history of land management in the corridor traveled by the Nez Perce Indians and subsequently, Lewis and Clark.

This book contains a fine bibliography of the many applicable written resources needed to understand the history and subsequent management of the trail corridor. The bibliography is especially useful in listing the important, but hard to locate, "gray" literature relating to the region.

Also included is the most thorough cartobibliography ever compiled for the Clearwater Mountains, including maps from collections throughout the region, and from the National Archives as well. These maps, coupled with careful use of early air photos, offer a great deal of useful and important information on the human modification of this region's great historical resources. No other scholars of the route of Lewis and Clark in Idaho have been as successful in this important map research as have been the Eastmans.

This book makes a useful contribution to our understanding of trail distances and the abilities of early travelers to cross a difficult landscape. It also offers some valuable insights into the types and locations of trails used by the Nez Perce and Salish people who crossed the area in search of buffalo, fish, and other trade goods. All of these contributions to our knowledge are informed by the thousands of hours spent in the field by the authors. The result is a valuable and important contribution to the scholarship of the adventures of Lewis and Clark. There are also some useful suggestions for the preservation of the remaining historical resources.

— Dennis Baird

Introduction

As the bicentennial of the Lewis and Clark Expedition approaches, identifying the explorers' route through Idaho becomes an important and controversial task. Nearly thirty-eight years ago Congress recognized the historical importance of the expedition's route, and the Lewis and Clark Trail Commission was established on October 6, 1964, as a five-year temporary body with advisory powers.[1] Congress asked the Commission to identify, mark, and keep publicly available the route traversed by Captains Meriwether Lewis and William Clark on their 1804-06 travels from the mouth of Wood River, Illinois, to the Pacific Ocean. As a result, "in 1978 the roughly 3,700 mile long Lewis and Clark Trail...was designated a National Historic Trail."[2]

Lewis and Clark's journey over the Lolo Trail and the Bitterroot Mountains in September of 1805 was one of the most difficult passages of the expedition. This portion of the trail covers approximately 150 miles from Lolo, Montana, to Orofino, Idaho. At the Weippe Prairie the trail forks, the 1805 fork westward goes to Orofino, and the 1806 eastward return route goes from Kamiah to the Weippe Prairie. This portion of the Lewis and Clark Trail crosses the Lolo National Forest, the Clearwater National Forest (where approximately 96 miles of the trail are located), private lands, and Idaho state lands. The trail through the Clearwater National Forest is spectacular, traveling through pristine high mountain peaks, ridges, and meadows.

The Lewis and Clark Trail in Idaho is the most valuable portion of the trail, in many places the trail has not been disturbed by man for almost 200 years. In the past, a number of researchers, including Ralph Space, a former supervisor of the Clearwater National Forest, have attempted to positively identify the trail, but it has proved to be a difficult task. The goal of identifying the trail has not been realized and many sections of the Lewis and Clark route have not been properly identified. This needs to be done so that visitors can experience the same difficult challenge that faced the Lewis and Clark Party. The visitor will want to walk where "they" walked, walk on unaltered trail tread straight up the mountains and across the ridges. Proper identification will also enable the United States Forest Service to protect not only the Lewis and Clark Trail, but also prehistoric and historic sites and travel networks. Furthermore, trail studies will help anthropologists to understand the Lewis and Clark trail as it relates to cultural migration and diffusion by Native Americans.

The Lewis and Clark Trail is only one trail in the "Lolo Trail System," which was defined by the Clearwater National Forest in 1985 as a corridor containing the following trails: the Lewis and Clark Trail, the Lolo Trail (or Northern Nez Perce Trail), the Bird-Truax Trail (minimal grading of the Lolo Trail, 1866), the Nee-Me-Poo (new name for the Lolo Trail, 1877 Nez Perce War route), and the primitive Lolo Truck Road (Road 500). This trail corridor on the Clearwater National Forest varies from 2 miles wide to less than ¼ mile wide and encompasses most of the five trails, which follow roughly the same route over much of the forest, and became a National Historic Landmark in 1960.[3]

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[1] Lewis and Clark Trail Commission, *The Lewis and Clark Trail, Final Report of The Lewis and Clark Trail Commission* (Washington, D.C.: GPO, 1969), 1. For in-depth information about the National Council of the Lewis and Clark Bicentennial, the official coordinator of the 200th anniversary of the expedition, see Sammye Meadows, "Partnering for the Lewis & Clark Bicentennial," *History News*, 56 (Spring 2001): 5-9.

[2] U.S. Forest Service, Northern Region, *Lolo Trail System Implementation Guidelines: Draft* (Orofino, ID: Clearwater National Forest, 1985), 1.

[3] Carol D. Shull, Chief of Registration, National Register of Historic Places, to David F. Jolly, Regional Forester, received 24 August 1993, gives the designated date of the Lolo Trail National Historic Landmark as October 9, 1960. (Copy in Eastman Papers (unprocessed), Special Collections and Archives, University of Idaho Library, Moscow, Idaho).

Ralph Space, retired forest supervisor of the Clearwater National Forest and author of several works on the Lolo Trail, defined the historical trails on the Clearwater National Forest as the Lolo Trail (also defined as the Northern Nez Perce Trail), the Bird-Truax reconstruction, and the Forest Service reconstruction of the Lolo Trail.[4] Space also further defined the Lolo Trail as one of the two main routes of travel for the Nez Perce to the buffalo herds in Montana and Wyoming.[5] Space acknowledged the historic importance of the Lolo Trail as "the oldest route across the Continental United States traveled by white man. Lewis and Clark first traveled it in 1805 and their journey alone makes quite a story."[6]

The oldest locations of the historic trails can be determined by examining historic maps. George B. Nicholson's 1867 map of the Lolo Trail identifies the trail (also defined as the Lou-Lou or Lo-Lo Trail) as coming from the Weippe Prairie to Musselshell Meadows, over Snowy Summit to Rocky Ridge and along the divide to Willow Ridge. At this juncture Lewis and Clark, coming up out of Hungery Creek, joined the Lolo Trail, and continued eastward on the trail into Montana.[7] The 1879 map of the Nez Perce Indian Campaign by the U.S. Army Department of the Columbia indicates the same location as Nicholson's map.[8] Forest Service maps (beginning with Lieberg's 1898 map of the Bitterroot Reserve and subsequent maps by the Clearwater National Forest up to 1942) also define the Lolo Trail as the route from the Weippe Prairie to Musselshell Meadows, over Snowy Summit and Rocky Ridge to Willow Ridge and on to Montana, on the historically correct Lolo Trail.[9]

In 1866 the Lolo Trail was rebuilt with minimal grading by Bird-Truax. From 1902 through 1907, the Forest Service also rerouted the Lolo Trail. In the 1930s, the Forest Service with the Civilian Conservation Corps program built the 500 Road which follows closely the original Lolo Trail. In the 1960s, the Forest Service began intense logging of the historic trails, which has continued to the present day.

However, the Forest Service has recognized the importance of identifying the trail, and has continued to conduct research and fieldwork over the years. This work led up to the field reconnaissance of 1983 and 1984 which was designed to locate trail tread and map the approximate trail system across the Clearwater National Forest.[10] Forest Service archaeologist Milo McLeod accomplished similar work on the Lolo National Forest in 1977 and 1978.[11] Accordingly, when the Forest Service defined the Lolo Trail as five trails it also concluded the trail was "probably never one trail prior to 1866, but was most likely a braided trail system covering a major ridge or drainage which sought out the path of least resistance." Thus, the braided trail myth was created. This researcher found that one reason for the existence of braided trails is the Forest Service's reconstruction and relocation of old historic trails. Ralph Space documented the braided trails created by the Forest Service in its relocation of the Lolo Trail in Siberia Creek.[12]

The Forest Service's labeling of the trail is confusing because there are not five separate trail treads on the ground. There is, however, one trail tread in places, and branches in others; some of these branches are historical and others were created by livestock grazing (sheep, cattle, and horses), hunters, game animals (deer, elk, and moose), or by Forest Service reconstruction of segments of the historical trail. William R. (Bud) Moore, a long-time resident and author, defined "the Lewis and Clark Trail as a single path across the Clearwater National Forest except

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[4] U.S. Forest Service, Northern Region, *The Clearwater Story: A History of the Clearwater National Forest*, by Ralph S. Space (Missoula, MT: U.S. Forest Service, 1964; edited and re-published Washington, D.C.: GPO, 1981), map inside back cover. Hereafter cited as *The Clearwater Story* (1981).

[5] Ralph S. Space, *The Lolo Trail: A History of Events Connected With the Lolo Trail Since Lewis and Clark* (Lewiston, ID: Printcraft, 1988), 1.

[6] Ibid., introduction.

[7] G. B. Nicholson, "Explorations and Surveys for a Wagon Road from Virginia City, Montana to Lewiston, Idaho" [map reproduction], State Historical Society of Wisconsin, Whi(X3) 51707, 1866, as reproduced by Dennis W. Baird, ed., in *With Bird and Truax on the Lolo Trail: Building the Virginia City to Lewiston Wagon Road, 1865-67* (Moscow, ID: University of Idaho Library, 1999), 22. Ralph Space differs with Nicholson, and places the trail junction of the trail at Sherman Saddle (Space, *The Lolo Trail*, 14).

[8] Robert H. Fletcher, "Department of Columbia Map of the Nez Perce Indian Campaign," in *Report of the Secretary of War, 1877*, U.S. House, 45th Congress, 2d Session, H.Ex.Doc.1, Pt. 2 (Washington, D.C.: GPO, 1877), insert.

[9] J. B. Lieberg, "The Bitterroot Forest Reserve Showing Land Classification" [map], 1" = 4/5 mile, 1898, in "The Bitterroot Forest Reserve." pp. 317-410, by John B. Leiberg, *Twentieth Annual Report of the United States Geological Survey to the Secretary of the Interior, 1898-99*, Pt.5. (Washington, D.C.: GPO, 1900), plate CXIV; and U.S. Forest Service, "Clearwater National Forest, Idaho [map]" (U.S. Forest Service, 1915, 1917, 1926, 1930, 1936, 1942).

[10] U.S. Forest Service, "Clearwater National Forest, Idaho [map] (U.S. Forest Service, 1986).

[11] *Lolo Trail System Implementation Guidelines*, 21.

[12] Ibid., 1.

for in the meadows."[13]

Current Forest Service management has not improved the situation as it has not defined the historic trails (both the Lolo Trail and the Lewis and Clark Trail) accurately, and manages the trails in bits and pieces, instead of managing them as a whole unit. Another problem lies in the practice of relocating the historic trails, which has intensified in the 1990s.

There are a number of sources the researcher must use if the Lewis and Clark Trail is to be properly identified. Lewis and Clark's journals, journals of the other members of the expedition, William Clark's maps, and both explorers' compass bearings all must be examined. Historical maps are a tremendous source of information, as they provide a chronological record of the landscape and how it has changed over time. The researcher has to be able to read the ground (which is becoming a lost art), and do a thorough research of the historical record. He has to determine if the trail tread was built by the Forest Service, railroad survey parties in the early 1900s, hunters, game, or Nez Perce Indians, while keeping in mind that the trail is an old Indian trail that was rebuilt in places by the Forest Service. Finally, fieldwork is essential, as the researcher has to apply to the landscape the knowledge learned from every resource to be able to identify the trail tread by its characteristics. (Indian trail characteristics will be examined in detail in Chapter VI, Methodology Used in This New Effort).

In this researcher's opinion there are major errors in the Forest Service and other researchers' locations of the Lewis and Clark Trail and campsites. For example, three general trail locations have been offered by various experts as the route across Fish Creek. The first route is the trail location supported by the Forest Service; it travels north of Dollar Ridge, then north along the side of the 500 Road. The historic trail is located on an old Forest Service trail built in the 1920s that parallels the 500 Road[14] The trail then leaves the main (500 Road) ridge and turns 90° east on an old Indian trail that travels down Hungery Creek. Most researchers disagree with this location because it does not match the journals and Clark's maps; the Forest Service would have benefited from studying old Forest Service maps to understand the history of how the trail tread that they chose appeared on the ground.

The second route is offered by Steve Russell, an Iowa State University researcher, who has placed the trail down the first ridge east of the Forest Service Trail 230 to Fish Creek Meadows.[15] This proposed location does not match the journals, Clark's maps, or Lewis's recorded compass bearing.

The third route of the trail, across the upper Fish Creek drainage, is agreed upon by several researchers, including Elers Koch, John Peebles, and Ralph Space. Elers Koch located the trail coming up out of Hungery Creek and across Fish Creek to Dollar Ridge in a fairly straight line.[16] John Peebles, a University of Idaho researcher, also located the trail coming up out of Hungery Creek and crossing the Fish Creek drainage similar to Koch's location.[17] Peebles admits his map may be as much as 1 to 2 miles off on the trail location from Windy Saddle to the top of Dollar Ridge because he did not discover trail tread. Ralph Space's location in Fish Creek agrees with Koch's and Peeble's locations. Space states in *The Clearwater Story* that "The party proceeded up Eldorado Creek, crossed it and then up the ridge between Dollar and Sixbit Creeks, over the main divide into the drainage of Fish Creek, where they found a small meadow and lunched. They then went over another divide into Hungery Creek and camped at a small meadow."[18] Chuck Raddon, retired Clearwater National Forest Trails Coordinator, is also of the opinion that the trail came up out of Hungery Creek at Windy Saddle, then went across Fish Creek to Dollar Ridge. While most researchers agree that Lewis and Clark crossed Fish Creek, questions remain. Where is the old trail tread? Is that old trail still on the ground?

Another disagreement on trail location occurs where the trail came off the dividing ridge between the Lochsa and the North Fork Rivers and went down into Hungery Creek. Steve Russell has the trail coming down a ridge from Green Saddle.[19] This location does not match the journal descriptions, mileage, or Clark's maps, but it is an easier route into Hungery Creek than the Willow

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[13] Bud Moore, *The Lochsa Story: Land Ethics in the Bitterroot Mountains* (Missoula, MT: Mountain Press Publishing Co., 1996), 1-12, 461; and interview by author, July 21, 2001. Moore first came to the upper Lochsa country in 1930 and spent several years there trapping and working for the Forest Service.

[14] Ibid., 20; and U.S. Forest Service, "Clearwater National Forest, Idaho [map]" (U.S. Forest Service, 1930).

[15] Steve F. Russell, "The Riddle of Hungery Creek, September 18-20, 1805, and June 16, 18, 25, 1806," *Idaho Yesterdays*, 44 (Spring 2000): 26.

[16] Elers Koch, "The Lewis and Clark Route Retraced Across the Bitterroots," *Oregon Historical Quarterly* 41 (June 1940): 164.

[17] John J. Peebles, "On the Lolo Trail: Routes and Campsites of Lewis and Clark," *Idaho Yesterdays* 9 (Winter 1965-66): 7 (map).

[18] *The Clearwater Story* (1981), 16.

[19] Russell, "The Riddle of Hungery Creek," 25.

Ridge route, as it bypasses the steep canyon downstream in Hungery Creek and avoids four potentially dangerous creek crossings during spring run-off from April through June. Russell's route is located on an old Nez Perce Indian trail into Hungery Creek, while Peebles located the trail coming down Willow Ridge.[20] The Willow Ridge route may not match Private Joseph Whitehouse's expedition journal, which describes the journey from Sherman Peak to Hungery Creek on Sept 19th, 1805, as follows:

> We descended down this Mountain, which we found dangerous for about 3 Miles, We then ascended another Mountain; which was as bad to ascend as any Mountain we had yet seen, this was very fataigueing to ourselves & horses,...We descended this Mountain & came down into a narrow Valley, where we found a run of water. We halted at that place to rest our horses & to eat some portable soup...We delayed about 2 hours— & proceeded on, & descended a mountain about four Miles, and arrived at a Creek...[21]

Peebles located this portion of the trail on the spine of Willow Ridge in the 1960s, and indicated on his map that his trail might be off as much as 1/2 mile on the Willow Ridge location, fully aware that his location did not quite match the journals.[22] Andrew Arvish, a retired Clearwater National Forest employee, located the trail going across the steep side of Willow Ridge. This trail crosses several creeks and matches the journals better than the route on top of Willow Ridge.[23] The Forest Service moved the trail back up to the top of Willow Ridge, as this route is easier to travel than the one on the side of the mountain. Is there another route from Sherman Peak into Hungery Creek that matches the journals and has straight-line Indian trail characteristics that has not yet been discovered?

This on-going research will cover the Lewis and Clark Trail from the Weippe Prairie east to the Montana State Line, covering the trail segments from camp to camp. In the year 2000, field research was completed from the Weippe Prairie to the June 16, 1806, camp (Horsesteak Meadow) in Hungery Creek, a distance of 36 1/2 miles as recorded by Lewis and Clark in 1805 and 37 miles for 1806. Field work for 2001 consisted of verifying segments of the trail from Willow Ridge to Packer Meadows. More intensive field work will be done in 2002. This report is the first of a series of reports, and its purpose is to locate and explain the findings of the trail and camp relocations, to relate to the reader the methodology and techniques used during the fieldwork, and to examine the Clearwater National Forest management of the historic trails.

Figure 1: Band of sheep going to mountain pastures along the Lolo Trail; Schultz's Ferry and Greer, Idaho, in background, 1920s (Photographer unknown, author's collection).

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[20] G. B. Nicholson, "North Nez Percé or Lo-Lo Trail, Idaho," [map reproduction], State Historical Society of Wisconsin, Whi(X3) 51708, 1866, in Baird, *With Bird and Truax on the Lolo Trail*, 13; and John J. Peebles, "Lewis and Clark's Route over the Lolo Trail," [map], In Peebles, "On the Lolo Trail," frontispiece.

[21] Gary Moulton, ed., *The Journals of the Lewis & Clark Expedition,* vol. 11, *The Journals of Joseph Whitehouse, May 14, 1804-April 2, 1806* (Lincoln: University of Nebraska Press, 1997), 322-323.

[22] John J. Peebles, "Lewis and Clark's Route over the Lolo Trail," [map], in Peebles, "On the Lolo Trail," frontispiece.

[23] U.S. Forest Service, *Following Lewis and Clark Across the Clearwater National Forest*, pamphlet and map (Missoula MT: U.S. Forest Service, 1971).

CHAPTER I

General Historical Background

The Lewis and Clark Trail is without a doubt the most important historic trail in the Pacific Northwest. Nez Perce Indians in prehistoric and historic times traveled the Lolo Trail from the Clearwater and Snake River basins to the buffalo hunting grounds in Montana. Pi Lú-Ye Kin described a typical trip in 1870 like this: "Under the leadership of Looking Glass and Shorn Head the Nez Perces started from the plains of Oyaip in June, 1870, one hundred and thirty tipis strong—from twelve hundred and fifty to fifteen hundred people. About two months were consumed in the journey to the buffalo country."[24]

This 150-mile route also played an important part in the historic Lewis and Clark expedition going west in 1805 and again on their return journey in 1806. While most of Lewis and Clark's travels overland occurred in Montana (where major highways and roads parallel most of the Trail), the Clearwater National Forest in Idaho has the longest number of miles of primitive unroaded Lewis and Clark Trail than anywhere else in the nation. It is the roughest piece of trail Lewis and Clark crossed, and it is still undeveloped in many places. The Idaho portion of the Lewis and Clark trail also has been the least modified. Portions of the trail in Fish and Hungery Creeks remain as they were almost two hundred years ago, as reported by *The Lewiston Morning Tribune* in 1984, stating that "during 1984 a field reconnaissance was made of the roughly twelve miles of the Hungery Creek drainage. This section of the Lewis and Clark Trail is widely considered to be the least disturbed section of the entire 3,700 mile route and the one most closely approximating conditions similar to those viewed by Lewis and Clark."[25] We have the opportunity to preserve the old route, mostly as it was when the Corp of Discovery struggled over the steep mountains on their journey west.

The Weippe Prairie area played a key role in the Lewis and Clark expedition, where the friendly Nez Perce helped the weary explorers after completing the most difficult part of the entire journey—the crossing of the Bitterroot Mountains. After meeting the Nez Perce, Clark hurried back to the Lewis party on September 22nd, 1805, and described their condition: "...walked up to the 2nd village where I found Capt Lewis & the party Encamped, much fatigued, & hungery, much rejoiced to find something to eate of which They appeared to partake plentifully."[26] The Nez Perce befriended and fed the white men, and took them into their lodges. The lodges on the Weippe Prairie numbered approximately 200 in 1805; Clark's map indicates five village sites on the prairie. The Weippe Prairie has been an important camas gathering area for the Nez Perce Indian Tribe for centuries. The exact meaning of the name Oyayp, Oyipe or Weippe has been lost, although some say it means beautiful prairie. Lewis described the Weippe Prairie on June 10, 1806, as a blue lake from the Camas blossoms. Anthropologist Madge Schwede described the meaning of Weippe or "Oyayp as referring to a place to gather; located in flat area to south of Weippe, Idaho."[27] Perhaps the best description of the Weippe Prairie was by Thomas Sutherland, a volunteer aide-de-camp with General O. O. Howard during the 1877 Nez Perce war, who remarked that "the first night

[24] Pi Lú-Ye Kin (Albert Thomas Moore), *The Life History of a Nez Perce Indian*, as told to Anthony E. Thomas, *Anthropological Studies* 3 (Washington, D.C.: American Anthropological Association, 1970), 7.

[25] Lewiston Morning Tribune, July 22, 1984, quoted in *Lolo Trail System Implementation Guidelines*, 22.

[26] Moulton, *The Journals of the Lewis & Clark Expedition,* vol. 5, *July 28-November 1, 1805*, 230.

[27] Madge L. Schwede, "An Ecological Study of Nez Perce Settlement Patterns" (M.A. thesis, Washington State University, 1966), 32.

we camped at the Wey-ipe (Nez Perce for a marshy place in the mountains), where camas was so plentiful that the path taken by our men was in many places actually white with that favorite Indian bulb."[28]

Sometime between 1806 and 1831 usage of the Lewis and Clark route from the Weippe Prairie to Eldorado, Fish, and Hungery Creeks (the southern branch of the Lolo Trail) was most likely abandoned. The northern branch was used by the John Work party of the Hudson's Bay Company in 1831, a route that went to Musselshell Meadows and over Snowy Summit to Rocky Ridge and met the southern branch near Sherman Peak. From here the Lolo Trail stayed on the main dividing ridge most of the way to Lolo Pass. The trail today is generally a single path, about 16 inches wide, indicating an old Indian trail with single path characteristics.[29] Further indications of the age of the trail are supported by a tree that was aged at over 100 years growing in the trail tread along the route known as the Lolo Trail (or the Northern Nez Perce Trail) that was rerouted in places by Bird-Truax in 1866 and in the early 1900s by the Forest Service.[30] Another name for the old ridge-top trail by the Nez Perce Indians is Khusahnah Ishkit or buffalo trail, more recently renamed the Nee-Me-Poo Trail.[31] All of these names refer to the same ridge-top trail, and this trail joins the Lewis and Clark Trail near Sherman Peak. This report will call this trail by its original historic name—the Lolo Trail—throughout the report.

The Hungery Creek route may have been abandoned, since we know John Work used the Musselsell Meadows-Rocky Ridge route in 1831. There are many factors that may have influenced the change in the Lolo Trail route. Infectious diseases like measles and smallpox reduced the Nez Perce population in 1780, 1800, and 1825.[32] In 1847, a measles epidemic took a tremendous toll: "Measles broke out among the Indians at Ahsahka, and 48 died. This was a large part of the population at the time."[33] The Lolo Trail could also have been abandoned as a result of pressure from the better-armed Blackfeet Indians. The Blackfeet, numbering approximately 20,000, acquired muzzleloader rifles from Canadian traders and drove the Flatheads and Shoshone Indians out of the plains of western Montana. The Blackfeet warriors traveled widely, to Walla Walla, Minnesota, and California on their raids.

It has also been said that this route was abandoned because the trail in this area was not a trail at all—that Lewis and Clark simply took a wrong turn and were lost in Hungery Creek. However, Lewis recorded that the expedition followed a discernable path up Hungery Creek on September 19th, 1805, stating that "the road was excessively dangerous along this creek being a narrow rockey path generally on the side of steep precipice,..."[34] Another possibility considered by some researchers is that snow had covered the portion of the Lolo Trail that is located on the divide going west to Rocky Ridge and on to Musselshell Meadows, which made Clark lose his way. Yet, Clark was able to find the trail under the snow, as on September 16, 1805 he recorded in the journals that "I walked in front to keep the road and found grat dificuelty in keeping it as in maney places the Snow had entirely filled up the track, and obliged me to hunt Several minits for the track."[35]

The Lolo Trail began to see more use during the fur trade era. In 1831, John Work of the Hudson Bay Company led his Snake River brigade east from Kamiah to the Weippe Prairie and on the Lolo Trail, bypassing Fish and Hungery Creeks. In 1850 Major John Owen acquired and rebuilt the St. Mary's Mission near present day Stevensville, Montana, and traveled the Lolo Trail a number of times, documenting his crossings on three separate occasions in 1852, 1857, and 1863. In 1853 Congress authorized the Secretary of War to find a feasible railroad route to the Pacific Ocean. As part of the northern survey, Lieutenant John Mullan of the U.S. Army traveled and

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[28] Thomas A. Sutherland, *Howard's Campaign Against the Nez Perce Indians 1877* (Portland, OR: A. G. Walling, 1878; reprint Fairfield, WA: Ye Galleon Press, 1980), 24. The white camas is poisonous; the blue camas is used for food by the Nez Perce Indians.

[29] These measurements were taken at Beaver Dam Saddle. In other locations the tread is a faint path, 16 inches wide and 3-6 inches deep.

[30] The tree was bored with an increment woodborer by Robbin Johnston, Archeologist, Clearwater National Forest.

[31] Elizabeth Wilson of Kamiah provided the name Khusahnah Ishkit to Ralph Space. See Space, *The Lolo Trail,* 1, 63. In 1891, Kew-kew-lu-yah (Billy Williams), a Nez Perce elder, told anthropologist Alice Fletcher that the trail had two other names: Nah wal (Lolo Trail), and Nak sore mah (Road to the Salmon). See Robert Lee Sappington and others, "Alice Cunningham Fletcher's 'The Nez Perce Country,'" *Northwest Anthropological Research Notes*, 29 (Fall 1995): 182-83.

[32] Allen Pinkham, "Informal Chats About Lewis and Clark's Journey Through the Kamiah Area," (talk given at the Third Annual Lewis and Clark Symposium, Lewiston, Idaho, June 30, 2001).

[33] John P. Harlan, "Writings About the Lolo Trail," *Clearwater (Orofino, Idaho) Republican*, 2 December 1921, 5, as transcribed by the Idaho State Historical Society.

[34] Moulton, *The Journals of Lewis & Clark Expedition,* vol. 5, 215. Lewis and Clark refer to trails as roads, routes, or paths.

[35] Ibid., 209.

mapped the Lolo Trail in September and early October of 1854. Mullan's opinion was that the trail was too rugged for a railroad. Another traveler, Richard Lansdale, a physician and Indian agent, recorded his travel east across the Lolo Trail in 1857.[36]

The discovery of gold by E.D. Pierce in 1860 meant a dramatic increase of traffic. The Lolo Trail to Weippe and on to the gold fields became an important route for the miners. The influx of immigrants prompted the Idaho Territorial government to petition Congress for funds to build a wagon road to the Bitterroot Valley in Montana, which resulted in a $50,000 appropriation by the federal government. Wellington Bird, a physician from Iowa was the supervisor of this project, and Major Sewell Truax served as chief surveyor. From the Weippe Prairie the Bird-Truax wagon road went east from Weippe over Brown Creek Ridge and southeast along Brown Creek to the Mussleshell Meadows. From Mussleshell Meadows, Bird and Truax could only improve the Lolo Trail to a width suitable for a primitive pack train. Bird wrote a short report to Col. J. H. Simpson, the Chief of Engineers in Washington, D.C., warning that no real "wagon road" could be built with the money on hand, there being only enough to "cut out and clear a trail of ten or twelve feet wide over which pack trains can be taken."[37] Bird and Truax performed the first recorded trail changes of clearing, widening, and minimal grading in places on the trail.

Perhaps the most dramatic use of the Lolo Trail occurred during the Nez Perce War of 1877. Chiefs Whitebird, Looking Glass, and Joseph, along with 750-850 Nez Perce people and 2000 or more horses fled from General O. O. Howard's troops across the trail. Howard wrote later of the physical demands the Lolo Trail presented, stating "It does not appear far to the next peak. It is not so in a straight course, but such a course is impossible. 'Keep to the Hog-back!' That means there is usually a crooked connecting ridge between two neighboring mountain heights, and you must keep on it. The necessity of doing so often made the distance three times greater than by straight lines, but the ground was too stony, too steep, the canyon too deep, to attempt the shorter course. Conceive this climbing ridge after ridge in the wildest kind of wilderness."[38]

A hunting trip planned by several young men ended in tragedy on the Lolo Trail in 1893. The Carlin Party, consisting of William P. Carlin, A.L.A. Himmelwright, John Harvey Pierce, Martin C. Spencer, and George Colegate traveled the Lolo Trail in mid-September. They went to the upper Lochsa River and camped near the present Jerry Johnson and Colgate Warm Springs. By October 10th, six inches of snow had fallen on their camp and more in the mountains. Alarmed, they hired Ben Keeley, Jerry Johnson's partner, to build rafts to attempt to float down the Lochsa River, despite Johnson's opinion that floating the river was suicidal. On November 3rd the party had to abandon their rafts near the mouth of Holly Creek. George Colegate, who had become gravely ill from uremic poisoning and could not walk, was left on the banks of the Lochsa River where he later died.[39] The rest of the party walked along the side of the canyon of the Lochsa River where there were no trails, arriving at Hellgate Creek where they met their rescuers. A map of the Clearwater Basin and adjoining territory was prepared in 1894, showing the route of the Carlin Party.[40]

One would expect that the trail tread would be a lasting mark upon the land, especially at the higher elevations where soil deposited every one hundred years is minimal, and that is evident when the 1930s aerial photos are examined. The old trails can still be observed 130 years after the Lewis and Clark Expedition on these photos, and the old historic trail tread can be found today on the high ridges along the Lolo Trail.

George B. Nicholson's opinion of the soil conditions along the Lolo Trail was that "from the nature of the soil a road in the Bitter Root mountains would last for years. I was almost going to say ages."[41] In the lower elevations that is not always the case. By 1902, when Professor Olin D. Wheeler, a member of the Minnesota Historical Society, came to the Weippe Prairie, he reported most of the prairie under cultivation. Where the prairie was

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[36] Dennis and Lynn Baird, *In Nez Perce Country: Accounts of the Bitterroots and the Clearwater After Lewis and Clark* (Moscow, University of Idaho Library, 2003).

[37] Baird, *With Bird and Truax on the Lolo Trail*, 8. George B. Nicholson, a civil engineer with the survey, produced several reports and maps critical to the historical record of the trail that can be found in this manuscript.

[38] Linwood Laughy, comp., *In Pursuit of the Nez Perces: The Nez Perce War of 1877, As Reported By General O. O. Howard, Duncan McDonald, Chief Joseph* (Wrangell, AK: Mountain Meadow Press, 1993), 138. A good example matching Howard's description can be found between Cayuse Junction and Rocky Point.

[39] Ladd Hamilton, *Snowbound* (Pullman: Washington State University Press, 1997), 190.

[40] A. L. Himmelwright, C. E., "Map of the Clearwater Basin and Adjoining Territory," 1" = 15 miles (New York: R. D. Servoss, engraver), in Heclawa [Abraham Lincoln Artman Himmelwright], *In the Heart of the Bitter-Root Mountains: The Story of "The Carlin Hunting Party," September-December, 1893* (New York: G.P. Putnam, 1895), back of book.

[41] Baird, *With Bird and Truax on the Lolo Trail,* 26.

plowed the trail followed by Lewis and Clark was obliterated. Fortunately, Ralph Space was able to locate bits and pieces of the trail on the prairie where the land had not been plowed.[42]

All changes to the trail are not man-made. Livestock grazing of sheep, cattle, and horses affected the Lolo Trail, and in some places braided trail tread from livestock grazing is still evident. Sheep were present on the Lolo Trail in 1917 or before, as recorded by the annual grazing report of the Selway National Forest.[43] The *Pierce City Miner* newspaper reported in 1905 that "Between 15,000 and 20,000 head of sheep were taken through there during the past week to the summer range."[44] Three of the sheep grazing ranges south of the Lolo Trail were located in the upper Obia Creek drainage (Hungery Creek), from Castle Butte to Saddle Camp, and from Howard Camp to Doe Creek just past Indian Post Office Lakes.

Sheep driveways were located on the northern branch of the Lolo Trail (Musselshell Meadows over Snowy Summit), the middle branch of the Lolo Trail (Musselshell Meadows over Austin Ridge to Pete Forks), and from Glenwood along the watershed divide between the Lochsa River and the Clearwater River.[45] Sheep were also present on Cook Mountain in 1922 when they were driven from Lewiston and Kamiah in flocks of up to 5000 sheep.[46] In the 1930s several bands of 5,000 sheep were all along the Lolo Trail. This was verified by Roy Cochrell, nephew of Albert N. Cochrell, who worked on the Clearwater National Forest in the early years of the 20th century:

> The sheep came across Schultz's ferry at Greer and went up the Lolo Trail. I was about ten years old when I took a job guarding the sheep from bear. We grazed all the way across the Lolo Trail and drove the band to Packer Meadows where the lambs were separated out. The lambs were taken to Missoula, Montana, where they were sold. We stayed on the top of the ridges and moved a mile or two a day. The sheep were strung out from one side of the ridge to the other feeding through the brush.[47]

One sheep drive on the eastern end of the Clearwater National Forest was along the Windy Ridge Trail just east of Cook Mountain to Hoodoo Pass, where the lambs were separated and driven to the railhead at Rivulet, Montana. The sheep drive on the historic Lolo Trail continued from Weippe to Packer Meadows, with the lambs separated and driven over Lolo Pass to Lolo and Missoula, Montana. The main flock was then driven back home on the ridge top trail, which coincides with the Bird-Truax map placement of the route and the Lieberg map.[48] Examination of the 1930 Clearwater National Forest map shows the trail on the ridge top that was used for the sheep drive, and this was highlighted on the map by an unknown Forest Service employee. Ralph Space estimated that "there were about 26,000 [sheep] in 1928 and 32,000 in 1929. The peak was reached in 1933 with about 35,000 head," but by 1949 those numbers had dropped to 2,000.[49] Effects of erosion from 44 years of grazing came to the attention of the Forest Service, who noted that "it is likely that much of the evidence of early trails along the route from Musselshell Meadows to Howard Camp can be attributed to the effects of sheep driving and grazing."[50] Erosion control measures were used to stabilize the effects, and the Forest Service proclaimed in 1985, "Trail 40 today appears as a 40-50 foot wide path through the Forest in certain locations, though trees 10 to 20 feet high and brush are established over most of the path."[51]

Cattle and horses also grazed on the trail, but in numbers nowhere near the sheep usage. Horses grazed on the Bald Mountain grazing allotment between 1955 and 1981. The effects of grazing are almost gone on the south side

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[42] Space, *Lewis and Clark Through Idaho 1805-1806* (Lewiston, ID: Tribune Publishing Company, 1963), 7.

[43] U.S. Forest Service, "Selway National Forest, Idaho" [map], 1915, Supervisor's file, with grazing legend added, G, Grazing-Selway, November 15, 1917, Annual Grazing Report. These unique maps are filed in the Archives drawer. Nez Perce National Forest Supervisor's Office, Grangeville, Idaho. Also, copies at the Supervisor's Office, Clearwater National Forest, Orofino, Idaho.

[44] *Pierce City (Idaho) Miner*, 30 June 1905, 8.

[45] U.S. Forest Service, "Selway National Forest, Idaho" [map], Washington, D.C.: U.S. Forest Service, 1921, with added grazing legend. These unique maps are filed in the Archives drawer, Nez Perce National Forest Supervisor's Office, Grangeville, Idaho. Also, copies at the Supervisor's Office, Clearwater National Forest, Orofino, Idaho.

[46] Albert N. Cochrell, "Past Remembrances: Some Notes," private papers furnished by Roy Cochrell, Clearwater National Forest Supervisor's Office, Orofino, Idaho (Copy in Eastman Papers (unprocessed), Special Collections and Archives, University of Idaho Library, Moscow, Idaho).

[47] Roy Cochrell, interview by author, 7 January 2001.

[48] Ibid.; and Nicholson, "Explorations and Surveys for a Wagon Road from Virginia City, Montana to Lewiston, Idaho,"[map reproduction], in Baird, *With Bird and Truax on the Lolo Trail*, 22; and Lieberg, "The Bitterroot Forest Reserve Showing Land Classification" [map], plate CXIV.

[49] *The Clearwater Story* (1981), 89-90.

[50] *Lolo Trail System Implementation Guidelines*, 82.

[51] Ibid. The author noted a 50-foot wide sheep drive trail between Howard Camp and Moon Saddle due east of benchmark 6441.

of Bald Mountain, where the trails can be observed by a keen eye on the hillsides. Here the deeply worn Lolo Trail tread is easily discerned from the faint trail treads created by the sheep and horse use. In 1985, Forest Service research discovered that "the trails are evident when walking across the hillside, but are being revegetated by native grasses."[52] Cattle trails in the Packer Meadows area are obvious in the meadows and bordering forest. While the Forest Service has halted grazing at Bald Mountain, Hungery Creek, and Packer Meadows, some of the effects are still visible, although this researcher examined the grass hillsides of Bald Mountain and found that the horse and sheep grazing trails had all but disappeared.

The forces of nature also had a hand in impacting the Lolo Trail. In the early 1900's there were very large forest fires on the Clearwater National Forest. Fires burned the Fish Creek drainage in 1910 and 1919. In 1929, fires burned across the Lolo Trail near Bald Mountain and between Saddle Camp and Moon Creek. In 1932, the fire re-burned Fish Creek and burned lower Hungery Creek as well as the lower half of Willow Ridge. In 1934, fires burned across Hungery Creek and up Doubt Creek.[53] In 1961, the Clearwater National Forest had fires in the Musselshell area, one at Ashpile Creek, Cayuse Creek, Gravey Creek, Orogrande, and Surprise Creek in the Wilderness Area.[54] Gravey Creek fire burned part of the lower eastern half of the drainage to the top of Horseshoe Peak. Fortunately, the Lewis and Clark Trail location by Ralph Space (in Gravey and Horseshoe Creek) was not burned.

The Lolo Trail can be best defined as a single path, located mostly on the divide between the Lochsa and the North Fork of the Clearwater River with a few modifications wrought by livestock, Bird-Truax, and the Forest Service. There was considerable use of the Lolo Trail in the 1800s and the tread was well worn. From Lewis and Clark to the Carlin Party there were those who traversed the trail and recorded their experiences in journals, providing opportunities to see the trail as it was before man and natural forces altered it. These forces have added alternate routes branching off and sometimes back on to the original trail tread. Many have noted its unique characteristics, but several observations succinctly capture the essence of the Lolo Trail. Bud Moore described the Lolo Trail in the 1930s as "only one trail except in the meadows."[55] General Howard called the Lolo Trail the "most terrible trail on the continent of North America."[56] And finally, Elers Koch said that "The Lolo trail became one of the most historic mountain trails in the United States" because of Lewis and Clark and the Nez Perce War of 1877.[57]

Figure 2: Band of sheep grazing in 1925 with the Seven Devils in the distance (Photo courtesy of the U.S. Forest Service, Nez Perce National Forest, no. 99-5245).

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[52] Ibid.

[53] *The Clearwater Story* (1981), 93-120; and a computer generated Fire History Map printed by the Clearwater National Forest, 11 December 2001.

[54] *The Clearwater Story* (1981), 120.

[55] Bud Moore, interview by author, 15 July 2001.

[56] Floyd R. Barber and Dan W. Martin, *Idaho in the Pacific Northwest,* (Caldwell, ID: Caxton Printers, Ltd., 1956), 7.

[57] Elers Koch. "The Lewis and Clark Route Retraced Across the Bitterroots." *Oregon Historical Quarterly* 41 (June 1940): 161.

CHAPTER II

Previous Efforts to Locate Routes of the Lewis and Clark Explorers

Considering the national fascination with Lewis and Clark, it is not surprising so many people have attempted to locate the trail, and the results have been as varying as the number of trackers. Professor Olin D. Wheeler, an employee of the Northern Pacific Railroad, guided by the famous grizzly bear hunter and guide, William H. Wright, came to the Weippe Prairie in 1902.[58] He photographed a cabin (Levi Goodwin Homestead, also referred to as the Larson ranch by Ralph Space) on a hill that he felt was the camp spot of the Lewis and Clark party from June 10-14 and June 22-24, 1806.[59] When Wheeler left the Weippe Prairie, he followed the Bird-Truax wagon road, which went up Heywood Creek northeast for about two miles before going due east over Brown Creek Ridge. Wheeler thought that Lewis and Clark's eastern route was to Mussleshell Meadows, then south across Lolo and Chamook Creeks to the vicinity of Snowy Summit, rather than over "the foot" of Brown Creek Ridge and on to Crane Meadows and Pheasant Camp.[60] However, Wheeler's route does not agree with the Lewis and Clark journals.

Reuben G. Thwaites published his edition of the Lewis and Clark journals in 1904 and 1905.[61] Thwaites relied upon Olin D. Wheeler and George B. Nicholson for the trail location. Nicholson's route follows most of the Lewis and Clark Trail east from the vicinity of Sherman Peak.[62] Nicholson also drew a map of what he thought was the trail and camp locations at Hungery Creek.[63] The Nicholson map shows the Lewis and Clark route leaving the main dividing ridge at Mt. Romeo (renamed Willow Ridge), crossing Doubt Creek, and going down the ridge from Mt. Juliet (Bowl Butte). The bottom of this ridge comes out west of Doubt Creek on Hungery Creek. But, like Wheeler, Thwaites' route does not agree with the Lewis and Clark journals.

John P. Harlan, former Clearwater County Assessor, searched for the Lewis and Clark route across the Bitterroot Mountains.[64] He wrote several articles from 1921 to 1935 about his adventures that appeared in two local newspapers, the *Clearwater Tribune* and the *Clearwater Republican*. Later, these reports were transcribed by the Idaho State Historical Society in a paper entitled "Writings about the Lolo Trail." Harlan was guided by Chub Wilson, a local Weippe packer, and placed the first Nez Perce village site near where the present National Parks sign is located near the John Opresik property, 1 1/2 miles southeast of Weippe.

Elers Koch, former Supervisor of the Lolo National Forest and later Assistant Regional Supervisor for the Northern Region in Missoula, Montana, also spent many

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[58] Olin D. Wheeler, *The Trail of Lewis and Clark, 1804-1904: A Story of the Great Exploration Across the Continent in 1804-6; With a Description of the Old Trail, Based Upon Actual Travel Over It, and of the Changes Found a Century Later* (New York and London: G. P. Putnam's Sons, 1904), vol. 2, 279-280, 282. Many original Wheeler photos are at the Newberry Library, but unfortunately not those taken of the Lolo Trail.

[59] Space, *The Lolo Trail,* 15.

[60] Wheeler, "Route of Lewis and Clark Across the Bitter Root Mountains in Sept. 1805 and June 1806" [map], in *The Trail of Lewis and Clark*, foldout map, frontispiece.

[61] Reuben Gold Thwaites, ed., *Original Journals of Lewis and Clark Expedition, 1804-1806* (New York: Dodd, Mead & Co., 1904), vols. 3-5.

[62] Wellington Bird to Brig. Gen. J. H. Simpson, 1867, in Baird, *With Bird and Truax on the Lolo Trail*, 12-21.

[63] Nicholson, "Explorations and Surveys for a Wagon Road from Virginia City, Montana to Lewiston, Idaho" [map reproduction], in Baird, *With Bird and Truax on the Lolo Trail*, 22.

[64] Harlan, "Writings About the Lolo Trail," *Clearwater (Orofino, Idaho) Republican*, 28 October 1921, 2.

years on the Lolo Trail locating the campsites. Working from Lolo, Montana to Orofino, Idaho, Koch located and named the following camps: Glade Creek, Snowbank, Lonesome Cove, and Sinque Hole. A pioneer researcher of the Lewis and Clark Trail, Koch created a map of the route and published his article entitled, "Lewis and Clark Route Retraced across the Bitterroots," in the *Oregon Historical Quarterly* in June 1940.

John Peebles, a University of Idaho engineering professor, also spent time walking the trail. His map details when he is absolutely sure when he is on the correct route, when he is unsure because the land doesn't match the journals (as at Willow Ridge), or when he can't find the trail (Fish Creek). He published his findings on the Lewis and Clark Trail and camps in the Idaho Historical Society journal, *Idaho Yesterdays*, in 1965 and 1966 (see Peebles map reproduced below).[65]

Ralph Space, a Weippe native and Supervisor of the Clearwater National Forest from 1954 to 1963, continued work on the location of the Lewis and Clark Trail. According to his daughter, Judy Space, her father's one regret was that he didn't walk the whole route. Ralph Space left a detailed map of the route he chose; however, the Clearwater National Forest reports this map was lost.

Robert (Bob) Schloss, retired Forest Service employee, named some of the Lewis and Clark camps. Schloss said that he only labeled the camps that came from Lewis and Clark's daily activities recorded in the journals. Schloss said that he enjoyed walking the trail with Andy Arvish, another Forest Service employee, 30 years ago.[66]

Andy Arvish, who took an interest in the Lewis and Clark Trail in the early 1960s, assisted Ralph Space. Arvish mapped the trail from Lolo, Montana, to the western boundary of the forest. He produced a map of the route in 1970, which was slightly modified and placed on the archaeologist's site maps.[67] Arvish also wrote management plans for the trail in which he recommended keeping the trail wild, protecting the trail tread, and barring horse use on the trail. His management plans for the Wendover Ridge logging sale in the 1960s also called for protection of the trail, but when he inspected the site during the logging phase, loggers had fallen trees across the road and he could not check the trail. This researcher later discovered a bulldozed skid trail the loggers had made on top of Wendover Ridge which destroyed the trail tread for approximately 1 mile. Through Arvish's efforts, in 1971, the Lewis and Clark Trail from Bradford Bridge to Mex Mountain was placed on the Clearwater National Forest map.

Arvish also put together a brochure (including maps), entitled *Following Lewis and Clark Across the Clearwater National Forest* (1971) that the Forest Service mistakenly attributed to Ralph Space.[68] When Space retired in 1963 the Forest Service hired him to work with Arvish on locating the trail. The pair used aerial photos to locate campsites on Hungery Creek and then flew in by helicopter and marked them with short metal posts. Unfortunately, they did not have enough time allocated to completely locate the trail, and the Forest Service also cut short their time on Willow Ridge locating the Cache Mountain site.[69] In addition, Arvish had a special aerial photo flight done for the trail and marked the trail with white X's. When this researcher inquired about the photos the Forest Service could not locate them, and they were also unable to locate 7.5 minute USGS topographic maps that Arvish also used to plot the trail. Fortunately, his trail locations are preserved on site maps maintained by the zone archaeologist which are not available to the public.[70] After his retirement in 1977, Arvish tried to get the district rangers interested in the location and protection of the trail, but they were generally not interested (see Arvish maps pages 21-26).[71]

In the early 1980s Duane Annis, Clearwater National Forest employee, took an interest in the trail and wrote up a management plan as a project for a six-week college class. The Clearwater National Forest decided to formulate Annis's plan as the draft Lolo Trail System Implementation Guidelines in 1985. The portion of the draft guidelines applying to historic trails was adopted as a

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[65] John J. Peebles, "Rugged Waters: Trails and Campsites of Lewis and Clark on the Salmon River," *Idaho Yesterdays* 8 (Summer 1964): 2-17; "On The Lolo Trail: Route and Campsites of Lewis and Clark," 2-15; and "The Return of Lewis and Clark," *Idaho Yesterdays* 10 (Summer 1966): 16-27.

[66] Robert Schloss, interview by author, March 19, 2001. Two more camps were named in 2001 on the Weippe Prairie by Forest Service employee Norm Steadman (Norm Steadman, interview by author, 10 November 2001).

[67] The site map data set is an unpublished 16 map series of 7.5 minute series topographic maps that are updated to indicate archaeological site location of the Lewis and Clark Trail across the Clearwater National Forest. Located at the Clearwater National Forest Supervisor's Office, Orofino, Idaho.

[68] *Lolo Trail System Implementation Guidelines*, 98.

[69] Andy Arvish, interview by author, 10 March 2001.

[70] Department of Agriculture, U.S. Forest Service, "Site maps,"[maps], Zone Archaeologist, (Orofino, ID: Clearwater National Forest).

[71] Andy Arvish, interview by author, 10 March 2001. It is interesting that Peebles, Arvish, and Space all came up with similar locations of the trail, and it is very helpful that Peebles indicated on his maps when he was unsure of the route.

reference on March 6, 1987, and verified as being in effect on December 28, 1999.[72] Karl Roenke, forest archaeologist from 1980 to 1989, depended upon Duane Annis to locate trail tread on the ground. Annis and Roenke had a difficult task, yet they did an outstanding job in finding trail tread for the entire route across the Clearwater National Forest.[73] In 1983-84, they completed their fieldwork and in 1985 Annis placed the trail along with campsites on U.S.G.S. 7.5 minute maps.[74] This process made changes to the trail locations in Eldorado Creek, Fish Creek, Gravey to Moon Creek, and in the Crooked Fork Creek areas. Small Prairie Camp was moved 1/4 mile up Eldorado Creek on the public map, and the new location was accepted by the National Register of Historic Places in 1993.[75] The silviculturist and landscape architects suggested how the trail corridor could be logged and still maintain scenic values. Regulations guiding trail management were also proposed.

By 1984, the Lewis and Clark Trail as well as the land 1/4 mile on either side of the trail had the highest protection possible under several federal laws (see Appendix A). Timber sales were planned and took place on the site locations of the trail. The Forest Service Forest Plan for the Clearwater National Forest set aside 16,175 acres in the travel corridors along historic travel ways, specifically Lolo Trail, Lewis and Clark Trail, Nee-Me-Poo Trail, and the Lolo Motorway. This was management area A6, defined as "a corridor approximately 1/2 mile in width (1/4 mile on either side) that encompasses the historic Lolo Trail system which consists of: the Lolo Trail, Nee-Me-Poo Trail, Lewis and Clark Trail, Bird-Truax Wagon Road and Lolo Motorway."[76] According to the plan, timber, range and wildlife management practices were modified in the historic trail corridor to "ensure that prehistoric, historic, and archeological sites and values are studied, preserved, or protected in accordance with cultural resource regulations."[77] In the 1980s two timber sales (Dollar Ridge and between Cedar and Lolo Creeks), unfortunately destroyed some of the trail tread; as a result the trail was relocated in those areas.

Steve Russell, an Idaho native and an electrical engineer at Iowa State University, began several years of fieldwork along the Lolo Trail the 1980s and 1990s. Except for two articles, one in *Idaho Yesterdays* and a presentation paper on the Nee-Me-Poo Trail, the results of most of his work remain unpublished.[78] Several of Russell's maps appear in James Fazio's book, *Across The Snowy Ranges, The Lewis and Clark Expedition in Idaho and in Western Montana.*[79]

One of the lasting results of this research locating the Lewis and Clark Trail is a number of historic maps. Elers Koch left a general map. Andy Arvish left a detailed historic trail site map, which is in the zone archaeologist's office, Clearwater National Forest, Orofino, Idaho. John Peebles and Ralph Space both left detailed maps. Duane Annis and Karl Roenke located the Lewis and Clark trail tread, and their trail location was placed on the discontinued blue line maps that were available to the public from 1984 to 1994.[80] The historic trails on the Clearwater National Forest move about on the Forest Service maps about every 10 years. These trails may or may not match the Lewis and Clark journals and maps. All these changing trails mapped by the Forest Service may or may not exist on the ground.

As an examination of these maps shows, each researcher placed some of the Lewis and Clark Trail

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[72] James C. Bates, Forest Supervisor, to Regional Forester, 6 March 1987; and James Caswell, Clearwater National Forest Supervisor to rangers, staff, recreation, and trails staff, 28 December 1999 (Clearwater National Forest Supervisor's Office, Orofino, Idaho), 1.

[73] Karl Roenke, telephone interview with author, winter of 2000; and Duane Annis, interview with author, winter of 2000.

[74] *Lolo Trail System Implementation Guidelines*, 20.

[75] The new landmark boundaries for the Lolo Trail were formally accepted on February 1, 1993. See: National Park Service, "National Register of Historic Places Registration Form, Lolo Trail," including Lolo Trail National Historic Landmark Map set, in John T. Drake, to Forest Supervisors, Lolo and Clearwater NFs, "Adjustments, Lolo Trail National Historic Landmark,"3 September 1993 (Copy in Eastman Papers (unprocessed), Special Collections and Archives, University of Idaho Library, Moscow, Idaho).

[76] U.S. Forest Service, Northern Region, *Forest Plan: Clearwater National Forest* (Orofino, ID: Clearwater National Forest, 1987), III-19.

[77] Ibid.

[78] Russell, "The Riddle of Hungery Creek," 25–31; and "Two Trails, Which One is the Nee-Me-Poo?" (Pioneer Trail Research Presentation, dated 2 October 1995, for presentation at the 1st Nez Perce National Historic Trail Conference Symposium, Lewiston, Idaho, 16-18 October 1995), Nez Perce National Historic Park Library, Spalding, Idaho.

[79] James R. Fazio, *Across the Snowy Ranges: The Lewis and Clark Expedition in Idaho and Western Montana* (Moscow, ID: Woodland Press, 2001), 8, 36, 48, 58, 104, 120, 146, 184.

[80] Blue line maps were the working maps that the Forest Service used to mark ongoing changes to roads and trails. These maps have been replaced by computer-generated maps that show four trails in the Lolo Trail corridor. These maps are generally not available to the public. The 1994 USGS maps include some (but not all) of the changes made to the historical trail on the blue line maps.

segments in different locations. These differences in the trail locations are reflected on each new generation of maps. The lack of access to aerial photos from the 1930s (lost to researchers until Dennis Baird tirelessly located and scanned the photos for the Idaho Historic Air Photo Project at the University of Idaho Library[81]), the rarity of copies of Clark's elk-skin field maps, and the relative newness of the maps from Gary Moulton's *The Journals of the Lewis and Clark Expedition* may explain why many researchers do not agree on the location of the Lewis and Clark Trail, and may also explain why even today there are lost segments of the trail.

Perhaps the best summation of the struggle to locate the Lewis and Clark Trail was penned by Ralph Space, who stated:

> Study and interest in the Lolo Trail System occurred sporadically from the beginning of Forest Service management in 1907 until the 1930's. During the 1930's, an interpretive sign program was initiated largely based on Elers Koch's study and location of Lewis and Clark camp sites along the Lolo Trail.[82]

In the past the best research was accomplished by Forest Service staff; Elers Koch, Andy Arvish, Robert Schloss, Ralph Space, Duane Annis, and Karl Roenke. Peebles' work was also outstanding. Elers Koch found most of the camps and laid the foundation for the location of the trail; in this writer's opinion he was the pioneer in locating the Lewis and Clark Trail. Ralph Space based much of his work on what Elers Koch had already done, and Robert Schloss named some of the camps. Finally, without the research the Forest Service has done in the past, the location of the Lewis and Clark Trail would have been lost.

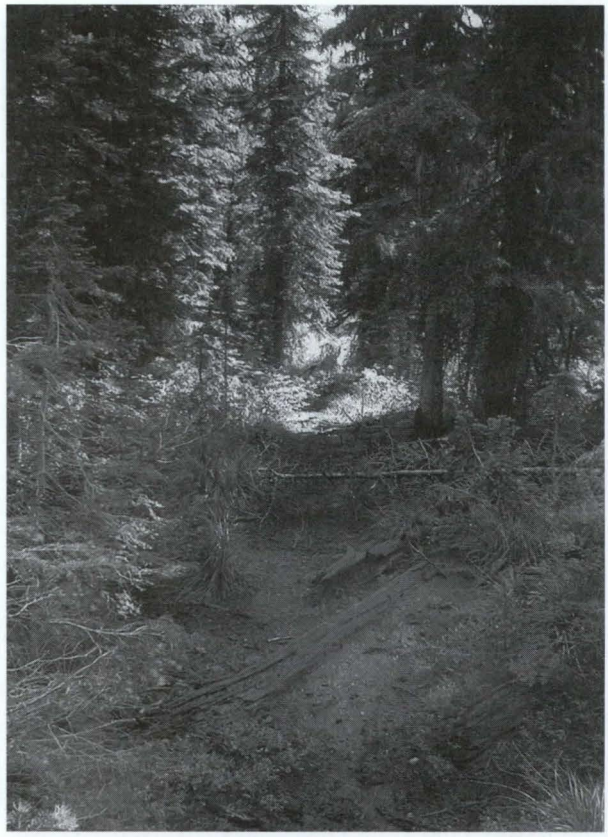

Figure 3: The Lolo Trail near Bald Mountain showing a single path (Photo by author, 11 July 2001).

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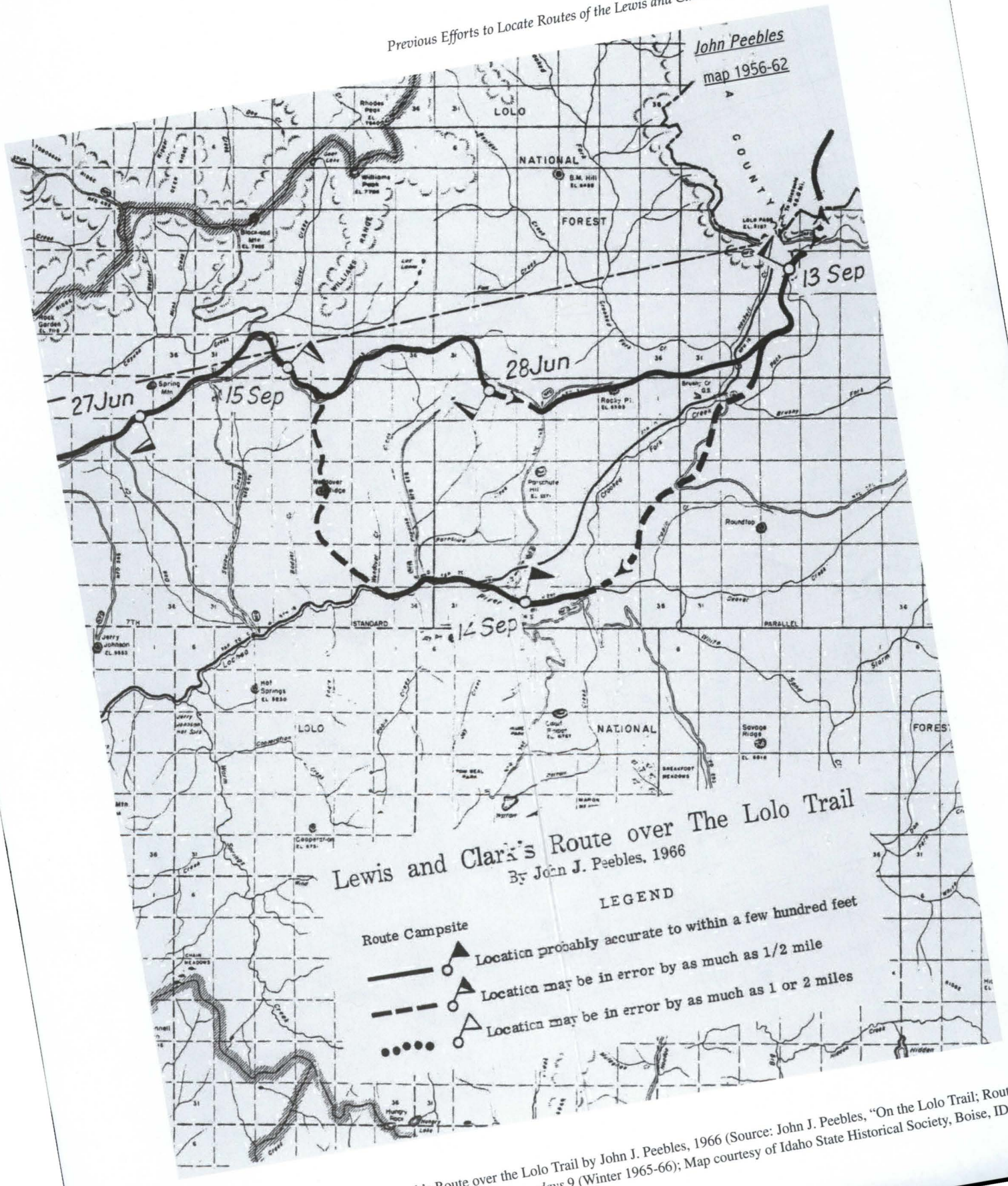
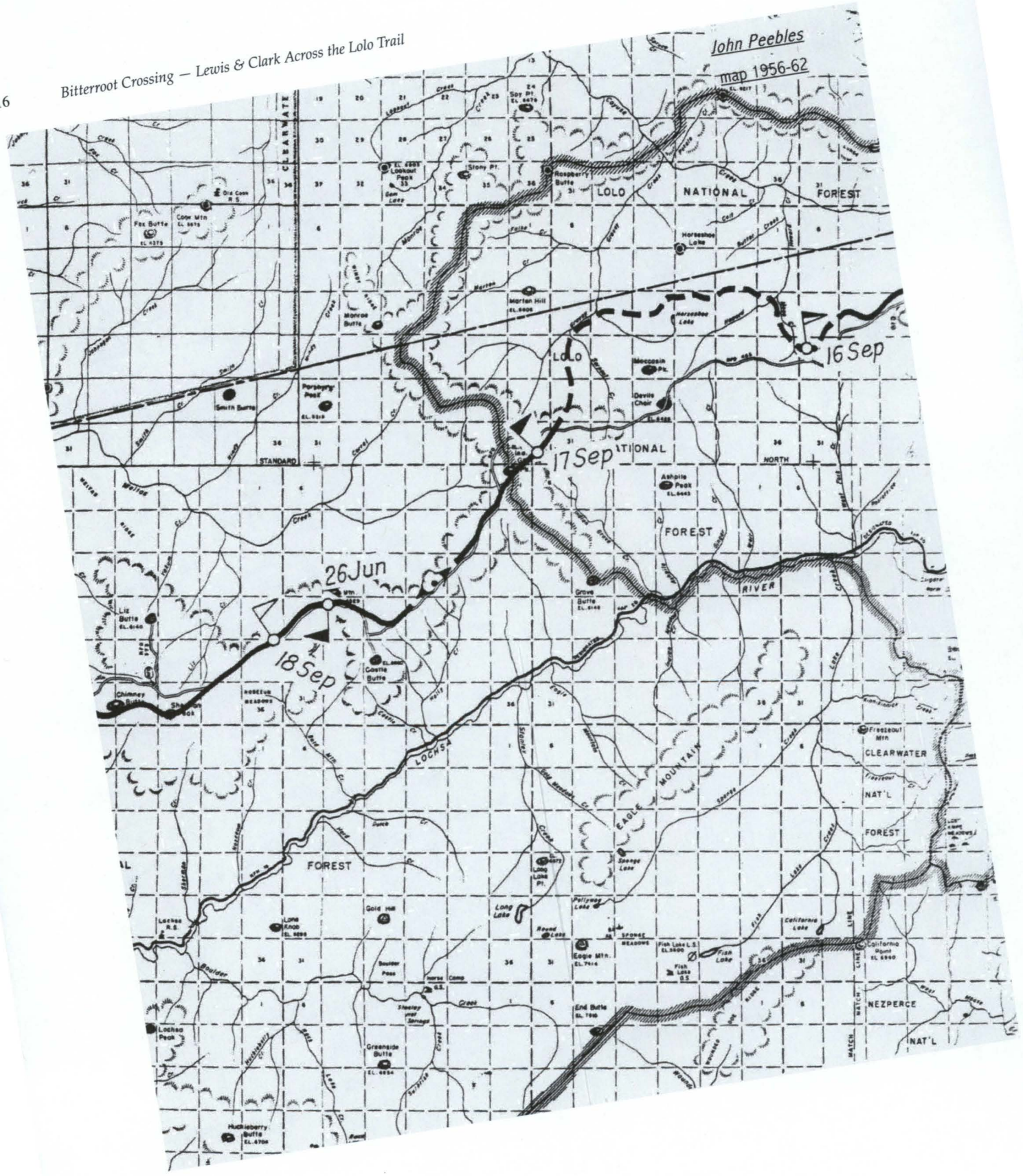
[81] *Project FA, 1937 Air Photos of the Powell Ranger District of the Clearwater National Forest* [computer file], Moscow, ID: University of Idaho Library, 1999; *Project FA, 1939 Vertical Air Photos of the Western Portion of the Clearwater National Forest: Fish Creek, Mex Mtn., Lolo Creek, Musselshell, Crane Meadow and Lacey Meadow* [computer file], Moscow, ID: University of Idaho Library, 2000; and *Historical Air Photos of the Lolo Trail and Vicinity, Lolo & Clearwater National Forests* [computer file], Moscow, ID: University of Idaho Library, 1999, copyrighted by UI Library.

[82] Space, *The Lolo Trail*, 53.



Map 1: Section 1 of 6 of Lewis and Clark's Route over the Lolo Trail by John J. Peebles, 1966 (Source: John J. Peebles, "On the Lolo Trail; Route and Campsites of Lewis and Clark," frontispiece, *Idaho Yesterdays* 9 (Winter 1965-66); Map courtesy of Idaho State Historical Society, Boise, ID).



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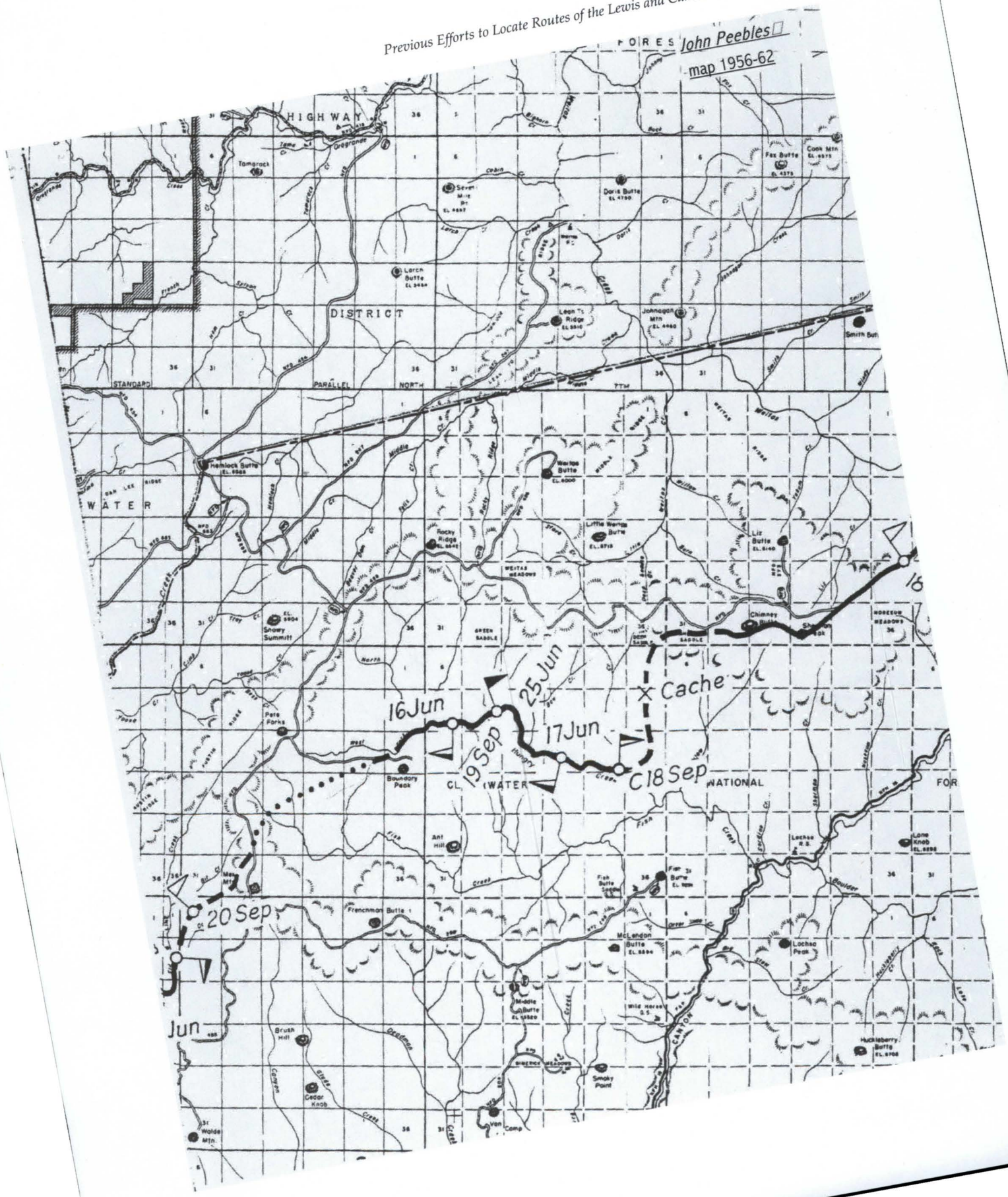

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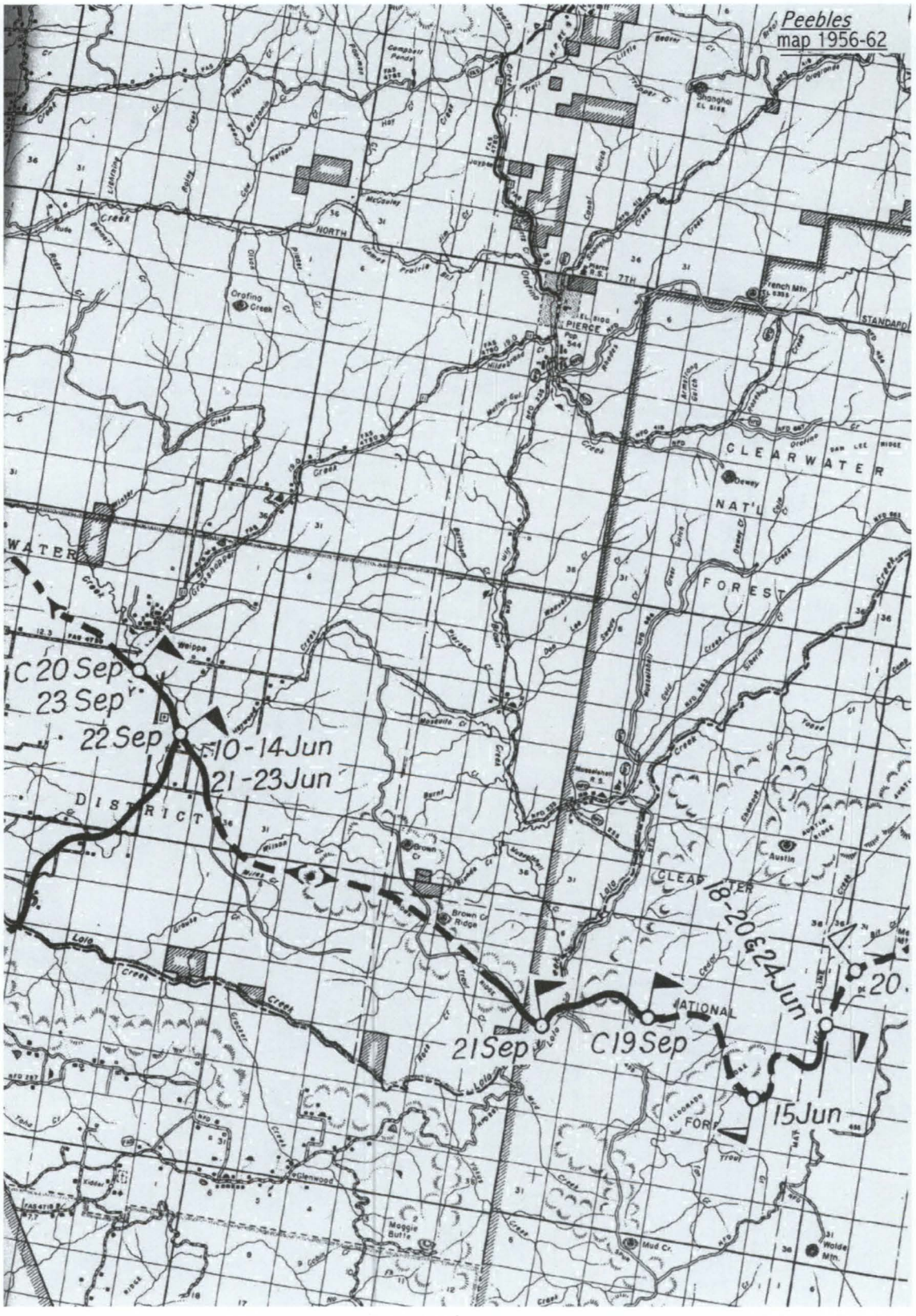

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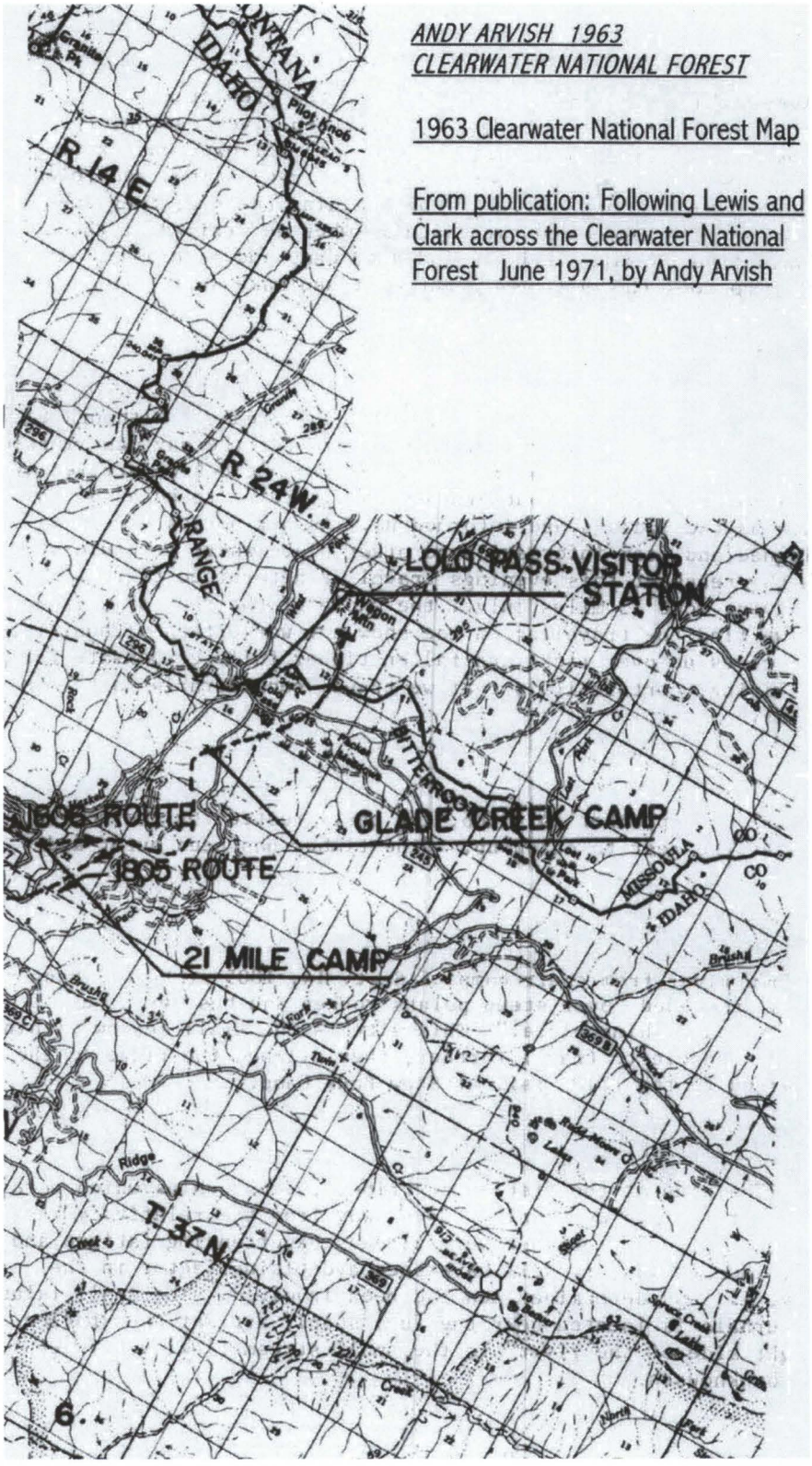
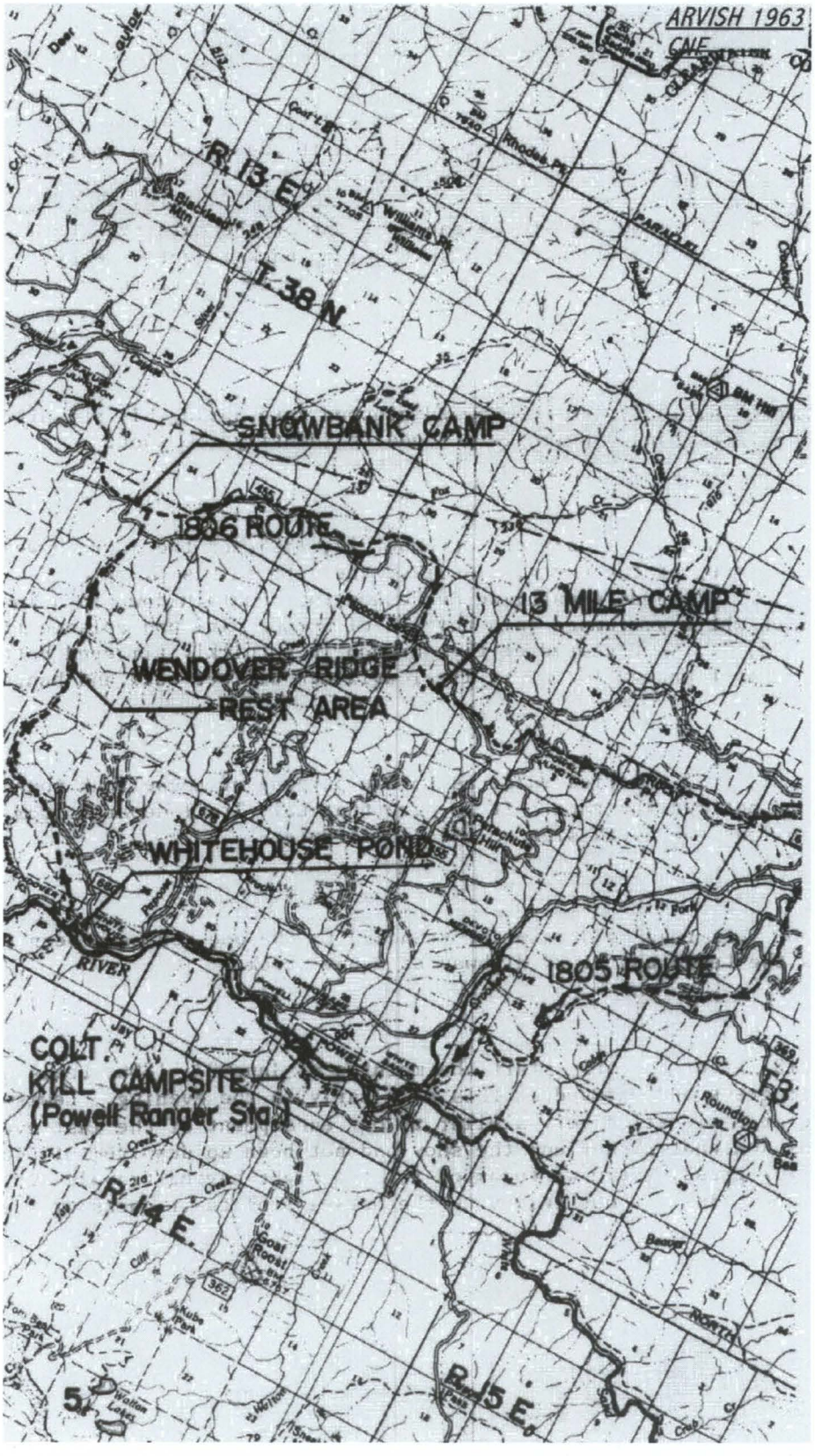


Map 2: Section 1 of 6 of Andy Arvish's Clearwater National Forest map from 1963. This map was used in the Forest Service pamphlet *Following Lewis and Clark Across the Clearwater National Forest*, 1971 (Source: Clearwater National Forest Supervisor's Office, Orofino, Idaho).



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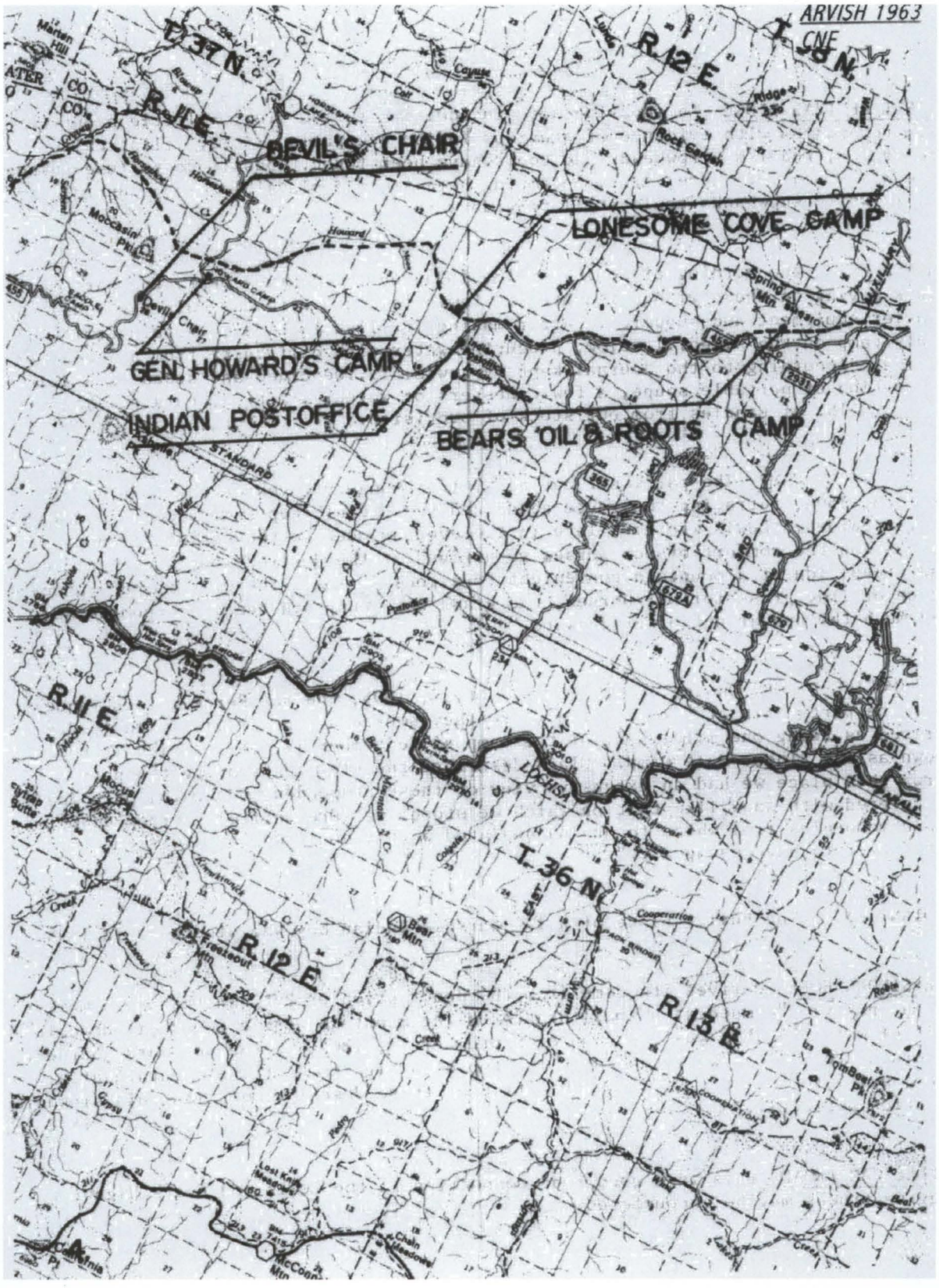

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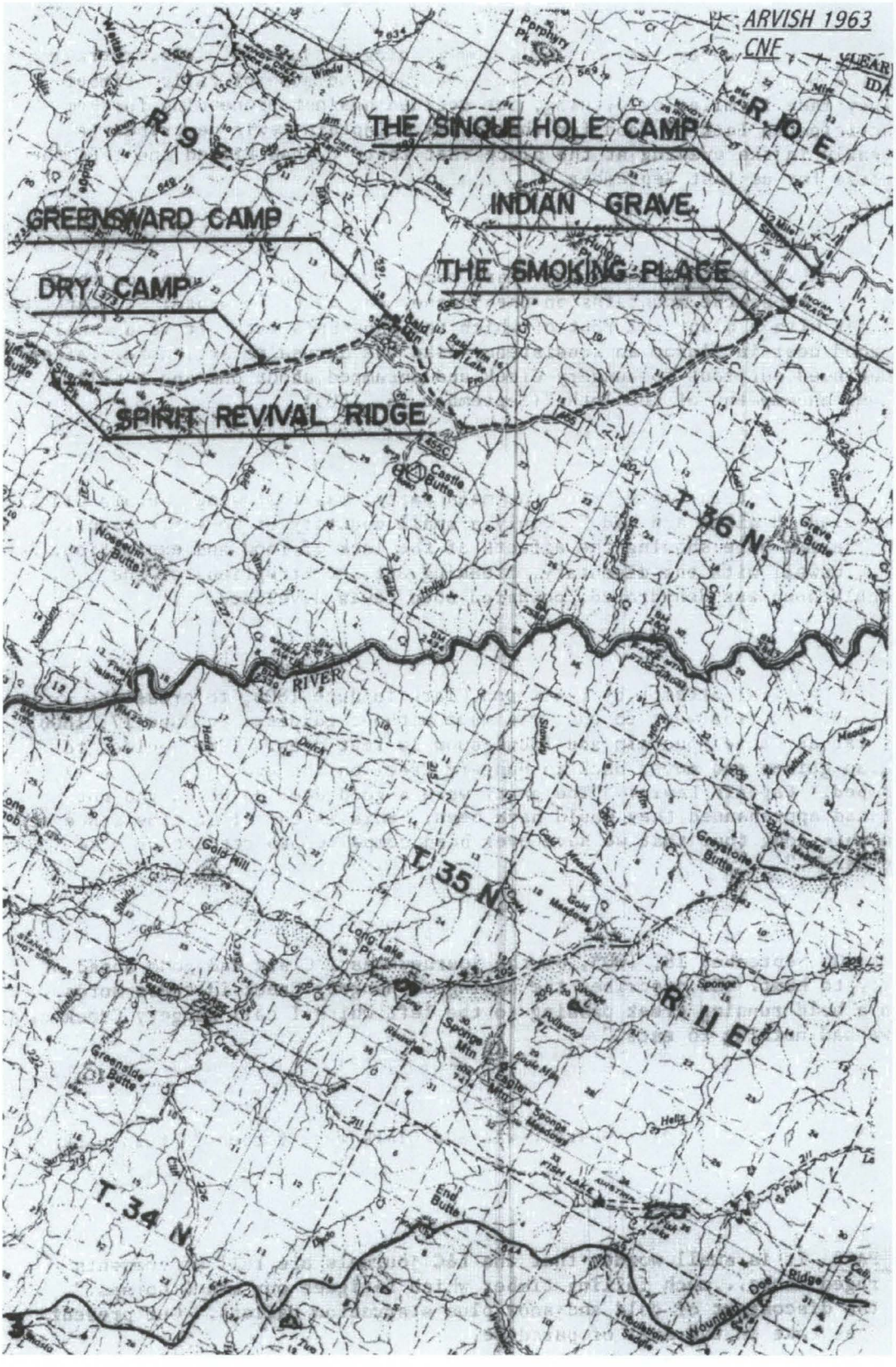

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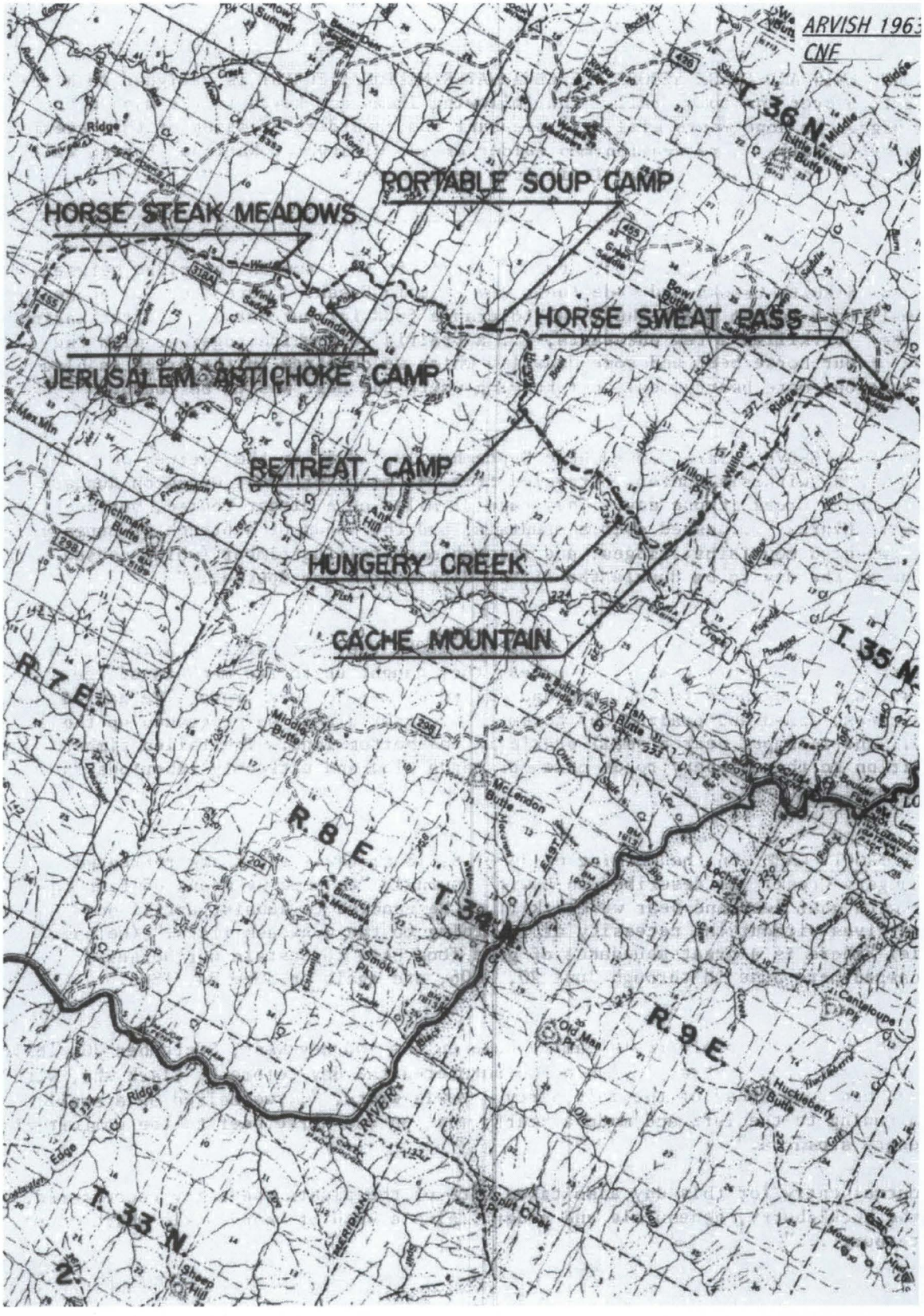

Section 5 of 6 of Arvish map.

Section 6 of 6 of Arvish map.

CHAPTER III

Why a New Effort Might Be Useful

With the publication of Gary Moulton's *The Journals of the Lewis and Clark Expedition*, the maps of William Clark are more accessible for researchers to use as a guide for locating the trail. This would seem to make the task easier, yet there is disagreement among researchers. Some investigators claim that Clark's maps are of no value because of their topographic distortion. Investigators have also disagreed with one another as to the actual travel corridor used by the Corps of Discovery, and how to apply Lewis and Clark's recorded mileage to determine how far they actually traveled.

Are the maps drawn by Clark useful? Can the mileage be used to determine the distance between campsites? Here is what historians, archeologists, and other experts have to say:

"The detail and accuracy of the Lewis and Clark Journals and associated maps provide a wealth of information about the Lolo Trail as well as accurate descriptions of the native inhabitants at initial contact."—Alvin Josephy[83]

"Not all of the 129 historic maps in the atlas came directly from the hand of Clark, the principal mapmaker, but all were closely associated with the expedition and most of them were Clark's handiwork. Being my first foray into expedition materials, I was amazed at the beauty, elegance, and precision of Clark's cartography. He had no apparent training, worked with crude and often unreliable instruments, and used dead reckoning for distances; yet one stands in awe of the accuracy and draftsmanship of his maps. They are models of cartographic excellence, admired and emulated by generations of explorers and map makers."—Gary Moulton[84]

"I found Clark's maps highly useful and accurate in locating the Lewis and Clark Trail from Lolo, Montana, to the Idaho State Line."—Milo McLeod, archeologist for the Lolo National Forest, Montana[85]

"The accuracy of the distances traveled each day by the expedition and their descriptions of campsites and topographic features are sometimes described in enough detail to be relocated today."—Ralph Space[86]

"Although Clark's maps can, in most cases, be regarded as hardly more than rough sketches, it is remarkable how the bends and islands shown in the larger streams can be correlated with the same features on modern maps. Frequently, the trails and campsites can be located quite accurately by the use of these maps. They are particularly valuable in determining on which side of a small stream the expedition traveled and camped."—John Peebles[87]

"Hungry Creek, it may be said here, is the creek formed by the confluence of Obia Creek and Fish Creek, which flows into the Lochsa River. The expedition struck it above the forks, on Obia Creek, followed up Obia Creek, and then crossed some of the upper branches of Fish Creek, but apparently used the term 'Hungry Creek' to apply to both forks. The Lewis and Clark map of this portion of the route, which is remarkably accurate, clearly shows the two forks of Hungry Creek, with their route up the north fork, or Obia Creek."—Elers Koch[88]

A common misconception about the maps is that the

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[83] Alvin M. Josephy Jr., *The Nez Perce Indians and the Opening of the Northwest* (Boston: Houghton Mifflin Company, 1997), 5.

[84] Moulton, "Journey's End," *We Proceeded On*, 26 (Nov 2000), 12.

[85] Milo McLeod, interview with author, 10 January 2001.

[86] U.S. Forest Service, Northern Region, *The Clearwater Story: A History of the Clearwater National Forest*, by Ralph S. Space (Missoula, MT: U.S. Forest Service, 1964. Edited and re-published, Missoula, MT: U.S. Forest Service, 1970), 43. Hereafter cited as *The Clearwater Story (1970)*.

[87] Peebles, "Rugged Waters: Trails and Campsites of Lewis and Clark," 8.

mileage recorded by Lewis and Clark is not accurate. Researchers make the mistake of trying to apply Lewis and Clark's mileage as a linear measurement on the ground. Lewis and Clark were recording mileage as time and effort to go from one point to another. One has to understand how to calculate mountain mileage, something this researcher has learned from a lifetime of traveling in the mountains. One linear mile on the map averages out to 1 1/2 ground (mountain) miles of time and effort to cover 1 linear map mile.[89] A good example of calculating mountain mileage would be George B. Nicholson's computations on the Southern Nez Perce Trail in 1866 for the Bird–Truax wagon road survey in his capacity as a civil engineer. Nicholson determined his mileage by timing the rate of travel on horseback, recording 18 ground miles from the Idaho-Montana state line to the Selway River.[90] This distance is 12 map miles (18 ÷ 12 or 1:1.5 ratio). He also recorded 6 ground miles from the Selway River to the top of Magruder Mountain. This distance is 4 map miles (6 ÷ 4 or 1:1.5 ratio). Nicholson's travel ratio of 1 map mile for every 1.5 recorded ground miles is consistent with this researcher's experience in using horses in mountain terrain and timing the rate of travel (or miles) on horseback.

However, on the Lolo Trail (or Northern Nez Perce Trail), Nicholson precisely measured mileage for the proposed wagon road with a surveyor's chain from the Weippe Prairie to Lou-Lou Forks for a distance of 5+ miles. From Lou-Lou Forks to Packer Meadows near the Idaho-Montana State line he measured the mileage by timing his horse.[91] His mileage was accurately recorded on marked "mile trees." Thomas Sutherland took notice of them during the Nez Perce War of 1877, remarking that the marked miles were much longer than one mile:

> About twelve years ago several thousand dollars were appropriated by Congress to have the Lolo Trail surveyed, and judging from the great distances between the mile posts, the engineers were in league with some one who wanted to get a further appropriation for a wagon road. In connection with the extraordinary length of these miles, they are nearly all straight up and down mountains.[92]

Sutherland experienced firsthand that it takes more time and effort to travel across the Bitterroot Mountains than to travel on flat ground. Bud Moore, who knows the landscape intimately as a retired Forest Service ranger and the author of *The Lochsa Story*, agrees that mountain mileage when traveling on foot or horseback is always further than the miles shown traveled on the map.[93]

Lewis and Clark recorded a ratio of 1 map mile for every 1.5 recorded ground miles from Packer Meadows to the Weippe Prairie, stretching that formula to over 2 miles recorded for every 1 map mile in extremely difficult terrain.[94] Understanding this concept is crucial to locating the trail. From the Sinque Hole Camp traveling west to Dry Camp (Lewis's route) and to Doubt Creek (Clark's route), each explorer separately recorded over 2 miles for every 1 map mile, a sure indication that this section of the trail is very difficult.

On the trip west from Glade Creek Camp to the first village campsite on the Weippe Prairie, Clark recorded 128 miles. This distance is 85.2 map miles, a ratio of approximately 1:1.5 (128 ÷ 85.2 = 1.50). Difficult terrain increased the ratio of recorded mountain miles to 2.31 recorded miles for every map mile. On easy ground, however, the ratio dropped. From Pheasant Camp to the first village camp site on the Weippe Prairie, September 22, 1805, Lewis records 10 ground miles for 8.6 map miles (1:1.16 ratio). As these recorded mileages and ratios show, the recorded mileage ratio changed from difficult to easy depending upon the difficulty of the terrain.

The researcher has to walk the correct location of the trail to know which mileage ratio to use in calculating the actual map miles from Lewis and Clark's recorded miles. The recorded mileage makes sense when the researcher experiences the same topography that Lewis and Clark crossed. Only then does the formula of time + effort equal recorded ground mileage really sink in. The 1 map mile = 1.5 recorded ground miles is a general average that can be applied to the recorded mileage of early mountain explorers. This is a concept every experienced mountain traveler understands, from old-time game wardens to old

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[88] Koch, "The Lewis and Clark Route Retraced Across the Bitterroots," 168. Obia Creek was later renamed Hungery Creek by the Forest Service as recommended by Ralph Space.

[89] Linear or map miles are measured on a map which is one dimensional or flat called flat map miles or linear map miles. A Silva map measuring instrument is used to measure the distance.

[90] Nicholson, "Explorations and Surveys for a Wagon Road from Virginia City, Montana to Lewiston, Idaho" [map reproduction], in Baird, *With Bird and Truax on the Lolo Trail*, 22.

[91] Ibid. A surveyor's chain is a 66 foot long metal tape that is used to measure the feet on the ground; 1 chain equals 66 feet, 80 chains equals 1 mile.

[92] Sutherland, *Howard's Campaign Against the Nez Perce Indians 1877*, 24.

[93] Bud Moore, interview by author, July 15, 2001.

horse packers. This researcher learned how to compute mountain mileage when going into an unexplored area with horses before beginning to study the Lewis and Clark route. When planning back-country horse patrols for the Idaho Department of Fish and Game, the author found this pre-planning critical to a successful trip.[95]

Learning how to interpret Clark's maps is a challenge because modern day concepts of how to travel across the Bitterroot Mountains do not apply. The researcher has to realize that Lewis and Clark were following an Indian trail, not a modern Forest Service constructed trail. The errors in many researchers' attempts to interpret Clark's maps can be explained by trying to see the trail through modern day glasses rather than 1800s glasses. The old Indian trails followed the best ground in a direct manner. Sr. M. Alfreda Elsensohn, founder of the historical museum at St. Gertrude's Convent near Cottonwood, Idaho, and a historian of Idaho County noted that:

> The old trails used by the explorer, trapper and trader, missionary, miner, and settler, were those which Indian sagacity and the instinct of native animals had selected as the best and easiest way over high and rugged mountains and through valley wilderness. Some of these old trails were very prominent in every step of history of the state...The Indian trails led to the hunting grounds and camas fields in various parts of the Clearwater country, as well as to the buffalo grounds in western Montana and northern Wyoming.[96]

Another pioneer recalled in 1939 that "the Indian was a born highway engineer. The native Indian trails were zigzag paths, with a straight course as the final objective. ...The routes were direct and no distance was lost. The unerring judgment of the Indian in this regard is worthy of our admiration."[97]

On the open plains the Indians could travel side by side; this was generally not possible on the Lolo Trail, a narrow single path most of the length. This researcher found only a few places on the Lolo Trail where horsemen could ride side-by-side. On the Lolo Trail the Indians rode mostly single file, as described by Thomas Sutherland, who wrote that "on leaving Kamiah...the march, which was single file on account of the narrowness of the path..."[98]

One of the best descriptions of what the Southern Nez Perce Trail on the open prairie must have looked like in the days of high traffic is by Sr. Elsensohn:

> One old history speaks of the Nez Perce Trail as the great highway of the Walla Wallas, Cayuses, and Nez Perces, eastward to the buffalo ranges to which they annually resorted for large supplies of meat. It was worn deep and broad, and in many stretches on the open plains and over the smooth hills, twenty horsemen could ride abreast in the parallel paths worn by the constant rush of the Indians from time immemorial. But the plow has long since obliterated much of this.[99]

Although the Southern Nez Perce Trail was worn deep and wide on the open plains, it was a single path in the mountains as described by George Nicholson.

Many researchers make the mistake of applying modern trail building specifications in determining the characteristics of an Indian trail. Their proposed Lewis and Clark travel routes ignore the basic principles of Indian travel. The Indian traveler went the shortest distance between two points; his trail tread was straight, and unlike Forest Service trails, it did not contour on a level grade in and out of the draws and canyons. The Indian had no modern Forest Service tools to build trails, nor did he bother to cut out fallen trees or brush. He was compelled to seek the routes of least resistance to travel. He traveled on the more open tops of the ridges, high watersheds, and over the tops of mountains. The Indian trail went straight up and straight down, and when he crossed a creek he often went straight up the side of a ridge for a short distance before turning and side-hilling, avoiding the thick vegetation found next to streams. Indian trails did not dogleg unless the path was blocked. An example of a deviation from a straight line can be seen at the Lolo Creek Camp Ground, where the Lewis and Clark Trail crosses Lolo Creek to avoid a steep basalt wall. In some cases the path diverted from a direct path to a meadow so horses could be fed and rested, an important factor to the Indian traveler. Several such places exist along the Lewis and Clark Trail

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[94] See entries for September 18 and 19, 1805 in Moulton, *The Journals of Lewis and Clark Expedition*, vol. 5, 211-215.

[95] See Chapter VI, "Methodology Used in this New Effort."

[96] Sister M. Alfreda Elsensohn, *Pioneer Days in Idaho County* (Caldwell, ID: Caxton Printers, Ltd., 1951) 2:421.

[97] Ibid., 391.

[98] Sutherland, *Howard's Campaign Against the Nez Perce Indians 1877*, 24. A good example of where the trail was wide enough to accommodate side-by-side travel is near Howard Camp, where the trail tread is a 15-foot wide path for about 300 feet and is located on a dry hillside under mature mountain hemlock trees.

[99] Elsensohn, *Pioneer Days in Idaho County*, 425.

that furnished grass for the expedition's horses.

The difference between Indian trails and modern Forest Service trails is easily identified. Forest Service trails are built with hand tools or motorized trail machines (used on Weitas Creek, Cook Mountain, and the down-river trails from Isabella Creek). After the Cook Mountain trail reconstruction was completed in the late 1980s, the trail was wide enough for small four-wheel ATV use. Modern Forest Service trails switchback up the hills rather than going straight up.

Recognizing the difference between the characteristics of Indian trails and Forest Service trails is like recognizing the difference between the old highway on Whitebird Hill that is narrow and full of curves and the new Whitebird Road, a modern highway that is straight and wide. Nearly everyone is familiar with the characteristics of an interstate highway and an old country road, but not everyone is familiar with the differences between a trail constructed by government agencies and an Indian trail. Without this knowledge, the researcher of the Lewis and Clark Trail will mistake modern trails for old Indian trails.

When using the Lewis and Clark maps in attempting to identify the trail, the researcher must understand that every entry and every map was recorded with care and for a reason. The maps were drawn so that trappers and settlers could follow the route of Lewis and Clark. Locating unique landmarks that match the journals is imperative to locating the correct trail. The Sinque Hole is a unique landmark; there is only one Sinque Hole. By matching the known point of reference of the landscape to the Lewis and Clark journals the researcher will be able to locate the trail accurately. For example, we know that the Corps of Discovery went up Wendover Ridge because there is only one Whitehouse Pond. No other landscape matches the journals.

To correctly locate the Lewis and Clark Trail, the researcher has to know how to interpret Clark's maps, how to apply the recorded mileage and remember to match the landscape to the journals. Understanding how Lewis and Clark recorded their ground miles for each day's journey is a key element in locating the trail. The researcher has to field test the proposed route by walking the ground to see if the terrain fits the physical description in the journals. Only when the primary source documents, the journals, and the maps match the landscape can the researcher be successful in identifying the correct route of the explorers. Combining this information with the knowledge of mountain horse travel and the differences between Indian trails and modern trails is another area that is crucial to the search. With this knowledge it can be determined where Lewis and Clark camped, changed directions, and how far they traveled each day. All the data collected makes sense and the pieces fit together when the researcher utilizes the primary source documents and walks the ground. And only when the pieces fit together and the trail is located correctly can the true location of the camps be established; locating the trail is essential to locating the camps.

Has the trail been located correctly or is it only an approximate route? The availability of new research materials, especially Gary Moulton's *The Journals of Lewis and Clark*, the elk skin journal maps from the Missouri Historical Society, and the recently discovered aerial photos from the 1930s allows the current researcher better tools to more accurately locate the Lewis and Clark Trail. Past researchers located the trail to the best of their ability, but using the Moulton journals and the 1930s aerial photos leads this researcher to believe half or more of the trail has not been accurately located, which puts several of the campsites in the wrong locations. With the bicentennial celebration of the trail on the horizon, the need to locate the real Lewis and Clark Trail through the Bitterroots—before it is lost forever—is more important than ever before.

CHAPTER IV

Historical Studies of the Trails of Central Idaho

In the early days, the Forest Service concentrated on cabin building, stringing and maintaining telephone lines, fire fighting, and trail construction. The Forest Service started their trail construction program by building primitive trails. Standards for trail construction improved, until about 1928 when it was decided that trails were to be built to a lower standard. Any trail that would get the crew to the fire was the goal—then the road building began. Major Evan W. Kelley became the supervisor of Region One in 1929 and his major fire control policies were to deliver manpower, supplies, and equipment to fires by extending a network of low-standard roads into the remote areas of the forest.[100] The Forest Service, and later the Civilian Conservation Corps, built the 500 Road along the North Fork-Lochsa River divide over the top of much of the Lolo Trail, following Major Kelley's policy but without regard to obliterating the trail. These early roads were called truck trails and they were very primitive.

Knowing when, where, and how Forest Service trails were built allows the researcher to find and follow the Lewis and Clark Trail and ensures that the pathfinder will not mistake an Indian trail for old Forest Service trails. This is why map history and trail history needs to be studied before the field research is started. *Early Days in the Forest Service* (four volumes of historical recollections published by the Northern Region of the U.S. Forest Service) is a good source that offers some insights.[101]

Early Days in the Forest Service, Volume 1

Hartley A. Calkins, a former employee in charge of road work, reported that the old Skookum trail (between Bald Mountain and Indian Grave Peak) was a seventeen percent grade in 1916, and that hiking was a hard, tough job with steep slopes of ten and twelve percent grades with thick brush and timber. On one trip to Elk City he made only six miles a day over those rough trails. Calkins also remembered following old trails built by miners during the gold boom, calling it "a poor excuse for a trail."[102]

American Forest and Forest Life published an article by Joseph B. Halm in 1927 that discussed the topography of the North Fork of the Clearwater River, stressing that "the entire drainage of the North Fork is through rough, mountainous territory, densely timbered with pine and fir, and few, save the occasional prospector or trapper, had ventured into this wild and almost trailless region."[103] Surveying a trail from Orogrande and "Asahka" (Ahsahka), Halm set out from Orofino with two university students, relating that "we outfitted with supplies and equipment for three months and the next day were...en route overland on our sixty-mile trip to the North Fork. We arrived at the old placer mining town of Pierce City, thirty miles out, just before dark...the old logs buildings there, mostly vacant and crumbling to decay, which had stood since the gold rush in the early 60's...At dawn the next morning we were off. Sixteen miles further on we

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[100] Moore, *The Lochsa* Story, 181.

[101] U.S. Forest Service, Northern Region, *Early Days in the Forest Service,* 4 vols (Missoula, MT: U.S. Forest Service, 1944, 1955, 1962, 1978).

[102] *Early Days in the Forest Service,* vol. 1, 20, 24.

[103] As quoted in *Early Days in the Forest Service*, 87.

reached the end of the wagon road. Pack horses took our outfit twelve miles down Orogrande Creek into the canyon of the North Fork to the "Bungalow," the end of the trail. The "Bungalow"...was merely an old log cabin with a large stone fireplace."[104] By the end of their three months of work Halm reported that "the traveler into the North Fork of the Clearwater today will find a much used auto road to the site of the old Bungalow, and a wonderfully scenic and well beaten trail, carved for miles through solid rock, traversing the entire length of the canyon to Asahka."[105]

Early Days in the Forest Service, Volume 2

Roy A. Phillips reported the status of trails in 1910, stating that "trails were few, those that did exist had been blazed out by trappers, prospectors, and miners."[106] Charlie E. Powell remembered a troublesome fire "on Rhoda Creek in the late summer of 1918, and the Elk Summit-Blodgett Creek trail was in heavy use. The trail on the Montana side at Blodgett Pass was dangerous and a few pack horses had been killed by falling off of it. Three of us were sent by ranger W. H. Young to improve the trail. We spent several weeks, as I remember, drilling rock by hand steel and blasting to widen the trail." Powell also reported "in October 1919 , I noticed a part[y] of Indians camped at Fales Flat [Southern Nez Perce Trail, Montana], on the West Fork, now called Nezperce Fork. This was the last time I ever saw them 'boxing' yellow pine either to use for medicine or to eat. There is still evidence of this on the 'cat faced' trunks."[107]

Early Days in the Forest Service, Volume 4

As Loyd Rupe recounted to Carl A. Weholt, about the early days on Elk Summit and the Lochsa, in 1910 "Major Fenn, Supervisor of the Selway Forest, assigned Adolph Weholt and his crew consisting of Loyd Rupe, Sam Weholt, Lou Lisne, and George Eckel, all of Harpster, to build a cabin and open trails in the Elk Summit area. . . . Grass grew belly deep. Deer were fat and plentiful," and the wildlife included goats, moose and grizzlies. He also recalled "the Blodgett Pass trail was another project assignment. It extended from Elk Summit, over the Bitterroots to Hamilton, Montana. When Loyd Rupe packed supplies over the trail, he found it necessary to reduce his packs to 35 pounds to keep the horses from tipping over backwards, as they clawed their way up the rock bluffs. The bleached bones of rolled animals could be seen far below—bleak evidence that an unfortunate hunter or trapper had met with disaster."[108]

Ray L. Hilding recalled that "In 1927 when I came to the St. Joe, there was no road at Avery or on the upper districts; and not many trails, just well-traveled main line horse trails. Most of the trails were constructed in the early 30's. In 1929, 1930 and 1931 the Roundtop District had a ten-man trail crew on the Little North Fork Clearwater River building trail from Rutledge Creek on down the river to mouth of Foehl Creek. Another 10-man crew was building trails on Sister's crew ('Sister' was the nickname of the crew head, a man who sounded like a woman on the phone at a time when only men were employed). Three other crews of five to six men were building trails on other parts of the District. Most of the lookout towers and buildings were constructed in 1930 and 1931 by Forest Service crews and in 1934 and 1935 by C.C.C. labor. The road to Avery from Wallace was connected in 1928, and travel from St. Maries to Avery by road was in 1933. Both roads were completed by Forest Service crews, but not entirely built by Forest Service. Until roads were developed, the Forest [Service] had about 140 head of pack stock – horses and mules. Each District had from 12 to 24 head, and about 30 head were used packing out of Avery."[109]

Major Fenn's Country: A History Of The Lower Lochsa, the Lower Selway, the Upper Middlefork of the Clearwater River, and Surrounding Lands.[110]

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[104] Ibid., 88.

[105] Ibid., 92.

[106] *Early Days in the Forest Service,* vol. 2, 22.

[107] Ibid., 30. The terms "boxing" and "cat face" refers to peeling of the cambium layer of pine trees.

[108] *Early Days in the Forest Service,* vol. 4, 116.

[109] Ibid., 147.

[110] Neal Parsell, *Major Fenn's Country: A History of the Lower Lochsa, theLlower Selway, the Upper Middlefork of the Clearwater and Surrounding Lands* (Kooskia, ID: Upper Clearwater-Lochsa-Selway Chamber of Commerce, 1986).

Neal Parsell wrote this historical overview of all lands administered by the Fenn Ranger Station on the Selway River, and it is a good source for information about trails, trail building, landmarks, and the U.S. Forest Service and its historical practices. Listed below are direct quotes from the text with page numbers.

"George Ring [one of the early forest rangers] and four men started clearing and improving the Lolo Trail in 1902, and in 1905 Ring supervised initial construction on the Selway Trail. The rangers took over several abandoned cabins, and built a few more...In 1905 the forest service was officially born."—page 10.

"The Bitterroot National Forest, as it originally existed, lasted one year...[In 1908] the Clearwater National Forest was proclaimed, which consisted of the northern part of the Bitterroot National Forest, including the Lochsa-Selway country, and part of the Coeur d'Alene National Forest..."

New rangers were then selected by competitive examination...in 1907. Each candidate had to supply three head of horses and tools to build trails and cabins, had to demonstrate that he knew how to use the tools and how to pack the animals using diamond hitches on the packs."—page 11.

"Starting in 1908 the Forest Service...began to get increased appropriations to build trails, roads, and other improvements; more men were needed. As one old timer put it; 'Those examinations really were effective. The written test eliminated the illiterates, and the field tests insured that we got experienced hands.'"

"Scheduled rail service was established at Kooskia in 1900, and several railroad companies began looking at the possibility of a route up the Lochsa and over Lolo Pass to Montana...[Railroad survey crews worked the Lochsa in the fall of 1906; and back again in the spring of 1907.] There were no trails along the Lochsa at all, and the Selway Trail ended at the Falls. The Lolo Trail was used, but any route possible was taken to get packstock from it to the river. Since the Forest Service had neither the money nor the manpower to build trails for the surveyors, the railroad hired their own trail crews and built barely passable trails and bridges along the Lochsa and up the Selway as far as Pinchot Creek...In the end, of course, no railroad was built."—pages 12-13.

"By 1910, forest service crews had completed a substantial number of improvements. The wagon road from Kooskia had been extended to Middlefork Ranger Station (Number One). Cabins had been built at Pete King...and other places...the first river trail to Moose Creek was finished in 1909. A trail up Rye Patch ridge [Pete King Trail] connected the Lochsa with the Lolo Trail. Rangers at Selway Falls and Cedars cabin at the head of Canyon Creek [Mex Mountain] could talk to the supervisor in Kooskia on the telephone."—page 13.

"The Selway National Forest...took in all of the Lochsa drainage and almost all of the Selway, about 1.8 million acres [in 1911]...A journey from the Supervisor's Office to the most remote ranger station took at least six days [by foot or by horse]."—page 14.

"Rye Patch Cabin was located about four miles up the ridge between Pete King Creek and Rye Patch Creek on what was then the main trail from Lowell [Pete King Trail] to the Lochsa-North Fork divide [Lolo Trail]. It was built before 1905, probably by rangers. A pasture (rye patch) was fenced to hold stock. Gradually the place fell into disuse, and by about 1920 it had been abandoned. All that now remains is the name of the creek."—page 15.

"The years 1911 to 1918 were not explosive fire years, and as much cabin, telephone line, and trail construction was done as appropriations allowed. The old Indian trails were improved by rerouting and regrading..."

"Early trails were built almost entirely by hand, although horse-drawn plows and graders came along later. Rock outcrops and bluffs were avoided if possible; all of the first trails had precipitous climbs and descents to get around these places. The objective was to get stock over the trails, nothing more."

"As the construction program progressed, trails came to be better designed. Locators tried to hold steady grades, and the construction crews could no longer go around bluffs and outcrops. They blasted through them."—page 17.

"[Beginning in 1920] with increased manpower and appropriations; construction of trails and lookouts accelerated...The toughest job of all, a 12-mile trail through Black Canyon on the Lochsa, was finished in 1927..."

"By 1924, the estimated carrying capacity of the Forest (the old Selway) was 23,500 sheep, although fewer were actually grazed. The season ran from mid-June through the end of September. Most of the range was in Fish Creek-Deadman Creek area and from Coolwater Ridge east. The sheep were trailed in over the top via the Lolo Trail or up the Middlefork Road. All of the permittees were big sheep outfits; few of them were local."—pages 19-20.

"By 1928, the essential trail network was in place, and 14 cable trams had been built to allow men to cross the rivers. But the Forest Service wanted to build even more trails. Construction standards were dropped; just about any trail a loaded pack string could traverse became acceptable. These 'way trails' ran everywhere in every direction, but most of them were never maintained and cannot be found today."—page 21.

"By the late 1920s, the Selway Forest owned and used

over 200 head of pack and saddle stock. However, even this was not enough in bad fire years. In 1930, several grazing permits were cancelled on Coolwater Ridge and a packstock reserve was established..."

"The deepening economic depression of the early 1930s made reorganizations and cost-cutting programs necessary throughout the Forest Service. All Forests eliminated ranger districts by combining them with others. On the Selway Forest the Bear Creek and Meadow Creek Districts were added to others in 1932. The Powell-Elk Summit country, which had been run out of Montana since 1915 but was still officially part of the Selway Forest, was combined into one district the year before. By the Spring of 1934, the Nez Perce Forest at Grangeville had five districts, the Clearwater at Orofino had four, and the Selway at Kooskia also had four."—pages 21-22.

"The Selway National Forest officially died on October 29, 1934...The Powell-Elk-Summit country went to the Lolo National Forest, which was more or less a formality since that Forest had administered it for 20 years. The Moose Creek District went to the Bitterroot National Forest and the Lochsa District went to the Clearwater..."

"The new Nez Perce-Clearwater boundary ran along The Lochsa-Lolo Creek divide from Woodrat Mountain east to the head of Canyon Creek, then to the Lochsa via Frenchman Butte and McLendon Butte, and then to Lizard Lakes by way of Huckleberry Butte and Stanley Butte."—page 33.

"By 1956, it was clear that the road [Highway 12] would be completed in the next few years. This made another forest boundary adjustment necessary."

"On July 1, 1956, all of the Nez Perce Forest lands north of the Middle Fork of the Clearwater and the Lochsa-Selway divide were transferred to the Clearwater National Forest."—page 40.[111]

The early 1900s were a time of great change on the Clearwater National Forest. Game trails and the few remaining Nez Perce Indian trails were replaced by primitive Forest Service trails built solely to get men to forest fires. By 1923 many of these trails were rebuilt, reflecting the "gentler" grade standards for trails. The Lewis and Clark Trail researcher should be familiar with the history of Forest Service trail building in the early years and be able to identify the character of these early trails.

[111] According to a ranger in the Lochsa District of the Clearwater National Forest, in 1962 the Lolo Forest boundary was changed to the Montana State Line (Cindy Lane, interview with author, 4 April 2001).

CHAPTER V

Map History and Historical Trails Study

The researcher must be familiar with past trail construction by the Forest Service so that he can determine who constructed the trail tread on the ground. Is the Lewis and Clark trail tread an old Indian trail that has been reconstructed by the Forest Service? Is the trail tread original Forest Service construction or is it an original Nez Perce Indian trail? The study of old Forest Service maps will help the researcher determine the history of trail construction and identify the builder of the trail. Early Forest Service history is sketchy, and only a few diaries kept by employees exist; so the maps are the best record of Forest Service trail building left today. The map history which follows provides the reader with a rough outline of trail building in the study area examined during the years 2000 and 2001.

The researcher must be aware that maps will show inconsistencies between what is shown on the maps and what is actually on the ground. One example is the 1936 map which shows a road from Boundary Peak to the Lochsa Ranger Station. This road was planned but only a few miles were built east of Boundary Peak. Later maps show the road ending at Boundary Peak.

Another inconsistency concerns old Indian trails. Lyle Wilson, former Forest Service trails builder,: "Nez Perce Indian trails with few exceptions were used for different purposes and traveled on steeper ground than Forest Service trails. The old Nez Perce Indian trails as a rule did not go across the landscape in a fashion that met Forest Service needs, so the Forest Service built new trails."[112] This is a Forest Service misconception not supported by the findings and research of several Forest Service archaeologist who found that Indian trails are located everywhere across the Clearwater National Forest and many were adopted by the Forest Service in the early 1900s. They also found year around Indian occupation sites located on the Forest. The Lolo Trail was one of the documented Indian trails adopted by the Forest Service. This Indian trail is still discernable from the Weippe Prairie to Montana and segments have been added to the Lolo Trail by the 1866 Bird-Truax trail building project and also by the early 1900s and more recent Forest Service trail improvement projects.[113]

One of the most important maps to study is the map of the Clearwater National Forest compiled from 1915 to 1917 by F. E. Bonner, J. B. Yule, and K. D. Swan of the Division of Maps and Surveys. Ralph Space had high praise for this effort, stating that "This map combined all previously known maps. The result was a contour map on a 1/2-inch and also a 1-inch to the mile scale. This was the best map of the Clearwater Forest that had ever been prepared. With revisions in 1926, 1936 and 1942 to show added roads, trails, lookouts, and telephone lines it served until maps prepared from aerial photos took its place."[114]

Note: Each map title is followed by a brief description of the important features of the map. C.N.F is used for the term "Clearwater National Forest." All maps mentioned here are cited in the map section of the bibliography.

C.N.F. 1985 Map.

This map is of the entire Clearwater National Forest and the study area is delineated by a box drawn on the map. The following maps show the general study area.

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[112] Lyle Wilson, retired Forest Service employee, interview with author, 10 March 2001.

[113] Dennis Baird, Editor, *With Bird and Traux on the Lolo Trail,3.*

[114] *The Clearwater Story* (1981), 44.

obtain "a map from the Forest Service," which does not accurately reflect Space's preference for the historic trail location.[121]

C.N.F. 1992 Map.

The Lolo Trail has been relocated—between Pete Forks and Pete Forks Junction on the 500 Road—for the first time on a Clearwater National Forest map. The placement on the map is the historic location of the old Pete King Trail, not the Lolo Trail. This is a drastic relocation since Trail 40 from Mussleshell over Snowy Summit to just north of Pete Forks Junction has been the traditional historic location of the Lolo Trail since before Bird-Truax. The Eldorado Trail is removed from the 1992 map.

C.N.F. 2001 Map.

The 2001 map reflects the no trail era. The Lolo Trail is now called the Nez Perce Trail on the map, or the Nee Me Poo on the ground. Both the Nez Perce Trail and the Lewis and Clark Trail are defined as "Approximate Historic Route, trail Non-Existent, not Suitable for Travel." However if the red dashed line shaded by the dotted corridor symbol is interpreted as a trail existing on the ground, then not all of the Nez Perce Trail and the Lewis and Clark Trail are non-existent. This is confusing because the legend would indicate both the existence and the non-existence of trail. Trail 58 (the old Eldorado Trail) is placed on the 2001 map for 1.5 miles. This old historic trail that dates back to the 1915 map was originally 11 miles long.

As this map history indicates, the loss of the historic trails on the Clearwater National Forest is gradual, constant, and progressive. The Pete King, the Eldorado, the Lewis and Clark, and the Lolo have all shared the same fate. This map history gives an overview of the progressive trail building by the Forest Service, points out changes in the maps, and the name changes of prominent landmarks. This map history also indicates changes in historic trail locations and dates those changes.

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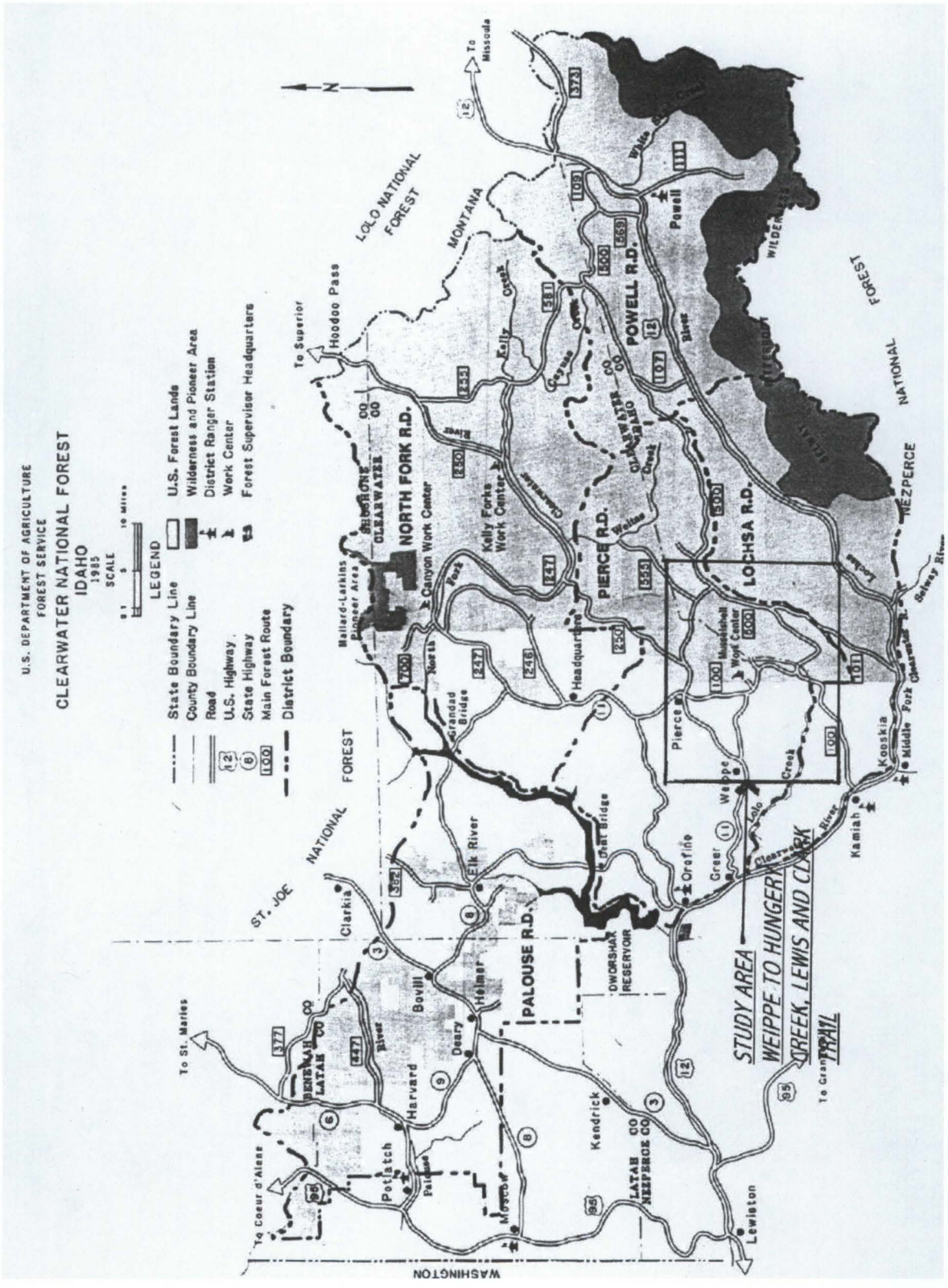
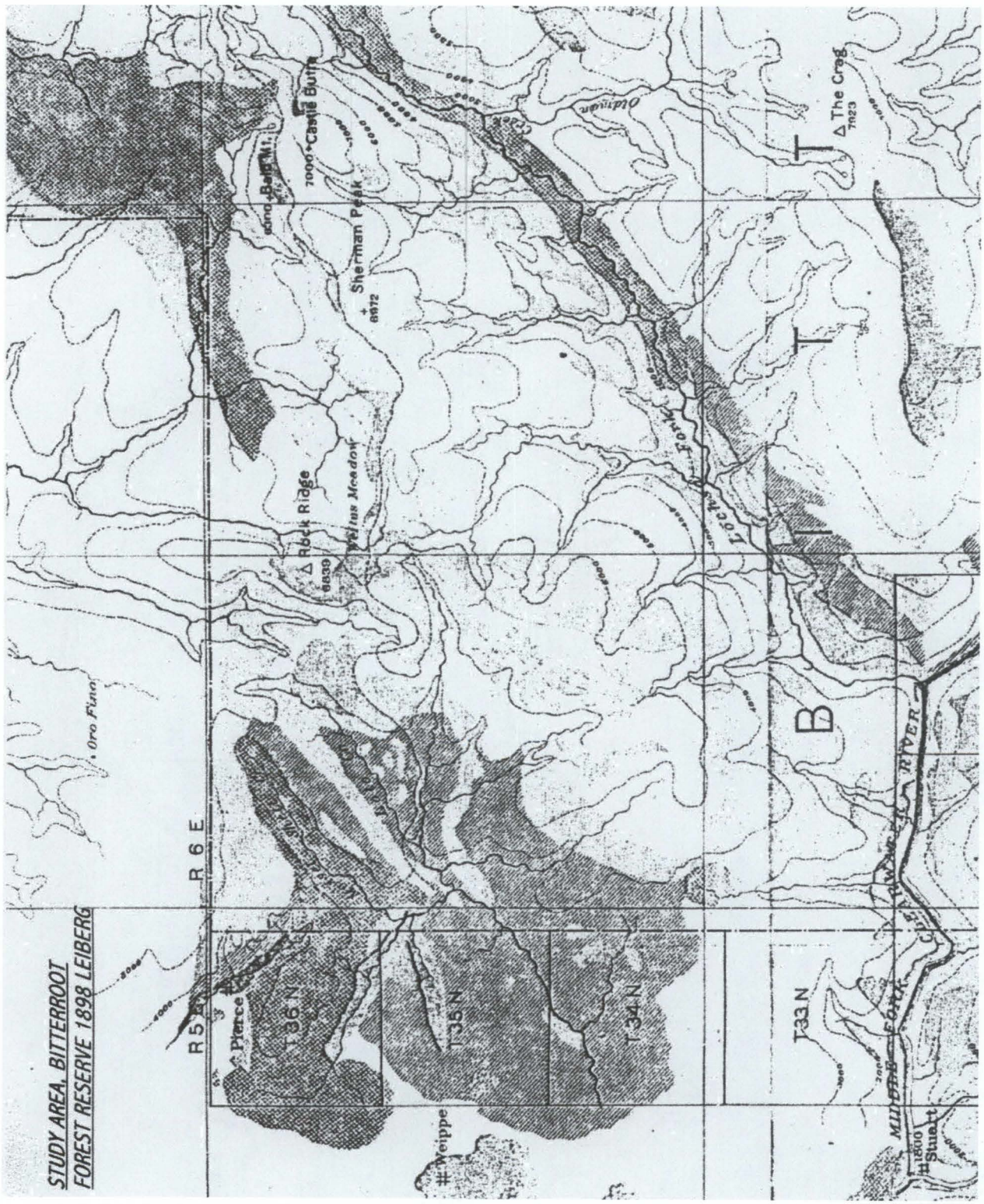
[121] Space, *The Lolo Trail*, 2d. ed. (Missoula, MT: Historic Montana Publishing, 2001), 121.

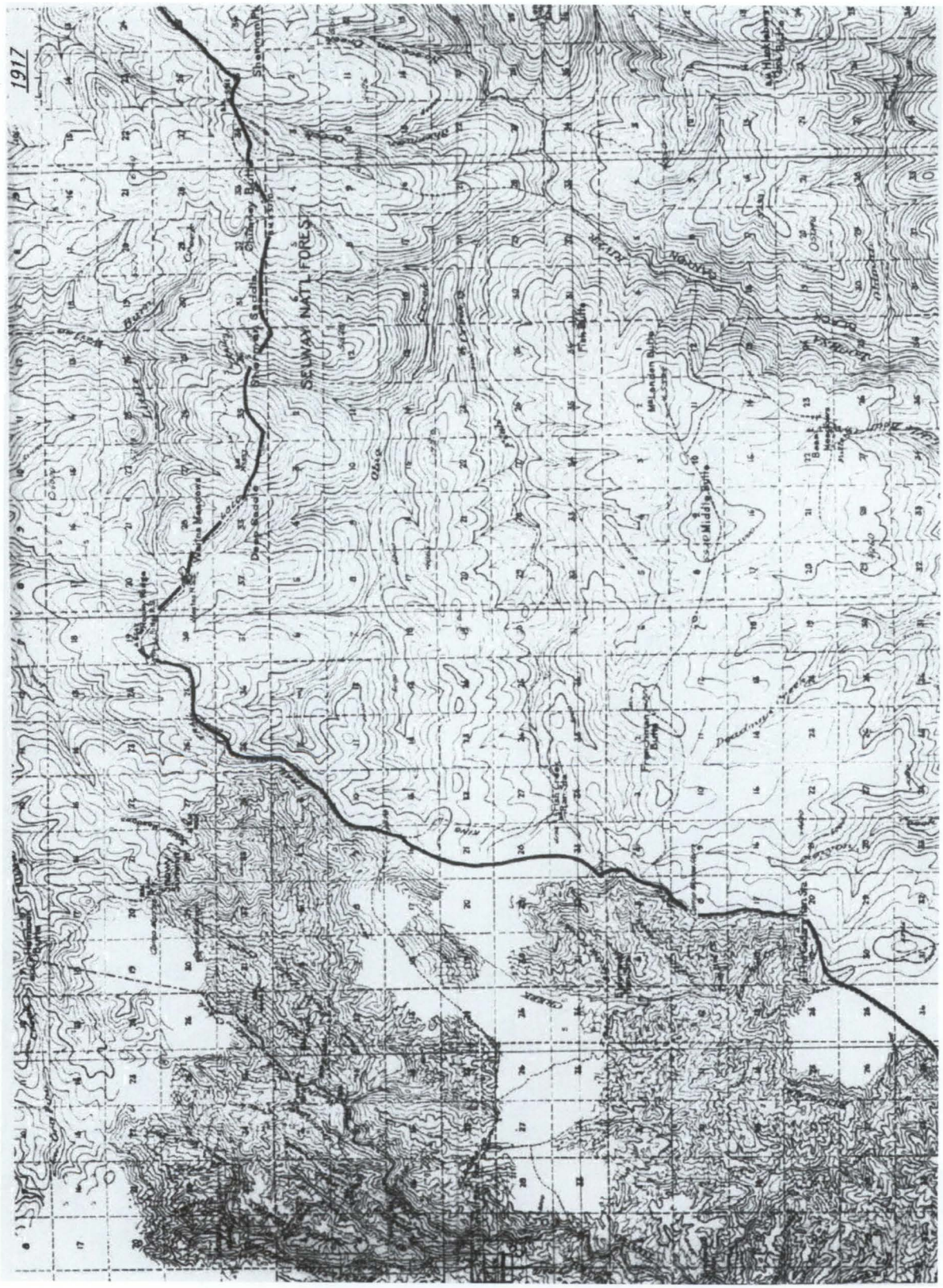


Map 3: Box indicates the study area of this research. (Source: U.S. Forest Service, *Environmental Assessment, Lolo Yoosa Timber Sale: Clearwater National Forest, Pierce Ranger District, Idaho County, Idaho* (Kamiah, ID: Pierce Ranger District, [1990]).

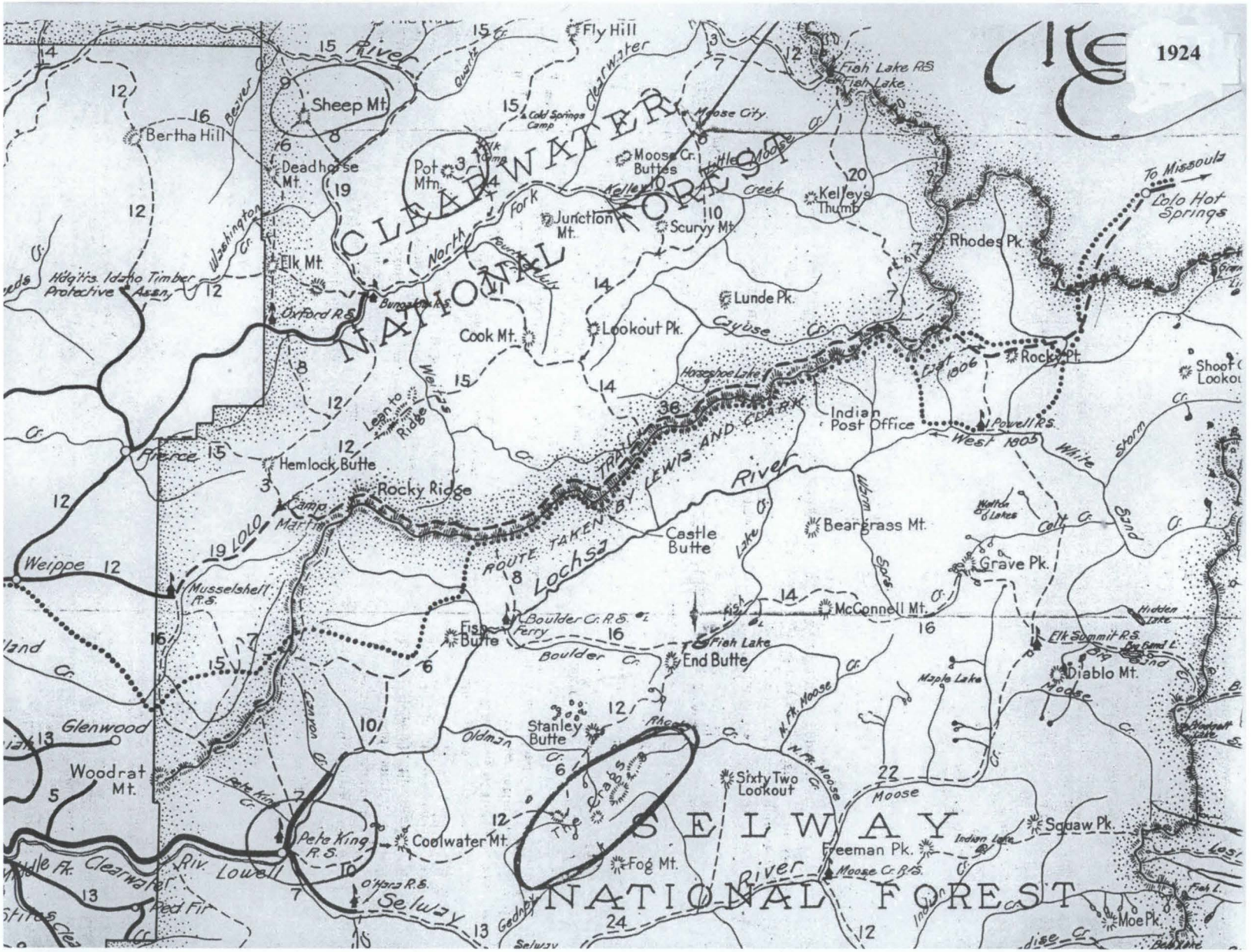


Map 4: The Lolo Trail from the Weippe Prairie to Castle Butte from Lieberg's 1898 map (Source: "Bitterroot Forest Reserve" [map], in *Twentieth Annual Report of the United States Geological Survey to the Secretary of the Interior, 1898-99*, Pt.5).

Map 5: This map shows the Lolo Trail from Musselshell to just north of Lolo Forks. The Lewis and Clark Trail from Bradford Cabin to the Pete King Trail is not on the map (Source: Clearwater National Forest, Idaho [map], U.S. Forest Service: Clearwater National Forest, 1915).

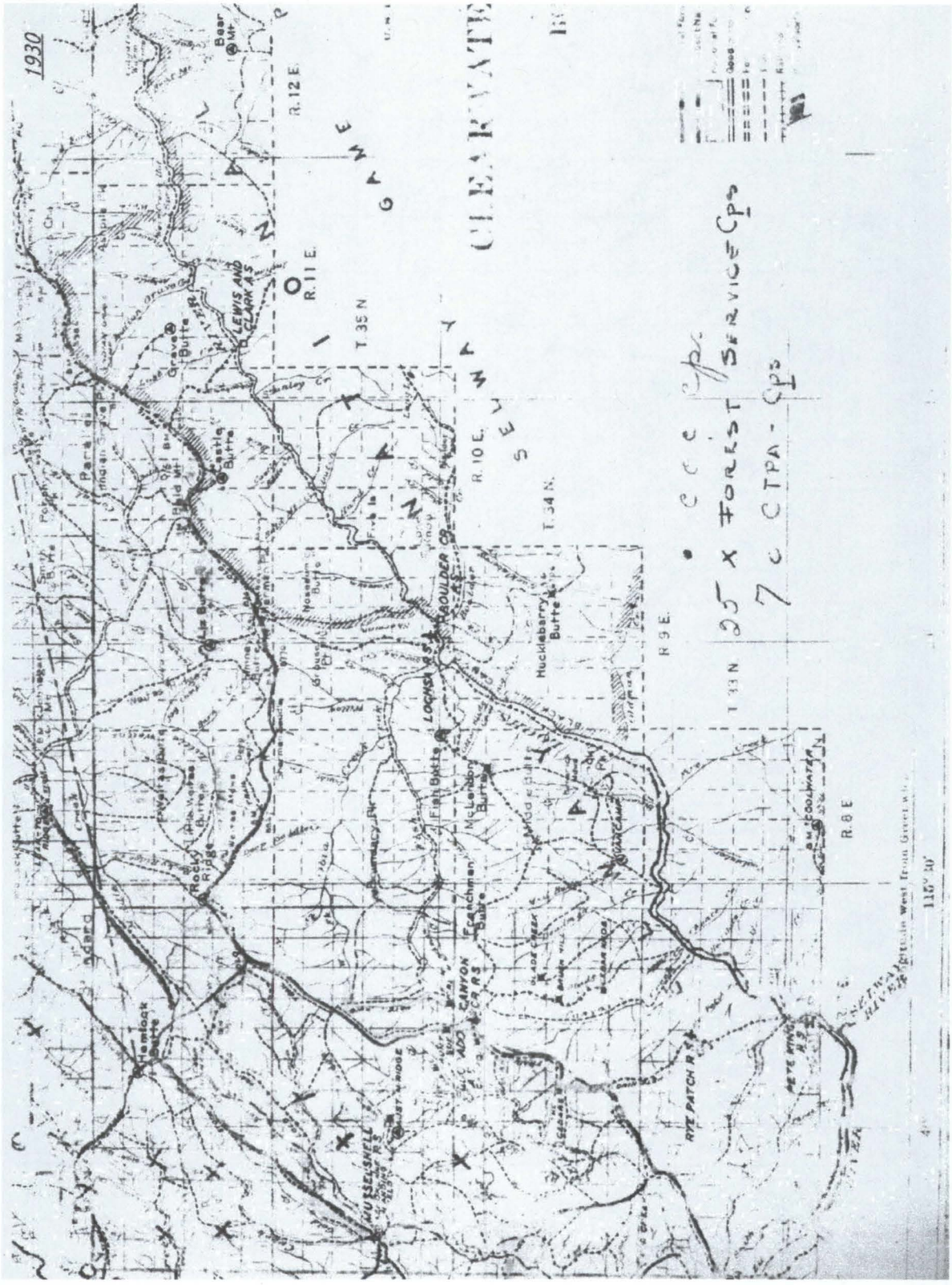


Map 6: The Lolo Trail from Musselshell to Chimney Butte. The Lewis and Clark Trail from Bradford Cabin to Sherman Saddle is not on the map (Source: Clearwater National Forest, Idaho [map], U.S. Forest Service: Clearwater National Forest, 1917).

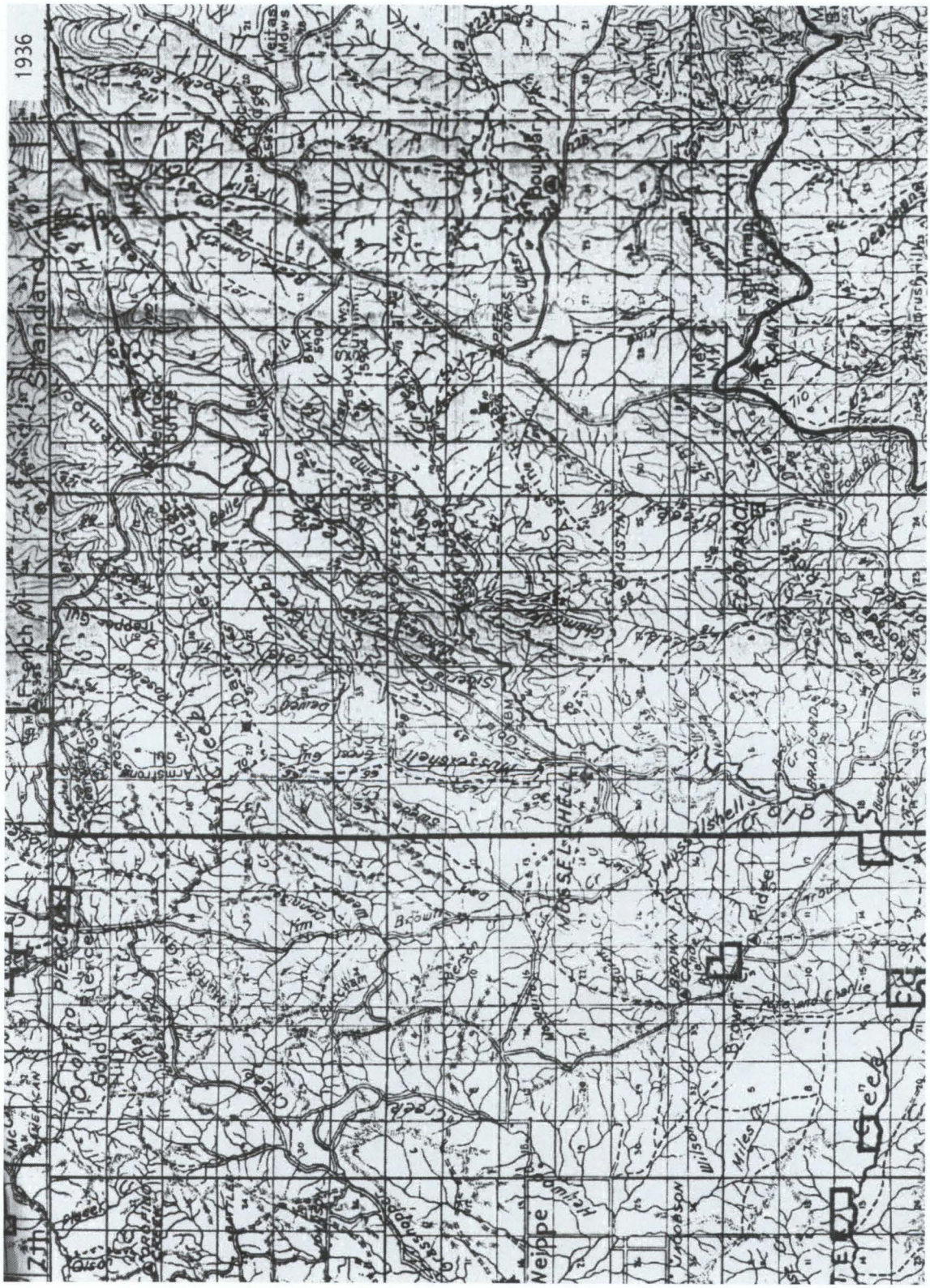


Map 7: The Lolo Trail from Musselshell to Lolo Hot Springs and the Lewis and Clark Trail from Weippe to Lolo Hot Springs (Source: Clearwater National Forest, Idaho [map], U.S. Forest Service: Clearwater National Forest, 1924).

Map 8: The Lolo Trail from Musselshell to Chimney Butte. The Lewis and Clark Trail is not n the map (Source: Clearwater National Forest, Idaho [map], U.S. Forest Service: Clearwater National Forest, 1926).



Map 9: The Lolo Trail from Musselshell to Devil's Chair near Camp Howard. The Lewis and Clark Trail from Bradford to Canyon Ranger Station near Mex Mountain has had several Forest Service trails built in the corridor (Source: Clearwater National Forest, Idaho [map], U.S. Forest Service: Clearwater National Forest, 1930).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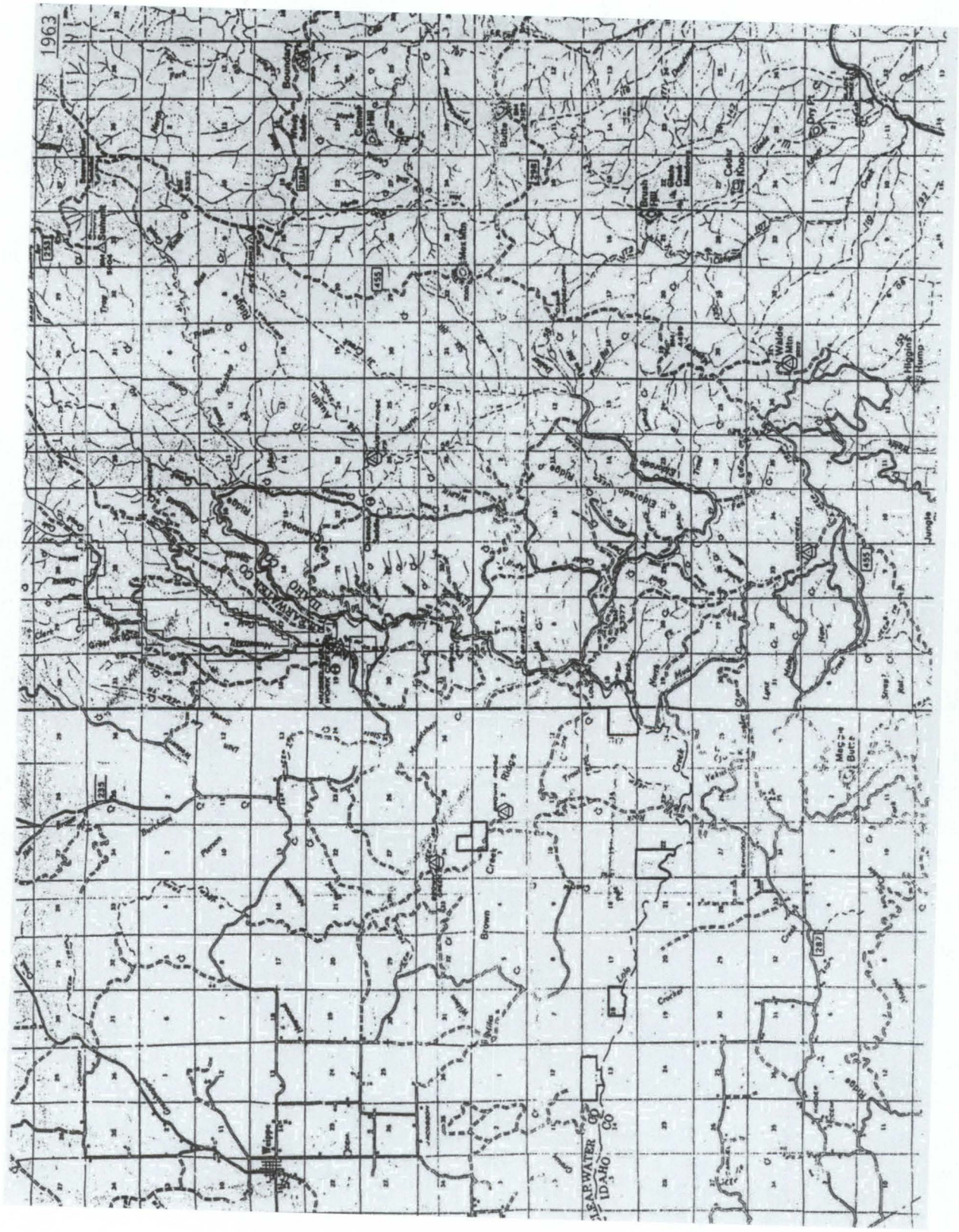
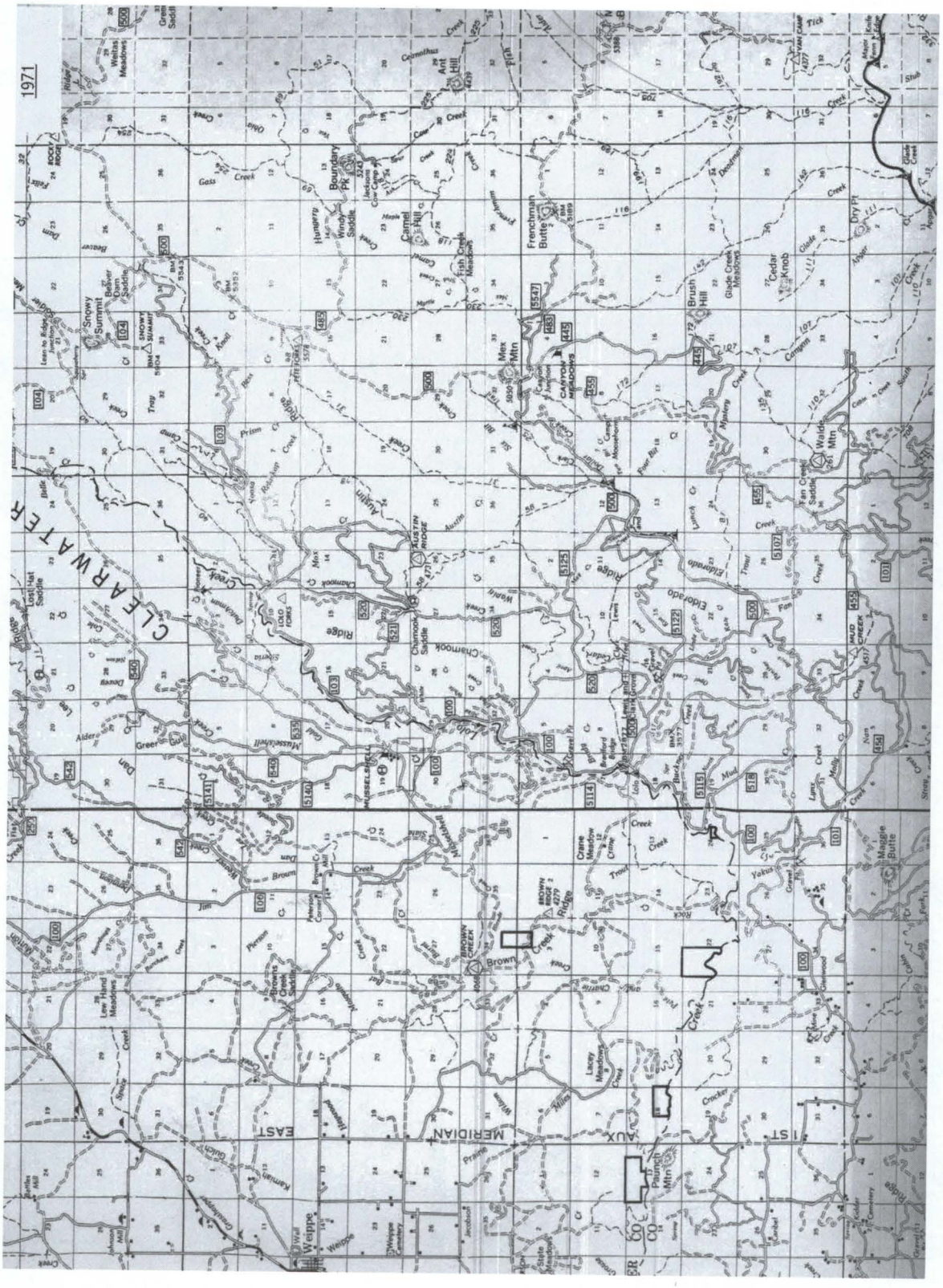
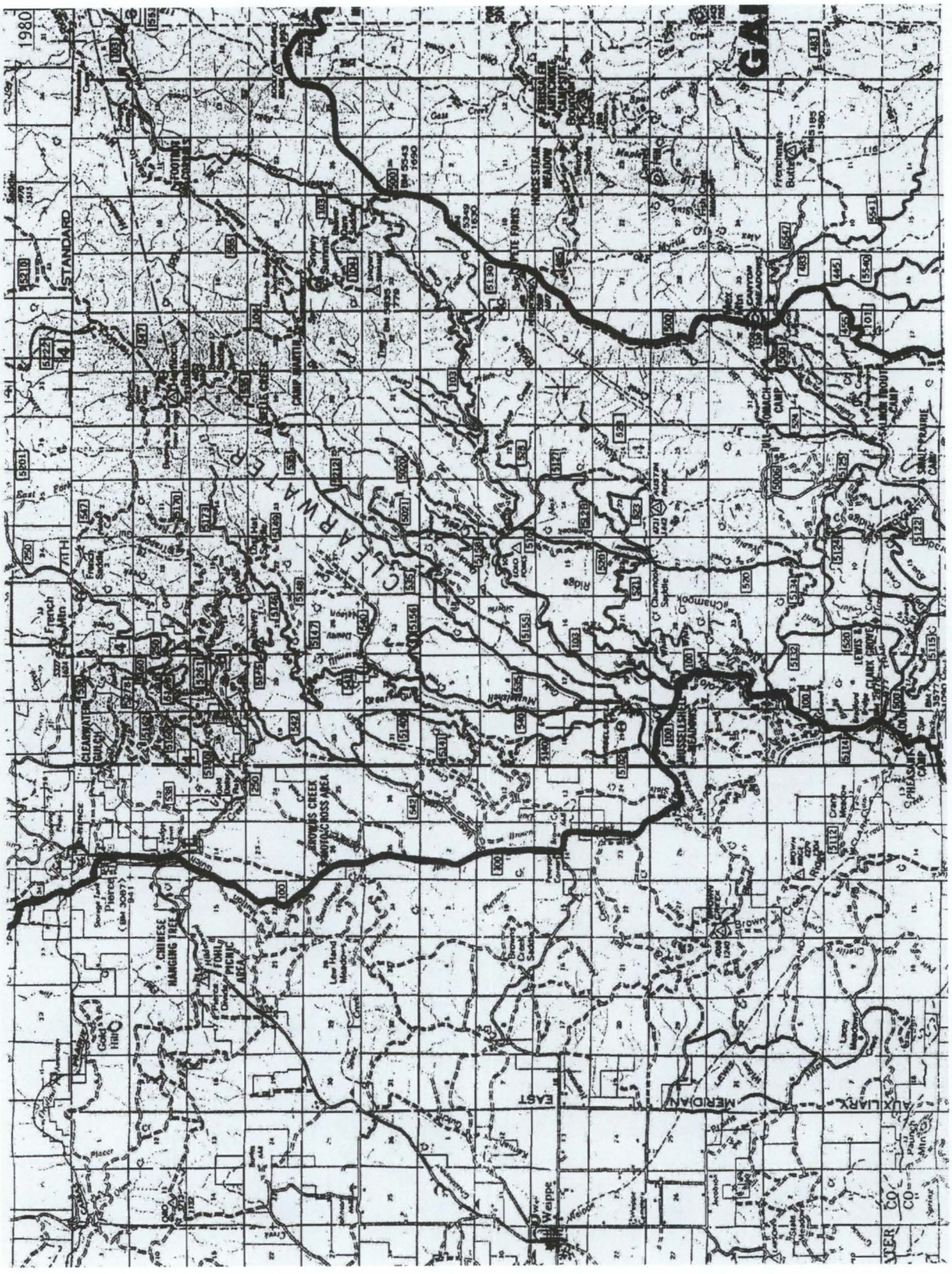
Map 11: The Lolo Trail from Musselshell to Noseeum Meadows. (Source: Clearwater National Forest, Idaho [map], U.S. Forest Service: Clearwater National Forest, 1942).



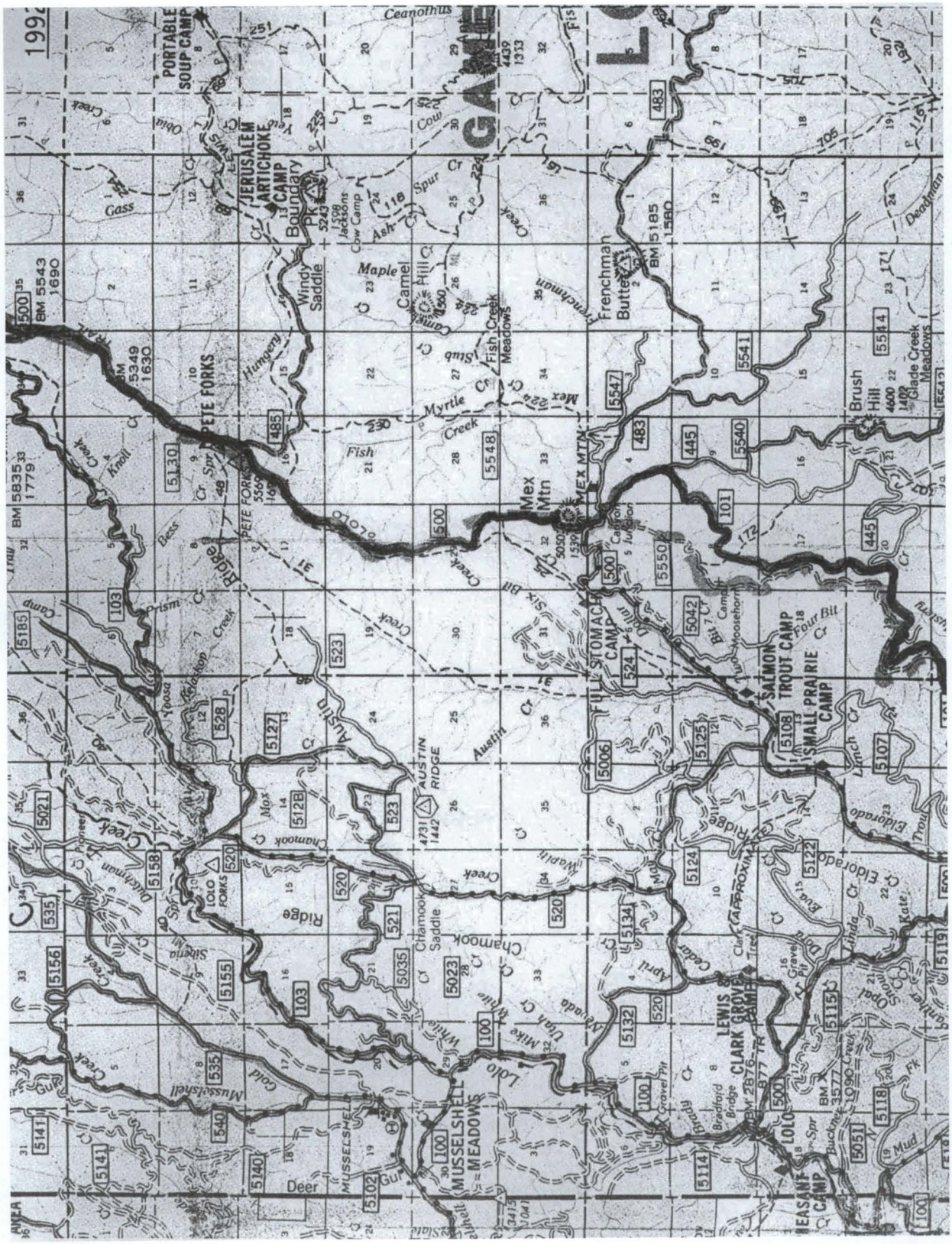
Map 12: The Lolo Trail was renamed Trail 40 from the Gold Creek Road to Camp Martin. (Source: Clearwater National Forest, Idaho [map], U.S. Forest Service: Clearwater National Forest, 1963).



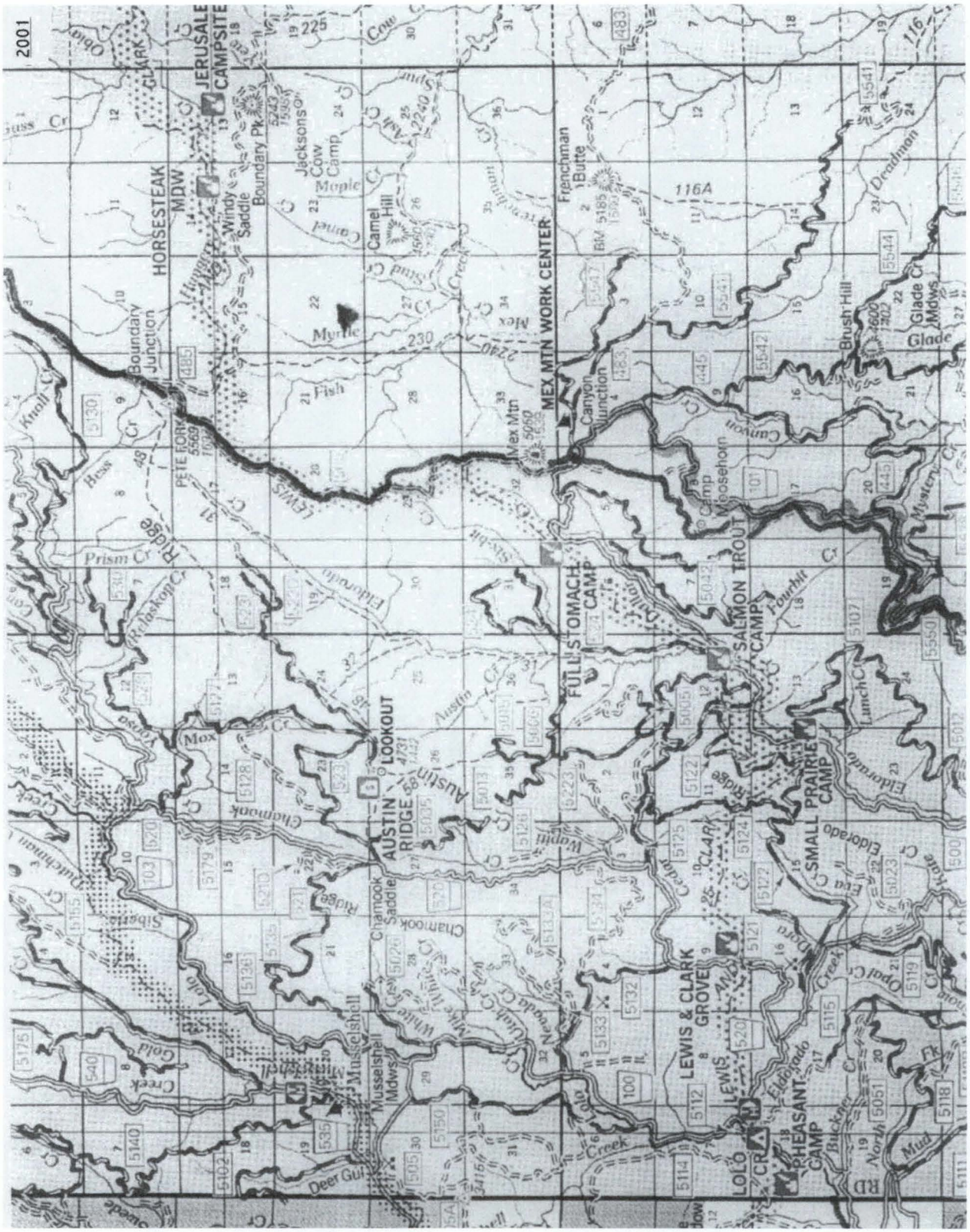
Map 13: Shows the Lolo Trail 40 from the Gold Creek Road to Camp Martin. The Lewis and Clark Trail from Bradford Bridge to just norht of Mex Mountain (Source: Clearwater National Forest, Idaho [map], U.S. Forest Service: Clearwater National Forest, 1971).



Map 14: Shows the Lolo Trail 40 has no changes from the 1971 map. The Lewis and Clark Trail from Weippe to Portable Soup Camp (Source: Clearwater National Forest, Idaho [map], U.S. Forest Service: Clearwater National Forest, 1980).



Map 15: From Mex Mountain north on the 500 Road the Lolo Trail is incorrectly labeled. The Lewis and Clark Trail from Bradford Bridge to Portable Soup Camp (Source: Clearwater National Forest, Idaho [map], U.S. Forest Service: Clearwater National Forest, 1992).



Map 16: The Lolo Trail 40 from Musselshell and continues towards Camp Martin. The Lewis and Clark Trail from Bradford to Jerusalem (Artichoke) Campsite and east beyond. The map legend indicates both the existence and nonexistence of the historic trails (Source: Clearwater National Forest, Idaho [map], U.S. Forest Service: Clearwater National Forest, 2001).

CHAPTER VI

Methodology Used in This New Effort

What methods should the researcher use to locate the Lewis and Clark Trail? The basic tools for a thorough, historically accurate search for the Lewis and Clark Trail are the primary source documents: the Lewis and Clark journals, William Clark's maps, the recorded mileage, and compass bearings. The next step is interpreting the maps and mileages and locating the route on old aerial photos. Field research of the landscape to interpret how the landscape matches the journals is the last step in the process. Experience as a woodsman following old trails and using horses in the mountains are invaluable skills useful in locating the route and campsites of Lewis and Clark. Working as a game warden for twenty-three years has taught this researcher that finding and following old trail tread is like tracking a game violator or a wounded animal. One learns by experience to read the ground.

The key to locating the Lewis and Clark campsites is discovering the correct trail, not an approximate route that Lewis and Clark traveled on. This cannot be achieved without understanding the characteristics of an Indian trail. What is an Indian trail, and how does this trail differ from Forest Service trails, game trails, or livestock trails? Have Indian trails been studied, or is this a new field of research? What has been written about them? One of the earliest references to Indian trails came from Samuel Black, the Chief Trader for the Hudson's Bay Company at Fort Nez Perces in 1829. Black responded to a questionnaire sent to all Chief Traders by the Indian Committee of the Natural History Society of Montreal. When asked "In what manner do the Indians reckon the points of the compasses, and by what means do they make their way through woods?" Black replied:

> East the rising Sun West the seting South the half day & the North the opposite or Cold Wind point. No woods in this district except some in the Mountains but the Indians make their way through the Country by some Marks within their Knowledge such as Eminences &c more over they have a Natural Talent or instinct particular to Indians and some Whites in going straight to a place & can point out the point or line it lies in at a great distance pretty correctly.[122]

Without using the straight-line concept, it would be difficult, if not impossible, to locate the Lewis and Clark Trail. The trail is an old Indian "express" way to the buffalo hunting grounds that went straight up and down the mountains. Early white travelers noticed this very distinct characteristic. Professor Oliver Marcy described his journey across the Lolo Trail in 1866, writing that

> The trail passed over the highest points on the divide and the angle of ascent and descent was sometimes very sharp. That of the East face of the first Butte down which we plunged through the Snow we measured with some care and found it to average 30° though in places it was steeper.[123]

Seasoned trappers, miners, and wanderers described an Indian trail in more colorful terms to Andrew Garcia, a Montana pioneer, when he inquired about the condition of the Skalkaho Trail:

> On my inquiry if the trail was good or bad, and how many days it took a pack outfit to make it over to the Bitterroot Valley, they said that none of them were over it, but from what they heard it was a hell-roarer and only fit for an Injun. Few if any white men used it. There was plenty of fallen timber and windfalls, and as the Injuns never cut a log out of their way, how they made it through, only themselves and the

[122] Black, Samuel, *"faithful to their Tribe & friends," Samuel Black's 1829 Fort Nez Perces Report*, edited by Dennis W. Baird (Moscow, ID: University of Idaho Library, 2000), 33.

[123] Baird, *With Bird and Truax on the Lolo Trail*, 39.

> devil knew. Across the divide on the west side going down the Skalkaho, it was steep and a long hard drag up or down to the bottom, in fact it was worse than trying to ride down a sixty-degree house roof, the trail following down the Skalaho straight as a gun barrel, without a twist or a turn through a rough broken up country of rock slides and through large granite boulders.[124]

What is an Indian trail? It's "straight as a gun barrel" and tough to travel.

This researcher used the following Indian trail characteristics in determining whether a section of ground could be considered as a possible route. They are:

1. The shortest distance between two points—the most direct route on good ground.
2. Trail tread can be found for most of the route (exceptions will be when the ground has been disturbed by logging, then usually only short segments can be found).
3. No meandering: one will see a straight path when looking at a small segment of the trail tread. Indian trails are almost always straight, can be almost vertical (straight up and straight down), exceed a 10% grade in many places, and follow the lay of the land.[125]
4. The trail covers the best ground and avoids thick vegetation, and is quite often found on top of a ridge or high on the side of a ridge near the top, where the draws are shallower and vegetation is less of a problem.
5. Pealed pine trees are found along the route (not a requirement but helps qualify the route).

Learning about the straight character of Indian trails has helped this researcher locate lost pieces of the Lolo Trail. For example, east of Lewis and Clark Grove the Forest Service version of the trail did not match the straight-line principle for Indian trails because it has a southern bow-shaped deviation in the straight line. Searching for a more direct, straight-line route led to the discovery of lost pieces of the historic trail, which ultimately led to the discovery of the precise location of Small Prairie Camp. The location the Forest Service has identified as Small Prairie Camp is an excellent example of where their trail does not follow a straight-line trail and thus identifies the camp in the wrong location (see map, following). Another example is the Full Stomach Camp location identified by this author that agrees with Lewis's September 21, 1805, description of traveling 1 1/2 miles September 21, 1805, where the party passed a large creek running to the left.[126] Clark's map shows the trail coming off the south side of Dollar Ridge (not the backbone) and striking Dollar Creek above its mouth (see maps, following). The straight-line segments located by the author were identified on aerial photos and verified on the ground. Neither the trail nor the camps can be located without understanding the straight-line Indian trail principle.

Another key to understanding an Indian trail is understanding the Nez Perce Indian and his horse—not the Appaloosa breed, but the old Spanish mustang horse, which could come in any color: buckskin, paints, whites and blacks, bay, sorrel, including, but not limited to the Appaloosa color. How does the Indian horse of the 1800s differ from the American horse of the same time period or the modern horse? On the western frontier during the nineteenth century, mustangs were in many ways preferable, as Texan folklorist, teacher, and author J. Frank Dobie describes:

> During the era of frontier wars, critics constantly asked, "Why cannot our cavalry on picked, grain-fed horses overhaul Indians on their scrub ponies?" Texas rangers on cow ponies of Spanish blood overtook them where grain-fed cavalry horses fell behind. In a phrase applied to some men, these ponies, whether ridden by Indian, cowboy or ranger, would do to ride the river with. They would stay—stay till hell froze over and a little while on the ice. ...this quality is good to remember out of times when it was the absolute for men who lived not only on but by their horses.[127]

What can the American mustang, scion of the old time Spanish mustang horses, tell us about Lewis and Clark, or an Indian trail? Knowing the type of horses available to

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[124] Andrew Garcia, *Tough Trip Through Paradise, 1878-1879*, ed. Bennett H. Stein (Boston: Houghton Mifflin Co., 1967), 440. The Skalkaho Trail is southeast of Hamilton, Montana, and although not on the Lolo Trail network, Garcia's recollection in a good example of the characteristics of an Indian trail.

[125] This contrasts with Forest Service trails, which are constructed according to a set of standards or guidelines. The required grade of modern constructed Forest Service trails is 10% or less for straight tread; climbing turns are used for slopes up to 20%, and switchbacks are used for slopes greater than 20%. See U.S. Forest Service, Northern Region, *Trail Construction and Maintenance Notebook* (Missoula, MT: U.S. Forest Service, 1997), 84-91.

[126] Moulton, *The Journals of the Lewis & Clark Expedition*, vol. 5, 225.

[127] J. Frank Dobie, *The Mustangs* (Boston: Little, Brown and Co., 1952), 61-62.

the explorers helps solve the mystery of how Lewis and Clark got from Sherman Peak to Hungery Creek. The Forest Service Willow Ridge trail route doesn't match the journal descriptions of Joseph Whitehouse in either mileage or landscape. Whitehouse recorded that:

> We descended down this Mountain, which we found dangerous for about 3 Miles, We then ascended another Mountain; which was as bad to ascend as any Mountain we had yet seen...We descended this Mountain & came down into a narrow Valley, where we found a run of water...We delayed about 2 hours—& proceeded on, & descended a mountain about four Miles, and arrived at a Creek, whose course run east, we pursued our way up the Creek on a bad piece of Road, some places running along close under the Mountain; which is high & steep on both sides of the Creek...[128]

The Forest Service route on Willow Ridge is an easy-to-moderate climb, certainly not a difficult one. But what other route is possible? The answer can be found in the straight-line Indian trail concept and Kiy.

Kiy was one of our horses, a registered Kieger Spanish Mustang horse, true to the old descriptions: full of endurance, "small, tough, ill-shaped, and had the disposition of 'Satan in Person.'"[129] Kiy was as wild as a wolf, as surefooted as a deer or an elk in the mountains, didn't dodge brush like a normal horse, but would travel in a straight line through it, running flat out at a dead run, and float straight as an arrow like only a gaited mustang can. Knowing Kiy's character and ability to climb like a mountain goat up the steepest hill helped this researcher locate the Indian trail from the watershed divide to Hungery Creek, a piece of trail that has been lost to researchers for almost 200 years.

As this research indicates, the original trail tread can indeed be found today despite the assertions of some Clearwater National Forest employees involved with managing the Lewis and Clark Trail. These managers say the original trail cannot be located, embracing the concept that "no physical trail existed."[130] On-the-trail research is a valuable tool, especially to seasoned trackers. Both this researcher and his wife, Mollie, have been finding and following old trails for years. After graduating from Ft. Lewis College in Durango, Colorado, with a B.S. in history, Mollie spent four summers in the San Juan Mountains breaking colts and following old abandoned mining trails. She described her experience to a Forest Service archaeologist in December, 2000, recalling that

> There is good trail tread on the ground, and then it fades to bits and pieces, or no trail. But if you keep going in the direction you're headed, and follow good ground, pretty soon you pick up lost trail tread. Then you find the mine, and you know that you found the old trail to the mine, not a game trail. These trails were over 100 years old, and it's not that hard to find trails from the 1800's. On the aerial photos, the Lewis and Clark Trail is only 130 years old. The Lewis and Clark Trail is on the aerial photos, and the old abandoned Lewis and Clark Trail in Fish Creek is still on the ground, just like the old trails to the mines.[131]

Once the researcher has pored over the primary source documents and acquired an understanding of Indian trails and horses, he can utilize more recent tools to help in his search. This researcher used the Forest Service 7.5 minute quadrangle maps showing the approximate Lewis and Clark Trail, examining them for any deviations from the route as described by the journals and maps. Descriptions from the journals of the trail, camping locations, mileage, and compass headings were also considered. Aerial photos were examined with the Abrams Model CD-1 stereoscope for trail tread that matched the journals. Global positioning system (GPS) readings of the trail tread to be investigated were taken from the photos and placed as accurately as possible on 1:24,000 scale topographic maps printed by the All Topo Maps™ computer program before conducting the field work. Map mileage was measured on the trail line using a Silva Map Measuring instrument. Only then was on-the-ground research made for trail tread. All possible routes were checked to within 1/4 mile on each side of the candidate route, and all trail junctions were mapped. Instruments used to locate the trail tread were the Brunton professional pocket Transit, the Brunton electronic compass with a magnetic distortion alarm, and the Garmin GPS 12 and GPS 12CX personnel locators. To locate GPS points on a topographic map, a Silva GPS card (Universal Transverse Mercator Projection) was used.

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[128] Moulton, The Journals of the Lewis & Clark Expedition, vol. 11, 322-323.

[129] Elwood L. Nye, "Marching with Custer, a day-by-day evaluation of the uses, abuses, and conditions of the animals on the ill-fated expedition of 1876," (*Veterinary Bulletin*, supplement to *Army Medical Bulletin* 35 (April 1941); reprint, Glendale, CA: Arthur H. Clark Company, 1964), 18.

[130] U.S. Forest Service, Northern Region, *Lolo Trail System Implementation Guidelines, 1990 Draft* (Orofino, ID: Clearwater National Forest, 1990), II-1. In this publication the Forest Service indicates that the trail "consisted of a general route based loosely on the fortuitous orientation of the ridges situated between the Lochsa and North Fork Clearwater River Drainages" (ibid.).

[131] Mollie Eastman, interview by author, 1 June 2000.

While the modern Brunton electronic compass allows more precise readings, Lewis and Clark achieved remarkably good results using the compass bearings from a pocket compass and a surveying compass. The magnetic north compass-bearing recorded by Lewis for the Fish Creek drainage is accurate when the declination is added for true north. By adding 18 1/2° (magnetic north for the Bitterroot Mountains in 1805), the actual bearing recorded was South 43 1/2° west, and this is the same reading this researcher came up with on the trail tread located on the ground in the Fish Creek drainage.[132] Lewis and Clark's recorded compass bearings on the Lolo Trail are not always as accurate as they are in the Fish Creek drainage because getting a good back-sight reading depended upon a clear view of a prominent landmark. Mineral deposits could have also caused errors in Clark's compass bearings, while poor visibility and timber presumably often obstructed their back view. Another factor is whether the daily journey was in a straight line or changed directions several times. A good example of changing directions is the Sept 15, 1805, journey from Colt Killed Creek Camp to Snowbank Camp. Clark records the compass bearing going down the Lochsa River as due west 90° (magnetic bearing, not true north bearing) and he records a change in direction which is NW 45° (magnetic bearing) when turning up Wendover Ridge.[133] Both bearings are accurate within 3-5 degrees. The trail up Wendover Ridge bends to the north and then again to the west. No compass bearings were recorded for these last two direction changes. For the most part Lewis and Clark's compass bearings are fairly accurate.[134]

Another factor that must be considered is the measurement of miles. Understanding how to calculate mountain miles was explained in Chapter 4, and the formula of time + effort = recorded ground miles was discussed. The average ratio to remember is 1:1.5, but this is not a hard and fast rule and varies from difficult to easy ground. With this in mind, are Lewis and Clark's recordings for mileage accurate? The trail miles from Portable Soup Camp to Pheasant Camp was recorded by Lewis as 30 miles (September 20-21,1805). The actual map mileage is 17 miles, which is a ratio of 1:1.76. This ratio is consistent with personal experience over many years of traveling in difficult terrain either by foot or horseback. Having personally walked this route, this researcher would agree with Lewis and Clark's mileage.

Does the formula (time + effort = recorded ground miles) alter for travel over snow? In 1806 Lewis and Clark traveled over hard-packed snow and recorded 53 miles from Bald Mountain to Packer Meadows. The map mileage is 39.5 miles, (53 ÷ 39.5) which gives a ratio of 1:1.34, which would indicate easy ground travel. This ratio compares favorably to the Bird-Truax party (which included Nicholson and Marcy), who also traveled over the Lolo Trail on hard-packed snow from Bald Mountain to Packer Meadows in 1866. Nicholson determined the mileage by timing his horse, and estimated 58 ground miles for the 43.5 actual map miles, which comes to a 1:1.33 ratio.[135] The Corps of Discovery and Bird-Truax have a nearly identical ratio computed in the same conditions for a similar distance. Clark commented that traveling over hard-packed snow is much easier and faster than traveling on bare ground over rocks and fallen timber.[136]

As these examples illustrate, the formula of time + effort = recorded ground miles varies according to easy, difficult, or snowy terrain. The following chart shows mileage recorded in 1805 by Lewis and Clark on the Lolo Trail and computes the difficulty of the ground travel. The next chart illustrates that Lewis's mileage agrees closely with Clark's when they traveled separately from the Sinque Hole Camp to the first village site on the Weippe Prairie. As the charts illustrate, Lewis recorded 76 miles and Clark recorded 74 miles.

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[132] The magnetic declination for the year 1805 between Missoula, Montana, and Lewiston, Idaho, was about 18 _ degrees east. See Peebles, "On The Lolo Trail: Route and Campsites of Lewis and Clark," 15.

[133] Moulton, The Journals of the Lewis & Clark Expedition, vol. 5, 206.

[134] This subject will be further examined in subsequent research.

[135] Baird, *With Bird and Truax on the Lolo Trail*, 22; and Nicholson, "Profile of The Lolo or North Nez Perce Trail over the Bitter-Root Mountains," [map], 1898 (Clearwater National Forest Supervisors Office, Orofino, Idaho). Bird-Truax took a different route in three locations between the Sinque Hole and Packer Meadows, which made their journey four miles longer.

[136] Moulton, *The Journals of the Lewis & Clark Expedition*, vol. 8, *June 10-September 26, 1806*, 60.

Lewis and Clark's Recorded Mileage, September 14-22, 1805

Camps	Recorded mileage	Map mileage	Recorded÷map mileage=travel effort ratio	
			equals effort	time/hrs (estimated)
Glade Cr. To Killed Colt Camp	17	11.5	1.48	(average)
Killed Colt to Snow Bank	12	11.2	1.07	11 (difficult[137])
Snow Bank to Lonesome C.	13	9.2	1.41	(average)
Lonesome C. to Sinque Hole	10	7.2	1.39	(average)
Sinque Hole to Dry Camp	18	7.8	2.31	8 (difficult)
Dry Camp to Portable Soup	18	13.0	1.38	6.5 (average)
Portable Soup to Full Stomach	15	9.5	1.58	7 (average)
Full Stomach to Pheasant C.	15	7.4	2.03	(difficult)
Pheasant Camp to First Village	10	8.6	1.16	(easy)
Totals	128	85.2	1.50 or average effort	
	RM	÷ MM	= Effort	

William Clark's Recorded Mileage, September 18-20, 1805

Camps	Recorded mileage	Map mileage	Recorded÷map mileage=travel effort ratio	
			equals effort	time/hrs (estimated)
Sinque Hole to Hungery Cr.	32	16.8	1.90	(difficult)
Hungery Creek to Cedar Cr.	22	18.5	1.19	(easy)
Cedar Creek to First Village	18	11.0	1.60	(average)
Totals	74	46.3	1.60 or average effort	
	RM	÷ MM	= Effort	

Another way to estimate the effort expended is to use the map mileage and the amount of time spent traveling in a similar formula. Effort gauged by time is essentially how long it takes to cover 1 mile. The average speed walking is 2 miles per hour, and the average for riding a horse is 3 miles per hour. When the Corps of Discovery covered difficult ground on foot, it took them 1 hour to go 1 mile. The chart below looks at some of times that Lewis and Clark recorded for their daily travels.

Daily Travel Time (estimated) Indicating Difficulty of Travel

Effort (Map miles ÷Time = Effort ratio[138])	
Colt Killed to Snow Bank Camp	11 ÷ 11 hrs. = 1.0 (difficult)
Sinque Hole to Dry Camp	8 ÷ 8 hrs. = 1.0 (difficult)
Dry Camp to Portable Soup Camp	13.0 ÷ 8 hrs. = 1.63 (average)
Portable Soup to Full Stomach Camp	9.5 ÷ 7 hrs. = 1.36 (difficult)

[137] Although the recorded mileage is almost the same as the map mileage, this section of the trail can be considered difficult in terms of effort because of the time factor, an estimated 11 hours of travel time. Portable Soup to Full Stomach Camp is similar.

[138] Travel times were estimated.

Methods of Finding Lost Sections of the Lewis and Clark Trail

The proposed route for fieldwork was broken into short segments of from 2 to 6 miles depending upon known trail locations. The segments were divided according to known points of reference and walked half way and back to the starting point, after which the other half of the route was walked halfway and back. Both halves of the proposed route had to meet in the middle. The route was checked for errors by doubling back to the starting point. GPS readings were taken at about 100 meter (or less) intervals and the results were plotted using the All Topo Maps™ program. Irregularities in the straight line were noted and a return field trip was made to check for errors. When the relocated trail route bypassed the U. S. Forest Service selected camp locations of Lewis and Clark, the new camp locations had to fit the journal description and the mileage had to fit the new routes. An average of eight hours per mile was spent walking the ground. In some cases, several trips were required to accurately find the entire piece of trail being studied.

The average width of the Lewis and Clark Trail tread is 40 centimeters or 16 inches unless the Forest Service has widened or rerouted the trail tread.[139] The trail tread varied from a deep ditch to just a narrow, faint path. In some places (about 50 feet in Fish Creek and about 100 feet in Hungery Creek) Engleman spruce trees have grown in the trail tread. This is a common occurrence in abandoned trail and can be seen on the old abandoned portion of the Pete King Trail between Pete Forks and the Boundary Peak 485 Road for about 600 feet. There is good discernable trail tread for most of the way from Pheasant Camp to Horsesteak Meadow Camp, which has been personally and extensively field researched. There is also good Lewis and Clark trail tread east of Horse Steak Meadow Camp, which will be examined in the future. In general, bits and pieces of lost trail tread occur only in a few places, but the gaps will get wider if the historic trail continues to be abandoned.

Summary

The Lewis and Clark trail is not just another trail in the Clearwater National Forest—it is unique. The Indian horse is not just another horse, but an unbelievable animal—unmanageable for most white people, but a very good mount for the Indian, a superior horseman. They rode a horse quite unlike our modern day horse, and they rode where few white men could follow, as documented by General Howard's observation that "We found an abrupt descent at the Lolo fork that none but old frontiersmen and Indians could ride down, so we slipped and slid, fell and scrambled up again."[140]

Knowing how to interpret and apply the mileage recorded by Lewis and Clark is another vital key to finding the trail on the ground. This can be calculated by using the formula Time + Effort = Recorded Ground Mileage. By walking the route the researcher can calculate the time and effort figures and also verify the miles recorded by Lewis and Clark.

The keys to locating the Lewis and Clark Trail are recognizing Indian trail characteristics using the straight-line concept, discovering where the Indian trail is located in relation to the landscape as described in the Lewis and Clark journals, and knowing the difference between a Forest Service Trail and an Indian trail. The researcher also needs to know how an old Indian trail differs from the modern trail, understand a mountain horse, and understand how the old mountain horse from the 1800s differs from the modern horse, and more specifically, how the Spanish mustang differs from the other breeds of horses in his ability to travel over steep, rough ground and do it without tiring. Captain Benjamin Bonneville, on leave from the U.S. army, visited the Nez Perce in 1833-34, and described them as superior horsemen, "Who own horses 14 to 14 1/2 hands high, stout built, and upon which the Indian gallop all day."[141]

Finally, the researcher must use accurate copies of the original source documents, the Lewis and Clark journals and maps. Gary Moulton did an excellent job of gathering all the Lewis and Clark journals and the maps, and this researcher highly recommends them. By applying the concept of time + effort = recorded ground miles to the journals and maps shows that Lewis and Clark were remarkably accurate cartographers. Lewis and Clark's mileage can be understood when timing the same route they took. When the journals and maps are disregarded or misunderstood then the findings of the researcher are subject to error.

In terms of importance the methods to use to locate the Lewis and Clark trail are as follows:

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[139] Trails used by horses or elk are about 16 inches (40 cm) wide. Deer trails are approximately 6-8 inches (15-20 cm) wide. The width is related to the width of the hips or placement of the feet by the animal.

[140] Laughy, *In Pursuit of the Nez Perces*, 135.

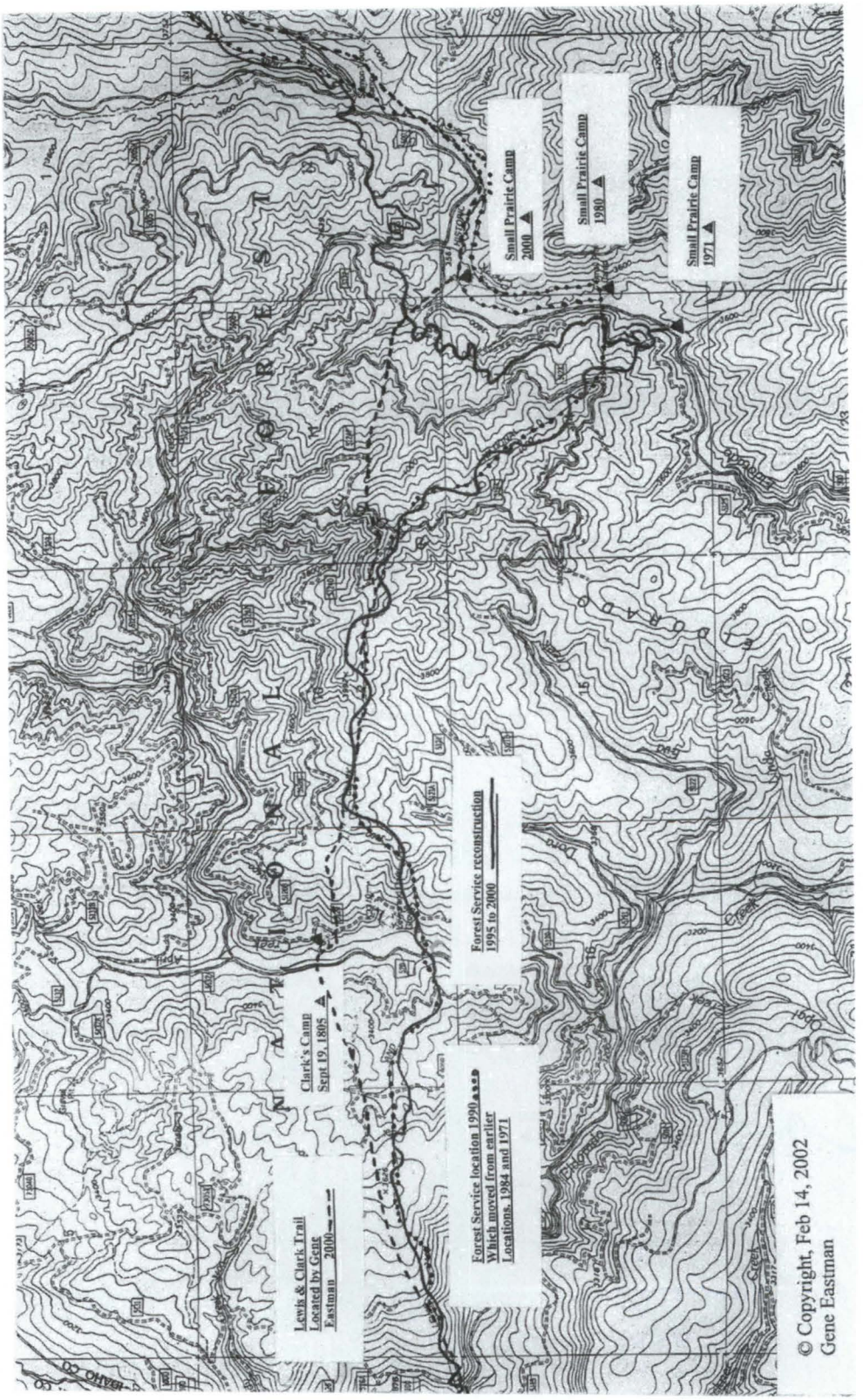
[141] Washington Irving, *The Adventures of Captain Bonneville* (Portland, OR: Binfords & Mort Publishers, 1963), 328.

1. The route description in the journals.
2. Clark's hand drawn maps.
3. Old and new Forest Service maps showing the trail and old Forest Service trails.
4. Indian trail characteristics.
5. Aerial photos.
6. Landscape
7. Recorded mileage.
8. Compass bearings.
9. Past researchers' locations of the trail.

Using these methods, this researcher looked at the general route location from the journal description and the Forest Service maps. The Forest Service location was compared to Clark's maps and checked for Indian trail characteristics. The errors were noted and aerial photos were examined for trail tread that matches the journals and Clark's maps. The trail tread was walked and the mileage was computed. Then, compass bearings were taken and compared with the journal entries. Application of the straight-line character of an Indian trail was essential, and prevents the researcher from being misled by a Forest Service trail that doesn't match the journals. And how can the researcher understand the trail, without understanding the horse the Indian rode? Maybe because these methods have been overlooked is why, even today, there are lost pieces of the Lewis and Clark Trail. When the trail matches the journals, then you know for sure that the lost trail has been found.

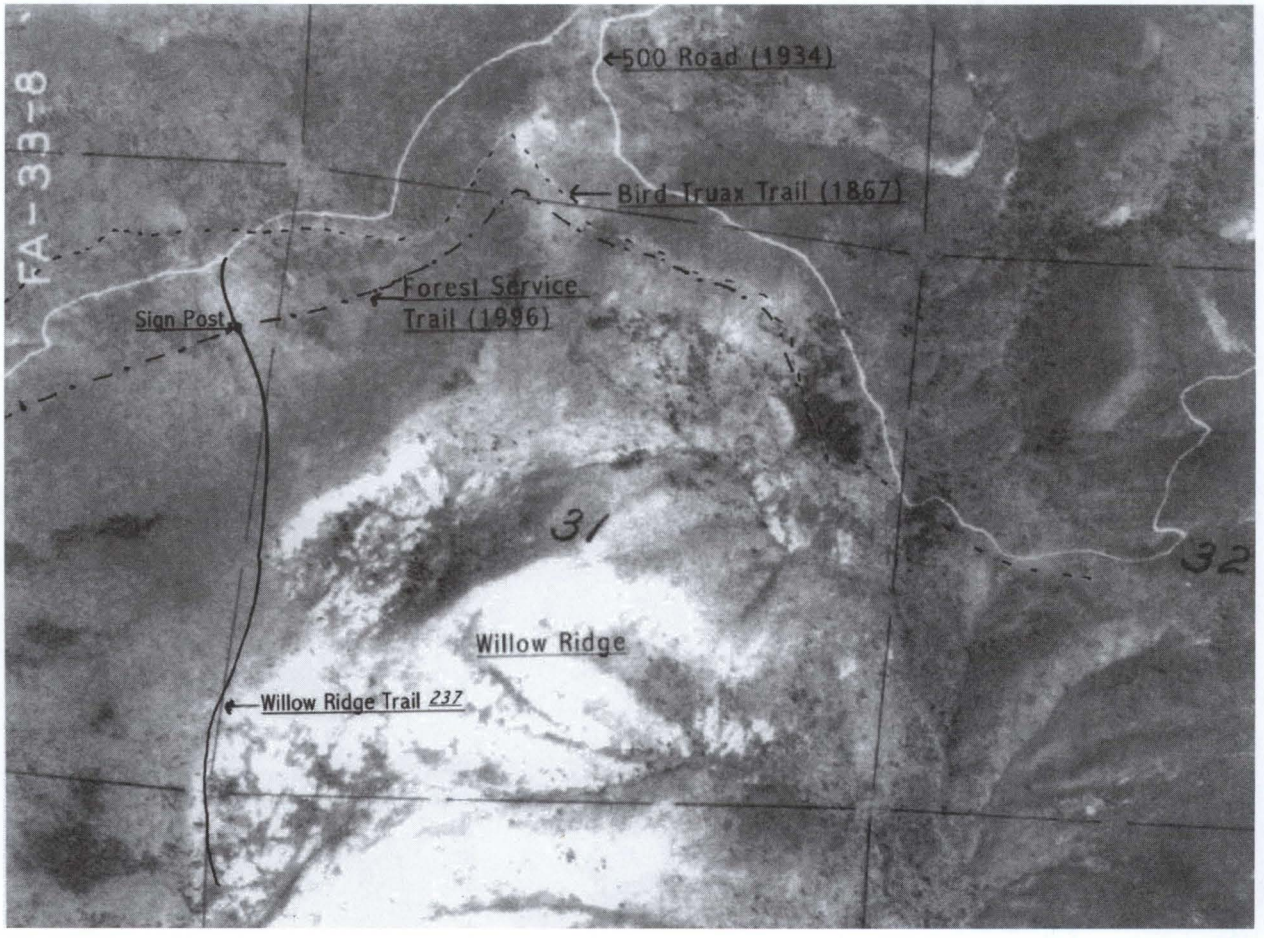
Zig-Zag Trail Construction

Figure 4: This zig-zag type of climbing turn rerouting depicted here is an example of Forest Service trail management from the 1920s to the present. Similar trail rerouting and reconstruction has occurred on the historic Lolo Trail (Trail 40) and an additional $3^{1}/_{2}$ miles of zig-zag reroutes are planned for the Lewis and Clark Trail on Wendover Ridge in 2002. (Source: U. S. Forest Service, *Trail Construction on the National Forests* (Washington, D.C.: GPO, 1923), 61.

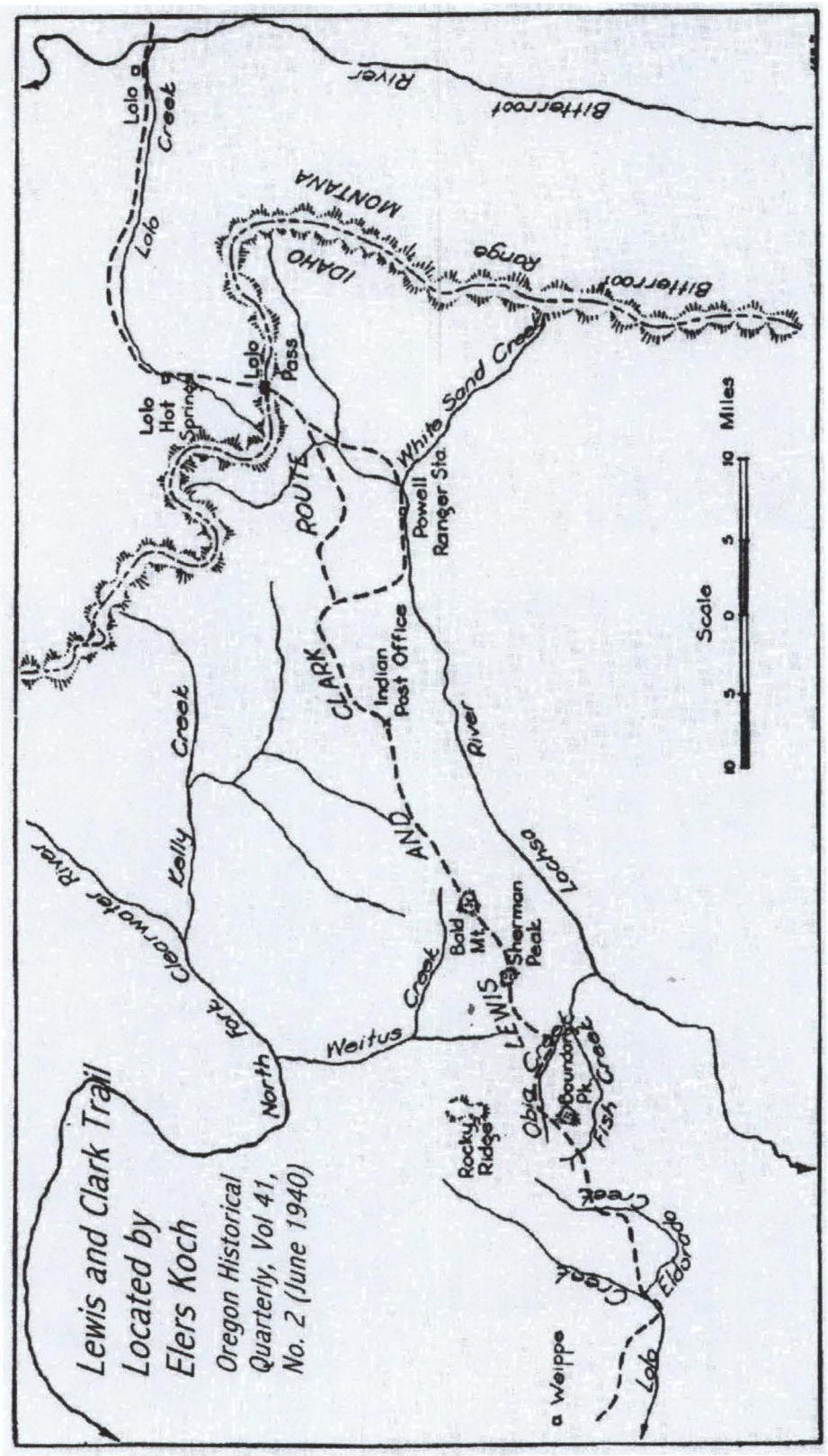


Map 19: Author's location of Small Prairie Camp (Base map is U.S. Geological Survey, "Musselshell, ID" [map], 1:24,000, 1994 edition).

Figure 5: Sign post and logos placed on the junction of the relocated Trail 237 (Willow Ridge and the new 1996 Forest Service Lolo Trail 40 (Nee-Me-Poo Trail) (Photo by author, July 2000).



Map 20: Aerial photo showing the old trail system and the new trail reconstruction. Location: T35N & 36N, R9E (Source: "FA33-8" [aerial photo], Project FA, 1937 Air Photos of the Powell Ranger District of the Clearwater National Forest [computer file] (Moscow, ID: University of Idaho Library, 1999).



Map 21: "The Lolo Trail Route of Lewis and Clark, 153 Miles in 12 Days, Across the Bitteroot Range" (Reprinted from "Lewis and Clark Route Retraced Across the Bitteroots," *Oregon Historical Quarterly* 41:2 (June 1940): 164. ©1940 Oregon Historical Society. Reprinted with permission).

CHAPTER VII

Summary

In the last 100 years there have been a number of Lewis and Clark Trail researchers: Olin Wheeler (on parts of the trail in 1898 and 1899, and over the entire Lolo Trail in 1902), John Harlan (1920s), Elers Koch (1915-40), John Peebles (1960s), Ralph Space (1960s and 1970s), Andy Arvish (1960s and 1970s), Robert Schloss (1969), Stephen Shawley (1984), Debra Doerflinger (1978-79) [142], Duane Annis and Karl Roenke (1983-84), Merle Wells (1988), and Steve Russell (1980s and 1990s). Each researcher contributed by locating a piece of the trail using the resources available at the time of their research. Today's researcher has the advantage of all the discovered Lewis and Clark journals as well as the maps from Gary Moulton's *The Journals of the Lewis & Clark Expedition*. Aided by the recent availability of 1930s aerial photos taken by the 116th Washington Air National Guard, today's researcher has the tools to ensure that the trail tread can now be more precisely located.

This researcher is adding his name to the list. A graduate of the University of Idaho with a B.S. in Forestry and a major in wildlife management, I am also a certified wildlife biologist by The Wildlife Society, and have conducted graduate-level research for the University of Idaho Cooperative Fishery Unit. Experience in surveying (local land surveys and a college surveying class) and summer school classes at the University of Idaho College of Forestry, studying timber cruising, trail and road construction, and aerial photo interpretation have broadened my skills. Twenty-three years as a game warden with the Idaho Department of Fish and Game gave me extensive experience in wildlife and fishery management. Spending much of my life working and recreating in the rugged mountains of Idaho have taught me how to evaluate sheep and cattle ranges, an understanding of trails created by stock; knowledge of big game habitats and game trails, and the ability to track both man and game by their sign left on the ground, skills which helped immensely in locating the Lewis and Clark Trail. Being familiar with the use of horses in the backcountry also helped in evaluating the daily travels of the Corps of Discovery. Applying these extensive skills and experiences to Lewis and Clark's recorded mileage, finding trail tread on aerial photos as well as on the ground, and then matching this information with the journals and maps proved to be the key to finding the trail tread. Having the right background and experience was integral to correctly interpreting the maps and journals.

There is much more to locating the Lewis and Clark Trail on the ground than studying the Lewis and Clark journals and finding trail tread on aerial photos. The researcher must also understand the Indian who made the trails and his horses. Indians traveled in a straight line and generally followed the ridge tops and open ground. They didn't build trail or clear out fallen trees and brush, as opposed to General O. O. Howard, who used fifty axemen to cut out the Lolo Trail before he could follow Chief Joseph, Whitebird, Looking Glass, and their bands in August of 1877. Indian trails are the shortest in terms of time and energy expended; navigated with the skill of a modern surveyor who uses his instruments to lay out a straight course. The Nez Perce Indians' instruments were woodcraft skills and knowledge of the landscape. The Indian horse was capable of going up and down the steepest mountains and traveling great distances without

[142] Debra Doerflinger researched the Lolo Trail from Lolo Pass to Cayuse Junction. See U.S. Forest Service, "Locating the Lewis and Clark Trail, Powell Ranger District, Clearwater NF, Summer, 1978," by Debra Doerflinger (1978; Clearwater National Forest Supervisor's Office, Orofino, ID); and "Locating the Lewis and Clark Trail, Powell Ranger District, Clearwater NF, Summer, 1978, 1979" by Debra Doerflinger (1979; Clearwater National Forest Supervisor's Office, Orofino, ID).

tiring. It was a cross between a mountain goat and a deer, whereas the horses General Howard rode were only suitable to use on good trail and lacked the stamina of an Indian Spanish mustang horse. The Nez Perce took their women and children where later the U.S. army wouldn't go—or couldn't go. During the 1877 war, the Nez Perce crossed the Salmon River at Billy Creek. General Howard retreated nearly 30 miles back to the more navigable Horseshoe Bend crossing.

In looking at past researcher's results, we must consider their knowledge of Indian trails. Does their proposed route travel over the landscape in the shortest distance possible and have Indian trail characteristics? Was the researcher being guided or was he familiar with the mountains? Did he know the difference between the capabilities of the Indian mustang and the ranch horse? Was he looking for a modern road rather than a primitive Indian road? What tools and information did the researcher have available and what tools did he use? Was he influenced by his concept of the landscape (round-about routes rather than the straight-over-the-top routes)? Did he make the trail fit preconceived ideas, old Forest Service trails, or easy travel routes?

To be successful in identifying the actual route and trail tread of the Lewis and Clark Trail, the researcher must utilize certain tools and have a historical understanding of trails, Indians, and horses. To begin, he needs to do a thorough research of old maps. This reveals the past history of trails, how they were constructed, and why and when they were built.

Second, understanding the character of the old trails is an absolute necessity or the researcher will be misidentifying trails. What is the character of the straight-line Indian trails, the Bird-Truax Trail, old railroad pack trails, early, transitional and modern Forest Service trails?

Third, the researcher needs to know the history of past Forest Service re-routing of the Lewis and Clark Trail. Did they keep records? If not, how can the researcher determine the historical trail tread from the newly constructed trail tread?

Fourth, a study of every past researcher's work needs to be conducted. Where was he sure of his location of the Lewis and Clark Trail and where was he unsure? How well did he document his work as to methods used to locate the trail and what are his qualifications?

Fifth, a thorough understanding of Clark's maps and how Lewis and Clark recorded their mileage is one of the keys to finding the route. When the researcher uses the primary source documents and the landscape fits the journal descriptions of the trail, then he is on the right track.

Sixth, the Lewis and Clark Trail cannot be accurately found without understanding the Indian Mustang horse. This is not the common mustang horse found on many of the wild horse ranges today, but the old time Spanish mustang, a horse that could go straight up and down the mountains. The modern ranch and pleasure horse is the family sedan compared to the mustang horse, which is the all-terrain vehicle, the deer and the mountain goat of the Bitterroot Mountains. Past researchers have often looked for the easy routes because they think "family car," not "all-terrain."

And seventh, the researcher needs to answer this question: Why is a new study effort needed? Although there have been many researchers in the past, they have not been successful in precisely locating the actual trail tread of the Lewis and Clark route. This has been for many reasons, but primarily because they lacked access to the aerial photos and the recently compiled journals and maps edited by Gary Moulton that include all journals of the men of the Lewis and Clark Expedition.

The knowledge of mountains, trails, horses, wildlife habitat, stock ranges, surveying, map and aerial photo interpretation, and investigative experience has helped this researcher locate what he believes to be the trail tread followed by the Lewis and Clark party. The bottom line is understanding how the builder of the Lewis and Clark Trail—the Nez Perce Indians and their surefooted mustang horses—traveled across the landscape.

Forest Service Management: An Editorial Comment

How has the Clearwater National Forest managed the historic trails? Are the managers of this historic landmark making decisions in good faith to protect the trails according to federal laws and guidelines? Has the public been informed and consulted before any management decisions are made? Has logging and rerouting of nearly 140 miles of the Lolo Trail since 1960 placed the trail in danger of being de-listed as a national landmark? Can the remaining 27 miles of unaltered Lolo Trail be saved from further changes?

The National Forests of the Northern Region, Living Legacy, describes management efforts on the Lewis and Clark Trail:

> In the past 5 years, Clearwater National Forest staff have scouted and flagged almost all of the original trail across the forest. "Take Pride in America" programs organized by the Forest in the years since 1988 have resulted in more than 650 person-days of volunteer labor that has cleared, marked, and maintained several miles of the route. In the next 5 years, they plan to have opened a route on which

one can still walk in the indented path that perhaps Lewis and Clark themselves trod.[143]

This management goal was never achieved. Instead, the trail segments in the best condition for travel on the Clearwater National Forest were buried and relocated. The Forest Service maintains that this policy was carried out to protect the trail, but a strong argument can be made that they disregarded Section 106 of the National Historic Preservation Act.[144] How can historic trail be relocated? When historic trails are relocated, what happens to their legal standing? When the character of a historic landmark is destroyed or buried, is it in danger of losing its legal protection and being de-listed as a landmark?

The Lolo Trail, just like Gettysburg, or any other national landmark, is history written on the ground. Landmark status means that the trail must be protected to the maximum extent possible in the same condition it was in when registered (1960). A landmark cannot legally be modified or changed to such an extent that it loses its character or imperils its landmark classification.[145] The Lewis and Clark and the Nee-Me-Poo Trails on the Clearwater National Forest are unique because they received protection as a landmark from Congress before they became national historic trails under the 1968 National Trails System Act (NTSA). This act does not supercede the more protective National Historic Preservation Act, which is the governing law for both historic trails within the Lolo Trail System. But, the evidence shows that the Lolo Trail National Historic Landmark was never protected by the Clearwater National Forest, and the documentation of this is recorded on the Forest Service maps since 1960 and on the ground itself.

The Forest Service seems to have two management policies for the historic trails that conflict with one another: staying off the historic trail to protect it and staying on the historic trail for the same reason. Staying off the historic trail means abandoning sections of the trail like Hungery Creek and Gravey Creek. The trail in Gravey Creek (which was located by John Peebles, Ralph Space, and Andrew Arvish) is an abandoned Forest Service trail that could be cut out and used by the public with very little effort. Staying on the historic trail sometimes means rerouting and relocating safe, non-eroded Forest Service trail tread, placing the hiker on brand-new Forest Service built trails. Wendover Ridge switchbacks are one example of this practice. At other times, staying on the trail means actual use of pieces of the historic trail, but the way these newly-constructed trails weave off and on the historic location, no one will know what is historic or not in 30 years.

When the Forest Service builds new trails, they alter the character of the landmark and create long-term problems. The Forest Service needs to look closely at all past changes and consider how the changes will affect the future of the landmark. Looking at the past means that the historic trails must be inventoried, and the changes that have been made since 1960 assessed as to their cumulative effects. Each small change may be insignificant in itself, but when all the small changes are added up, then the cumulative effects of 40 years of mis-management are very apparent when one considers only approximately 27 miles of unaltered trail in the Lolo Trail National Historic Landmark exists today.

How should portions of the historic tread subject to erosion or that contain sensitive heritage sites that need to be protected from the public be managed? Should the historic trails be marked and walked upon or should they be rerouted to protect the character of the trails, thus preventing erosion and loss of artifacts? If the trails are rerouted, should they be marked as the original trail routes or should they be marked as bypass routes? Currently the Forest Service is creating new trails and labeling them with trail sign post logos as the original historic trails, which conflicts with the National Trails System Act, which recommends that "deviations be noted on site."[146] Building new trails and burying the original trail is also in conflict with the National Landmark Statutes, and is not an acceptable management practice in the landmark. The Forest Service has stated that they are rerouting historic trails to protect artifacts; in fact, they have rerouted trails where they have done no heritage inventory. As for the erosion argument, a close examination of reroutes indicates that for most of the trail, erosion is not a problem. Trails have been rerouted where there was no erosion, and the original historic trail located by Ralph Space can no longer be seen.

Safety is another argument used to justify current management practices, yet safety is not mentioned in the

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[143] U.S. Forest Service, *The National Forests of the Northern Region, Living Legacy—*, by Robert D. Baker and others (College Station, Texas: Intaglio Inc., 1993), 230.

[144] See Title 36 *Code of Federal Regulations*, pt. 68.3, 800.4, and 800.5, 2001 ed.

[145] "Special requirements for protecting National Historic Landmarks," Title 36 *Code of Federal Regulations*, pt. 800.10 (a), 2001 ed. This section also states that "the Agency Official, to the *maximum extent possible* undertake such planning and actions as may be necessary to minimize harm to any National Historic Landmark that may be directly and adversely affected by an undertaking" (emphasis mine).

[146] U.S. Forest Service, Clearwater National Forest, *Nez Perce (Nee-Me-Poo) National Historic Trail Interpretive Strategy (Draft)* (Missoula, MT: U.S. Forest Service, 1993), 3.

Landmark statutes as an issue. Ironically, in one instance safety problems were created near Bradford Bridge by moving the trail to the extreme edge of two drop-offs by building switchbacks to achieve a 10% grade. Other National Landmarks realize hazards cannot be eliminated if they conflict with the integrity of the landmark. The visitor to the ancient cliff dwellings at Mesa Verde is warned that he should be in good physical condition to climb Indian-style wooden ladders and should not suffer from vertigo. Providing modern wooden stairs or escalators would alter the character of the Mesa Verde landmark. In the same manner the historic Lolo Trail should not be altered to provide a modern setting. The visitor should understand that he and his horse should be in good physical condition to follow the steep rugged paths that Lewis and Clark trod. It is important to remember that one of the main attractive characteristics of the trail is in its challenge to the visitor and walking on the actual historic trail tread—not a new Forest Service reroute. The integrity of the feel, the atmosphere, and the setting can only be experienced by following the ancient paths.

Finally, the policy of trail relocation must be reexamined to maintain the trail's integrity. It was recommended that the trail from Lolo Campground to Small Prairie Camp (the 1980 location) be marked and a pathway cleared with no tread constructed.[147] Instead, this section was relocated in several places and part of the historic trail tread located by Ralph Space was destroyed by logging and clear-cutting. The relocation of this section of the trail is documented in the 1990 trail description book, stating that "from the junction of Roads 100 and 500 near the Lolo Campground (Bradford Bridge) to Road 500 near Mex Mountain; much of this trail is slated for relocation and extension in the future."[148] This management practice is in direct conflict with the Lolo Trail Guidelines.

The Forest Service needs to look closely at all past changes (especially cumulative effects) and consider how the changes will affect the listing or de-listing of the Lolo Trail as a historic landmark before any new projects are even considered. It would be wise to listen to experts who have been in this situation before and follow their example. Milo McLeod, Lolo National Forest archaeologist, spent two years studying approximately 30 miles of the Lolo Trail in Montana, and discovered quickly that he "could not provide correct cultural resource input for the Missoula district's project planners without first obtaining a detailed knowledge of the Lolo Trail."[149] In his management recommendations, McLeod stated that to meet federal regulations:

> The Lolo Trail should be managed in such a way as to not create any long term adverse impacts to the trail, its environment and the related sites and/or features. Careful attention to project planning detail will be essential so as to not sacrifice or jeopardize any future management options for the Lolo Trail...It should be remembered that the Lolo Trail and its related sites individually may appear insignificant. However, when they are viewed together they represent a very significant part of our national, state, and local heritage.
>
> Not unlike other cultural resources, the Lolo Trail and its sites are nonrenewable. Once a portion is damaged or destroyed it is lost forever.[150]

McLeod's guidelines are also applicable to the Lolo Trail in Idaho, and the Clearwater National Forest managers should also realize that the Lolo Trail is nonrenewable and that great care should be taken to preserve the trail in its historical setting.

Looking Ahead

This book is just the first of a series of reports to answer three questions: where is the L&C Trail? How did this researcher locate the Trail? And, finally, how has the Forest Service managed the trail in the past and in the present, and how will it manage in the future? This volume lays the historic foundation and discusses methodology. Later work will continue the project and cover the Lewis and Clark trail from the Weippe Prairie and beyond.

The Lewis and Clark Trail on the Clearwater National Forest is unique because there is a 90-mile section of the trail on the forest. This total could conceivably grow to 130 miles if management includes the 30 miles from Lolo, Montana, to Packer Meadows and the 10 miles of Idaho State Land and private land on which more of the trail is located. When the Nez Perce Trail (Nee-Me-Poo Trail) is added, the opportunity exists to have a continuous travel

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[147] *Lolo Trail System Implementation Guidelines, Draft*, 44.

[148] U.S. Forest Service, "Trail Descriptions - Pierce Ranger District," R1-91-12, 1990 (Clearwater National Forest, Kamiah Office, Pierce Idaho), 2.

[149] U.S. Forest Service, Lolo/Bitterroot National Forest, *Lolo Trail Study Draft Management Guidelines*, by C. Milo McLeod (n.p.: Lolo National Forest, 1982), 46.

[150] Ibid., 46.

route of 162 miles of primitive Indian trail that could be traveled by foot or by horse.[151] Perhaps now is the time for the Forest Service to honor the trail by viewing it like John Harlan did in 1921:

> It is with a feeling of sobered respect I take up the history of this old trail. For trail it is in the most restricted sense of the word, and as such will ever remain. Its climatic conditions and inhospitable fastnesses forbid its being broadened into a highway. No modern railway project will ever degrade its lofty and rugged nature by girding its rock ribbed ridges with bands of steel. Nor will any idle swain glide o'er its course lolling [on] the soft upholstery of a modern touring car...
>
> They who again would travel this trail, must do so in the same old fashioned manner. They...must be men or women who are willing to sacrifice or forego comforts and conveniences of modern travel. Posses some of the physical stamina and grit of the long ago. Of this I am exceedingly glad. May it ever remain a trail sacred to the memory of the early untutored Red man and the intrepid pathfinding explorers. A remaining link of primitive highway between the present and the past.[152]

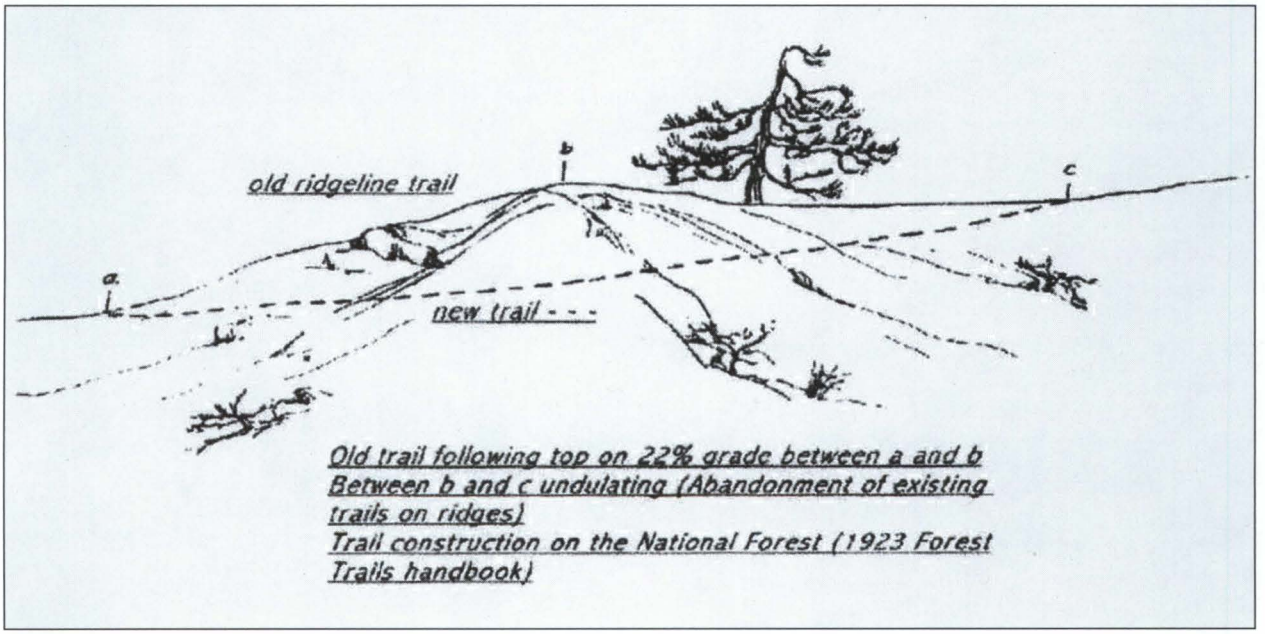

Figure 6: Saddle-to-saddle relocation of trail tread by the Forest Service (Source: U.S. Forest Service, *Trail Construction on the Naional Forests,* 60).

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[151] Lewis and Clark's 1805 route is 76.6 miles from Packer Meadows to the western boundary of the Clearwater National Forest (including the Plum Creek land on the eastern end of the trail), and the 1806 route adds an additional 13 miles. The Northern Nez Perce (Nee-Me-Poo) trail adds another 32 miles from Sherman Peak to Musselshell Meadows, bringing the total to 130 miles on the Clearwater National Forest.

[152] Harlan, "Writings About the Lolo Trail," *Clearwater (Orofino, Idaho) Republican*, 9 September 1921, 1.

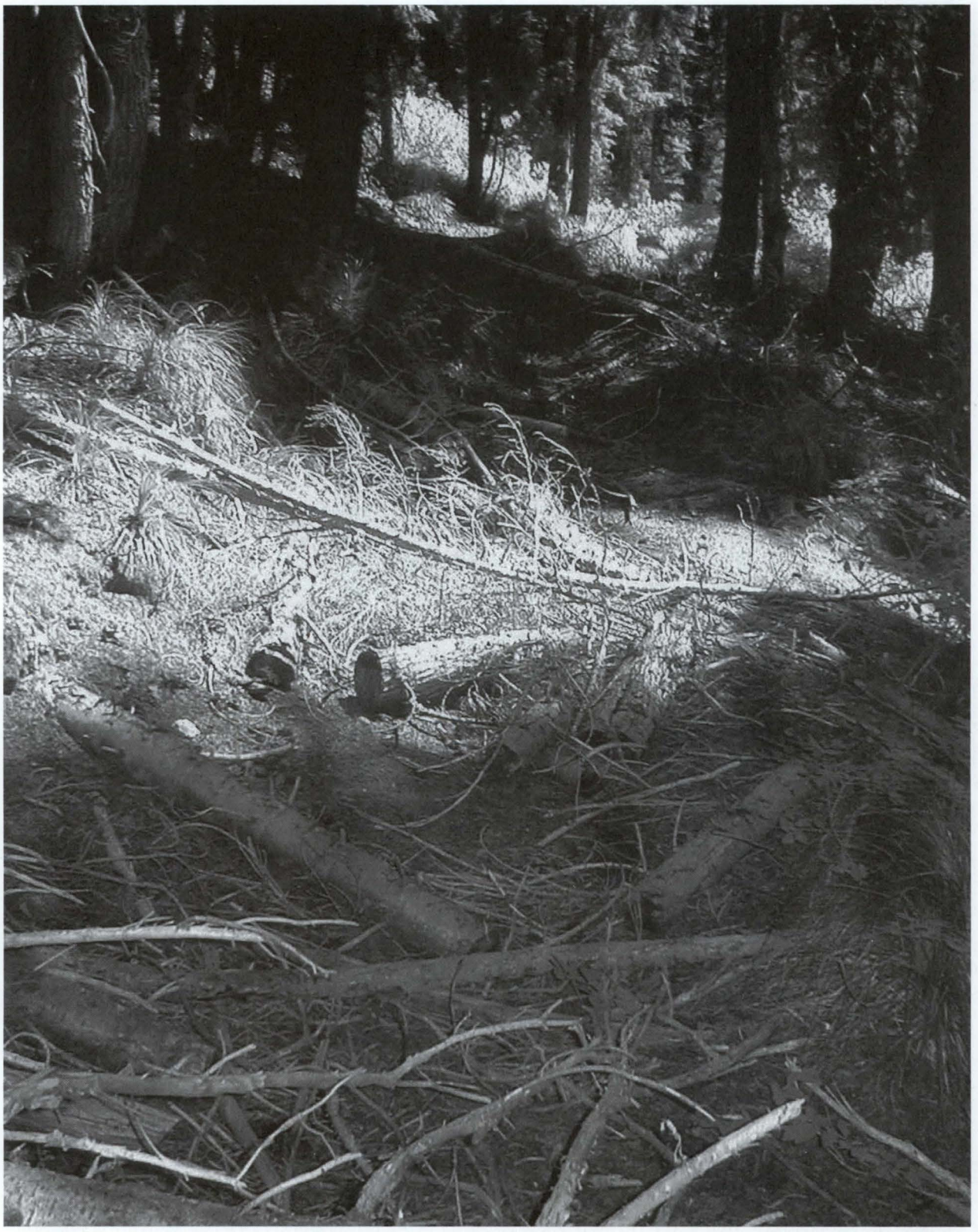

Figure 7: Original Lolo Trail east of Weitas Meadows. Trail tread covered with woody debris in the 1990s by the Forest Service and not approved by the Idaho State Preservation Office (Photo by author, July 2001).

Figure 8: Original Lolo Trail east of Weitas Meadows, abandoned and covered with brush by the Forest Service in the 1990s. A new trail was built after 1990 parallel to the original (Photo by author, July 2001).

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The resources essential for documenting the history of the Lolo and the Lewis and Clark Trails include early-day travelers' accounts of the trails and their maps and diaries. The historic trails researcher has to understand the builders of the trail—the Nez Perce Indians—along with understanding their methods of travel and the use of the mountain mustang, for these are the keys to properly locating the trail. The researcher also needs to be aware of the Bird-Truax work and the Forest Service alterations of the historic trails by rerouting and regrading. Looking at past researchers' results will help the researcher understand where others were sure of the location of the trail and where they were not. The oral interviews conducted with Forest Service personnel and local residents covered past uses along the Lolo and the Lewis and Clark Trails by the Nez Perce Indians, sheep cattle, hunters and horses.

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7.5 minute series topographic blue line maps used by the Forest Service from 1984. Computer –generated USGS 7.5 minute maps known as Cartographic Feature Files (CFF) of Weippe North and Weippe South were obtained from the Forest Service and donated to the University of Idaho Library, where they can be viewed on the "Inside Idaho: Interactive Numeric & Spatial Information Data Engine" at http://inside.uidaho.edu/. Maps with GPS gridlines were designed by the author using the MyTopo.com computer program and printed by MyTopo.com, Red Lodge, Montana.

Oral History

Persons interviewed were Roy Cochrell, Frank (Joe) Larson, William R. (Bud) Moore, Leonard Powers, Judy Space, and Lyle Wilson. Cecil Sanford (retired Game Warden 1948-1975; and Forest Service employee 1926-1934), interviewed 1968.

Information was gathered from the following Clearwater and Lolo National Forest Service personnel:

Steve Armstrong, archaeologist, Nez Perce National Forest, Grangeville, Idaho.
Andrew Arvish, recreation (retired).
Duane Annis, recreation (retired).
James Caswell, Forest Supervisor (past).
Larry Dawson, Forest Supervisor.
Jeff Fee, archaeologist, Kooskia, Idaho (retired).
Linda Fee, recreation.
Doug Gober, North Fork District Ranger.
Mark Hill, zone archaeologist (resigned).
Robbin Johnston, archaeologist.
Rolland Kehlet, trails coordinator (retired).
Tom Keller, recreation & trails, Clearwater National Forest (retired).
Larry Kingsbury, archaeologist, Payette National Forest
Cindy Lane, Pierce and Lochsa District Ranger.
Milo McLeod, archaeologist, Lolo National Forest, Missoula, Montana.
Pete Minard, Programs manager for Technical Services and Ecosystem Management (retired).
Charles Raddon, trails coordinator (retired).
Karl Roenke, former archaeologist, Clearwater National Forest.
Robert Schloss, recreation (retired).
Steve Staub, engineering (retired).
Norm Steadman, engineering (Lewis & Clark Historian).
Jack Stratton, archaeologist (resigned).

APPENDIX

Laws and Guidelines Governing Historic Landmarks

Historic Sites Act of 1935: P.L. 74-292; 49 Stat. 666; 16 U.S.Code 461 et seq. This is the first Federal law to establish historic preservation as a policy of the Federal government. It gave the Secretary of the Interior general authority to take steps to preserve National Historic Landmarks through National Park Service management if necessary. Regulations for the National Historic Landmark Program can be found in Title 36 CFR, part 65 (1983 ed.), "National historic landmarks program—final rule." See also *Federal Register*, vol. 48, no 23, pages 4652-4661.

Protection of Historic and Cultural Properties: Title 36 CFR, ch. VIII, part 800 (2001 ed.). This regulation spells out the steps to manage and protect National Historic Landmarks.

National Historic Preservation Act of 1966, as amended: P.L. 89-65, P.L. 96-515; 80 Stat. 915; 16 U.S. Code 470 et seq. This act expanded the register authorized in the Historic Sites Act of 1935 to include properties of state and local significance. Renamed the National Register of Historic Places, it includes National Historic Landmarks under Section 110(f). Regulations that implement this provision are contained in Title 36 CFR, part 800.10 (2001 ed.).

National Trails System Act of 1968: P.L. 90-543, Section 3(3) as amended. This Act states that the purpose of the National Historic Trail designation is to identify and protect the historic route of the trail and its remnants and artifacts for public use and enjoyment.

National Environmental Policy Act of 1969 (NEPA): P.L. 91-190. The fundamental intent of NEPA is to guarantee full disclosure of environmental impacts so they can be considered in decision making.

Mining in National Parks Act of 1976: P.L. 94-429. Section 9(a) of this act requires that whenever a mining operation threatens a National Historic landmark in or out of a National Park on public or private lands, the Secretary of the Interior is to ask the Advisory Council on Historic Preservation (ACHP) for its review and recommendations. The recommendations from ACHP would include alternative measures that may be taken by the United States to mitigate or abate irreparable loss or destruction of the national historic landmark.

National Forest Management Act of 1976 (NFMA): P.L. 94-588. Section 5 of NFMA requires the development and maintenance of comprehensive inventories of Forest resources. Section 6 requires the development and maintenance of forest land and resource management plans.

American Indian Religious Freedom Act of 1978 (AIRFA): P.L. 95-341. This act requires the protection and preservation of American Indians' inherent right of freedom to believe, express, and exercise their traditional religions. This includes access to sites, use and possession of sacred objects, and the freedom to worship through ceremonials and traditional rites. Section 2 states that federal agencies will evaluate their policies and procedures in consultation with native traditional religious leaders to protect and preserve Native American religious cultural rights and practices.

Archaeological Resources Protection Act of 1979 (ARPA): P. L. 96-95; 16 U.S. Code 1b. The purpose of this act is to protect archaeological resources and sites on federally managed lands. It prohibits the unauthorized disturbance of archaeological sites and artifacts, imposes fines and jail sentences for violations, and establishes a process to issue permits for authorized disturbances and collection by qualified individuals and organizations.

Is the trail tread of the Lewis and Clark historic trail an archeological site? The qualifications require that the proposed site be associated with events that have made significant contribution to our history; be associated with historical people; has native American Indian trail characteristics, or that it may be likely to yield information important in prehistory or history. The Lolo and the Lewis and Clark Trail meet these specifications, and thus

the trail treads are an archeological site that needs the utmost protection.

Lewis and Clark National Historic Trail: Comprehensive Plan for Management and Use: U.S. Department of the Interior, National Park Service (January 1982). It provides a framework from which to develop specific management approaches for the Lewis and Clark Historic Trail.

Idaho Code 1984 Sec. 18-7027 and Sec. 27-501 (Indian Grave Bill): This state law was passed to protect marked and unmarked human graves on private and public lands in Idaho. It is a misdemeanor to disturb graves and a felony to excavate graves with the intent of selling skeletal material or grave goods.

Lolo Trail System Implementation Guidelines, Draft: U.S. Forest Service, Clearwater National Forest, 1985. Provides the framework for managing the historic Nee-Me-Poo and the Lewis and Clark Trails on the Clearwater National Forest, and was intended to be a supplement to the Forest Plan. While outdated in terms of vegetation and road management, this document contains abundant information on history, laws, and other references. It provides excellent guidance in heritage resource and recreation management (beginning on page 21) and guidelines on managing specific trail segments (beginning on page 43).

Clearwater National Forest Plan: U.S. Forest Service, Northern Region, 1987. This plan describes the management of Historic Corridors (see Chapter III).

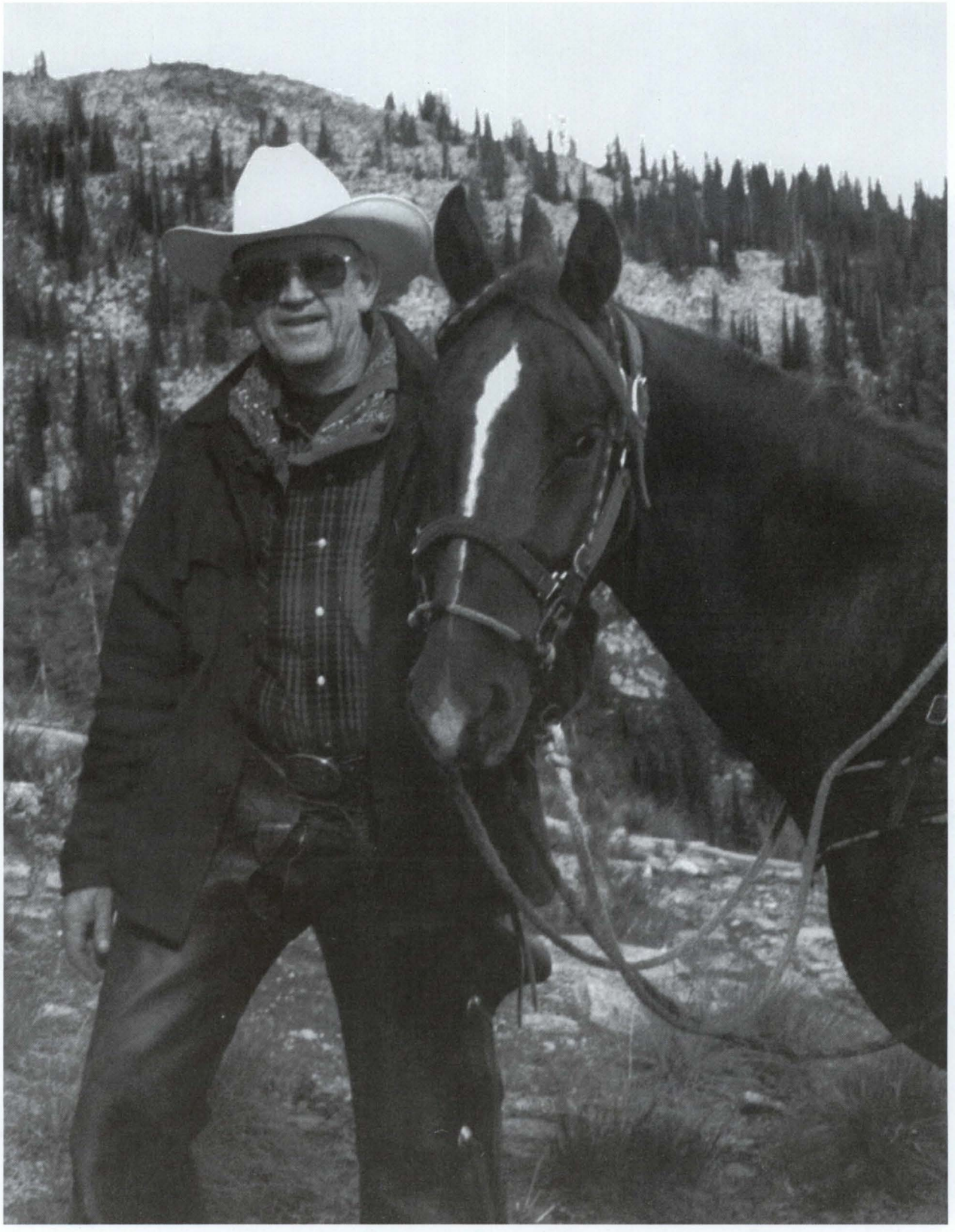

Figure 9: Gene Eastman on the abandoned old Indian ridge-line trail to McConnell Mountain with his Morgan horse Sonny (Photo by Mollie Eastman, 17 September 1998).

Few people probably know the Lolo Trail better than Idahoans Gene and Mollie Eastman, and they argue convincingly that the Lewis and Clark routes designated by the National Forest Service do not, in fact, trace the explorers' actual steps. They point out that Lewis and Clark kept mostly to ridge lines and followed Indian trails, "which followed the best ground in a direct manner" and did not employ switchbacks, *Bitterroot Crossing* reviews the long history of the Lolo system and the many attempts to locate the actual route taken by Lewis and Clark, beginning with Olin D. Wheeler in 1902.

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